Toyota TechStream / PCS & VCH Data
For IPTM 2021 Symposium on Traffic Safety

This presentation developed by

Brad Muir

of

CRASH DATA SPECIALISTS

WWW.CDR-TRAINERS.COM

Edited for time and Presented by

Rick Ruth

of

Institute of Police Technology and Management

IPTM

FOUNDED 1980

Toyota TechStream / PCS & VCH Data

- Brad Muir developed this presentation for his training classes and has presented it at regional conferences like WATAI
- Brad lives in Canada and the border is closed, he cannot come to the US thanks to Coronavirus restrictions
- He has graciously allowed it to be presented here at the IPTM Symposium by Rick Ruth, saving Rick from duplicating the effort!
- Attendees will receive a PDF

of the 270+ slide presentation

For time reasons not all slides

Will be presented

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Brad Muir ACTAR #1596

- 30 Years Police Experience
  - 21 Years as Reconstructionist with Ontario (Canada) Provincial Police
  - Crash Recon Unit since 1995
  - Supervised of one of 4 crash teams
- CDR Background
  - 300+ CDR/EDR Crash Tests
  - Using CDR data since 1997
  - CDR Instructor since 2003

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Crash Data Specialists

- Formed in 2011
- Using CDR and Crash Testing for over 19 years
- Experience pre-CDR from 97+

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Presenter Intro

Rick Ruth

- Ford Motor Company Product Development Engineering - Airbag Design and Field Performance 1973-2006
- Ford's lead field user of EDR's, Ford EDR policy committee, Ford Rep to SAE & EDR Committee, Ford Rep to Bosch CDR
- 2007-present, Ruth Consulting LLC – Airbags and EDR in Recon
- Teach EDR tech, EDR Analyst 1 & 2 classes for IPTM and EDR for SAE. Presents regularly at national and regional conferences on reconstruction, on EDR's.
- Does research on EDR accuracy – 19 publications
- Helps prosecutors and cops nationwide in EDR cases – mostly for free (charge for written reports and testimony).
- Takes civil cases in airbags and EDR aspect of Recon – mostly when referred by reconstructionists he has trained, who recognize when specialty expertise is needed.

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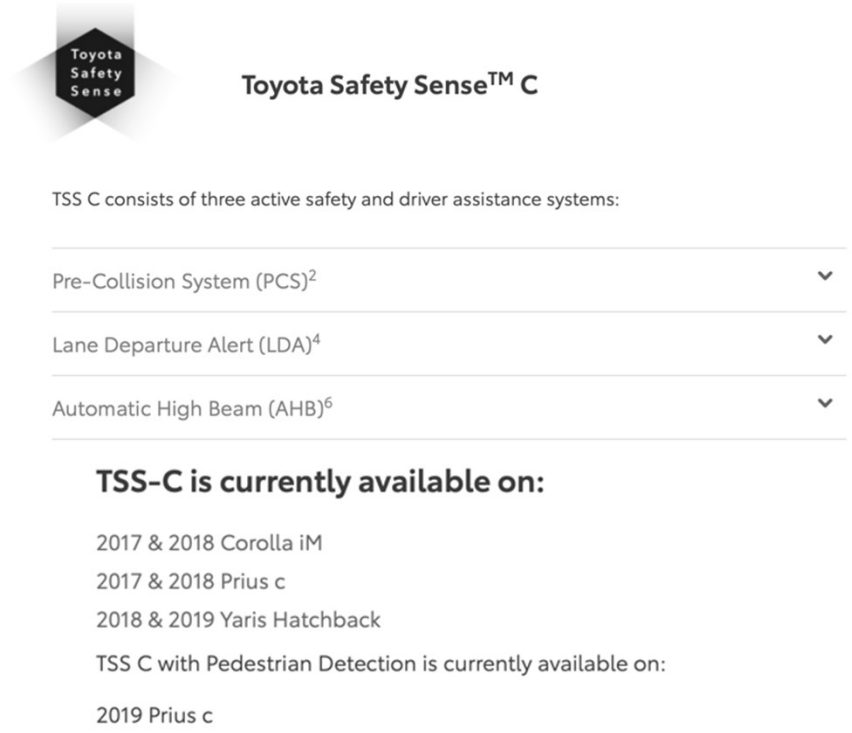
Toyota Safety Sense Data Retrieval

Toyota Safety Sense C

- First widely available version of Toyota's advanced safety system
- Some older Lexus vehicles had a similar system
  - Began in 2007
  - Widely available as of 2015 on Lexus branded vehicles



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Toyota Safety Sense P

Toyota Safety Sense™ P

TSS P consists of four active safety and driver assistance systems:

- Pre-Collision System with Pedestrian Detection (PCS)[3]
- Lane Departure Alert (LDA)[4]
- Lane Departure Alert with Steering Assist (LDA w/SA)[5]
- Automatic High Beam (AHB)[6]
- Dynamic Radar Cruise Control (DRCC)[7]

TSS-P is currently available on:

2017 - 2020 Avalon
2018 & 2019 C-HR
2018 - 2020 Camry & Camry Hybrid
2018 & 2019 Corolla
2018 & 2019 Highlander & Highlander Hybrid
2017 - 2020 Prius
2017 - 2020 Prius Prime
2018 Prius v
2017 & 2018 RAV4 & RAV4 Hybrid
2020 4Runner
2018 - 2020 Sequoia
2019 & 2020 Sienna
2018 - 2020 Tacoma
2018 - 2020 Tundra

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2016 Land Cruiser	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Standard	6156 – 5Dr. Wagon 4WD V8
2016 Avalon and Avalon Hybrid	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Optional on Limited and Limited Hybrid models with Toyota Safety Sense™ P option (TY)	3554 – 4Dr. Sedan Limited V6 3514 – 4Dr. Sedan Limited Hybrid
2016 RAV4 and RAV4 Hybrid	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Standard on Limited and Limited Hybrid. Optional on XLE and XLE Hybrid models with Convenience Package (CP) and on SE with Advanced Technology Package (AT)	4450 – 5Dr. Wagon Limited 4452 – 5Dr. Wagon Limited AWD 4454 – 5Dr. Wagon Limited Hybrid 4440 – 5Dr. Wagon XLE 4442 – 5Dr. Wagon XLE AWD 4444 – 5Dr. Wagon XLE Hybrid 4470 – 5Dr. Wagon SE 4472 – 5Dr. Wagon SE AWD
2016 Prius	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Full-Speed Range Dynamic Radar Cruise Control (DRCC)	Standard on Prius Three Touring and Prius Four Touring. Optional on Prius Three and Prius Four with Advanced Technology Package (AT)	1226 – 5Dr. Sedan Three Touring 1228 – 5Dr. Sedan Four Touring 1225 – 5Dr. Sedan Three 1227 – 5Dr. Sedan Four
2016 Prius *c*	**TSS-C** • Pre-Collision System (PCS) • Lane Departure Alert (LDA) • Auto High Beams (AHB)	Optional on Prius c Three and Prius c Four with any additional package: Prius c Three with Deluxe Moonroof and Safety Package or Wheel and Safety Package. Prius c Four with Upgraded Wheel and Safety Package	1205 – 5Dr. Hatchback Three 1207 – 5Dr. Hatchback Four

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VEHICLE	TSS-C or TSS-P	FITMENT / AVAILABILITY	MODEL NUMBERS
2017 Land Cruiser	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Standard	6156 – 5Dr. Wagon 4WD V8
2017 Avalon and Avalon Hybrid	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Dynamic Radar Cruise Control (DRCC)	Standard	All Model Numbers
2017 Corolla		Standard	All Model Numbers
2017 Highlander and Highlander Hybrid		Standard	All Model Numbers
2017 RAV4 and RAV4 Hybrid		Standard	All Model Numbers
2017 Prius	**TSS-P** • Pre-Collision System (PCS) with Pedestrian Detection function • Lane Departure Alert (LDA) with Steering Assist function • Auto High Beams (AHB) • Full-Speed Range Dynamic Radar Cruise Control (DRCC)	Standard	All Model Numbers
2017 Prius Prime		Standard	All Model Numbers
2017 Corolla iM	**TSS-C** • Pre-Collision System (PCS) • Lane Departure Alert (LDA) • Auto High Beams (AHB)	Standard	All Model Numbers
2017 Prius *c*		Standard	All Model Numbers
2017 Yaris		Standard	All Model Numbers

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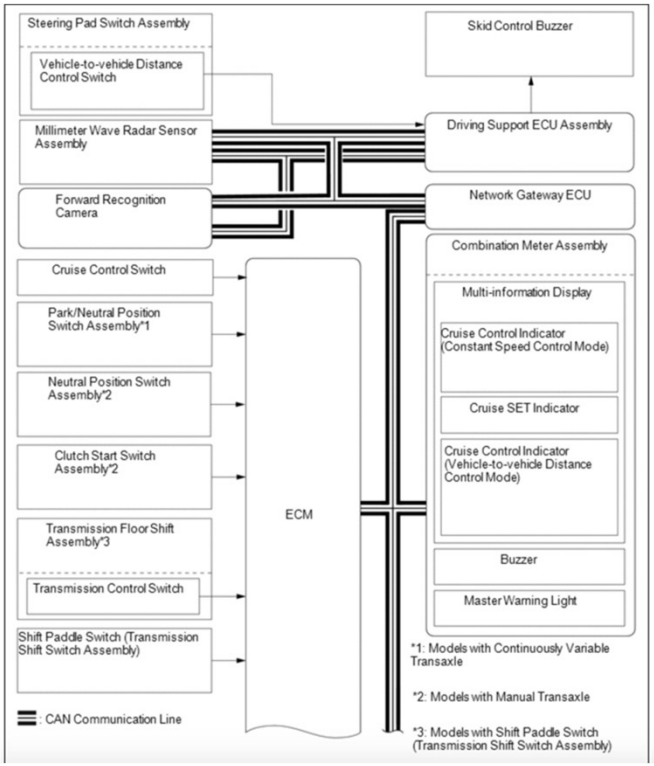
TSS 1.0 System Diagram

- TSS-P
- TSS-C
  - Forward Recognition Camera
  - Millimeter Wave Radar (some)
  - Network Gateway
  - Driving Support ECU
  - ACM
  - Skid Control ECU
  - PCM / Combination Meter



TSS 1.0 – Different Systems

- DENSO System:
- Camera
  - image storage location
- Driving Support ECU
  - FFD Data related to PCS operation
- ACM
  - Vehicle Control History (VCH) storage location
- PCM – Vehicle VIN

- Continental System:
- Camera
  - image storage location
- Millimeter Wave Radar
  - FFD Data related to PCS operation
- ACM
  - Vehicle Control History (VCH) storage location
- PCM – Vehicle VIN

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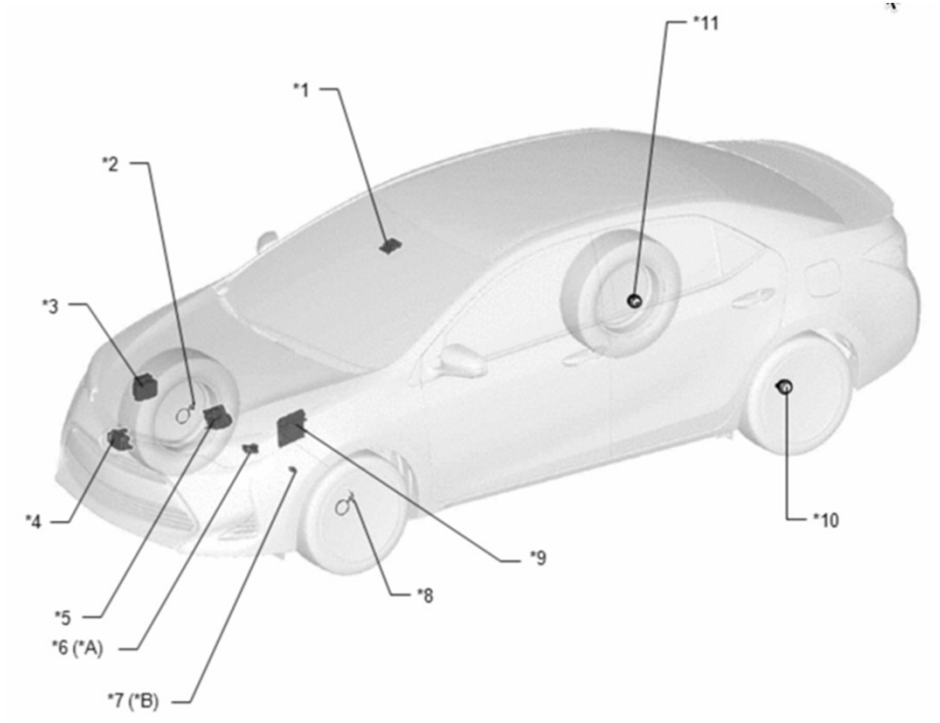
2019 Toyota Corolla – TSS 1.0



*A	Models with Continuously Variable Transaxle	*B	Models with Manual Transaxle
*1	FORWARD RECOGNITION CAMERA	*2	FRONT SPEED SENSOR RH
*3	BRAKE ACTUATOR ASSEMBLY • SKID CONTROL ECU	*4	MILLIMETER WAVE RADAR SENSOR ASSEMBLY
*5	THROTTLE BODY ASSEMBLY • THROTTLE POSITION SENSOR • THROTTLE CONTROL MOTOR	*6	PARK/NEUTRAL POSITION SWITCH ASSEMBLY
*7	NEUTRAL POSITION SWITCH ASSEMBLY	*8	FRONT SPEED SENSOR LH
*9	ECM	*10	REAR SPEED SENSOR LH
*11	REAR SPEED SENSOR RH	-	-

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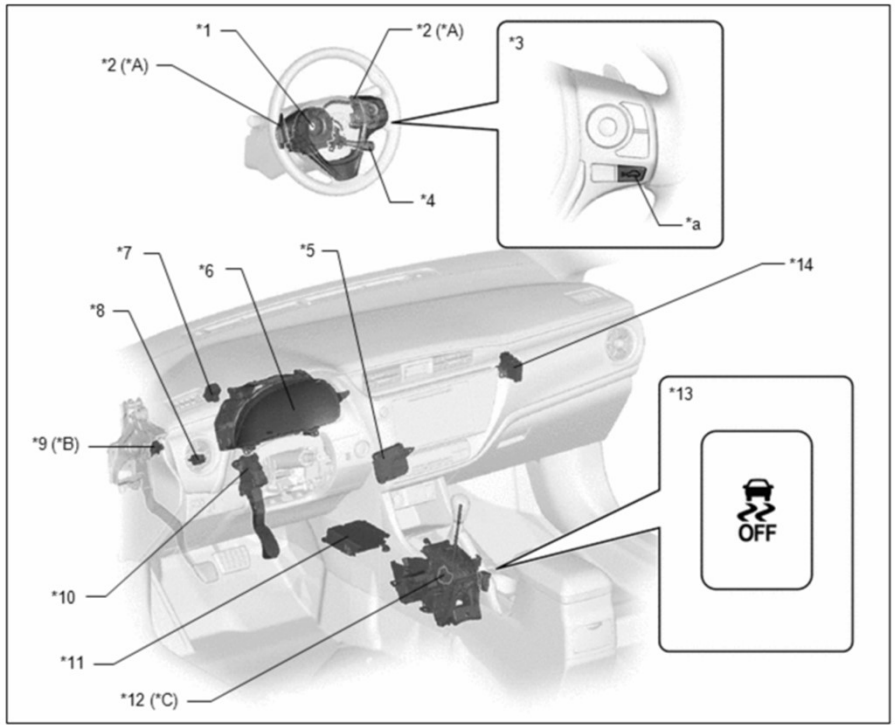
2019 Toyota Corolla – TSS 1.0



*A	Models with Shift Paddle Switch (Transmission Shift Switch Assembly)	*B	Models with Manual Transaxle
*C	Models with Continuously Variable Transaxle	-	-
*1	SPIRAL CABLE WITH SENSOR SUB-ASSEMBLY • STEERING SENSOR	*2	SHIFT PADDLE SWITCH (TRANSMISSION SHIFT SWITCH ASSEMBLY)
*3	STEERING PAD SWITCH ASSEMBLY	*4	CRUISE CONTROL SWITCH
*5	DRIVING SUPPORT ECU ASSEMBLY	*6	COMBINATION METER ASSEMBLY • MULTI-INFORMATION DISPLAY • MASTER WARNING LIGHT • BUZZER
*7	SKID CONTROL BUZZER	*8	STOP LIGHT SWITCH ASSEMBLY
*9	CLUTCH START SWITCH ASSEMBLY	*10	ACCELERATOR PEDAL SENSOR ASSEMBLY
*11	AIRBAG SENSOR ASSEMBLY • YAWRATE SENSOR	*12	TRANSMISSION FLOOR SHIFT ASSEMBLY • TRANSMISSION CONTROL SWITCH
*13	VSC OFF Switch	*14	NETWORK GATEWAY ECU
*a	Vehicle-to-vehicle Distance Control Switch	-	-

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TSS 1.0 – TSS-C and TSS-P Overview

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PCS Camera – Still Image Sequence

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Toyota Safety Sense 2.0

Toyota Safety Sense

Toyota Safety Sense™ 2.0

TSS 2.0 builds on TSS P and consists of six active safety and driver assistance systems.

- Pre-Collision System with Daytime/Low-Light Vehicle and Pedestrian Detection, plus Daytime Bicycle Detection [3]
- Lane Departure Alert with Steering Assist and Road Edge Detection (LDA)[5]
- Automatic High Beam (AHB)[6]
- Lane Trace Assist[8]
- Full Speed Range Dynamic Radar Cruise Control (DRCC)[7]

TSS 2.0 is currently available o

2019 Corolla Hatchback

2019 AND 2020 RAV4 & RAV4 Hybrid

2020 Corolla & Corolla Hybrid

Coming soon on more 2020 MY models

TSS 2.0 Overview

TSS 2.0 Overview

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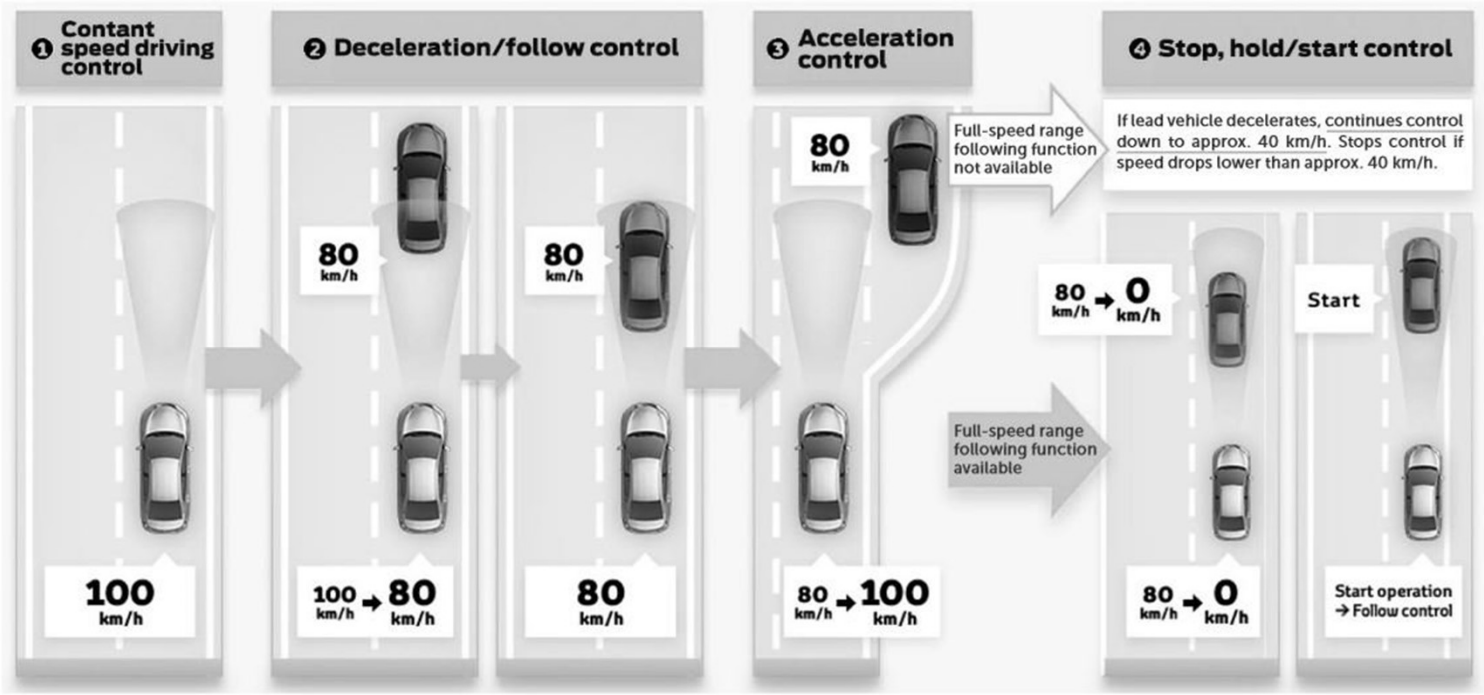
TSS 2.0 Radar Cruise Features



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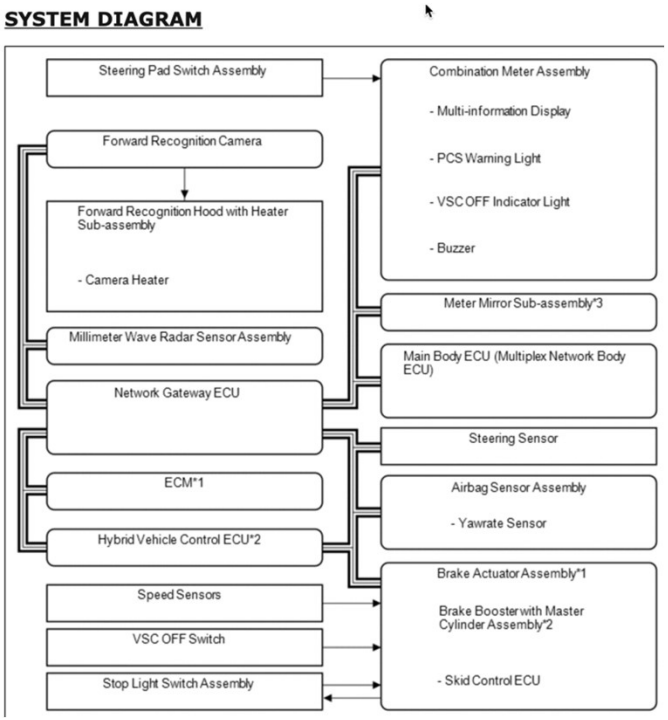
TSS 2.0 System Diagram

- TSS 2.0
  - Forward Recognition Camera
  - Millimeter Wave Radar (some)
  - Network Gateway
  - Main Body ECU
  - ACM
  - Skid Control ECU
  - PCM / Combination Meter



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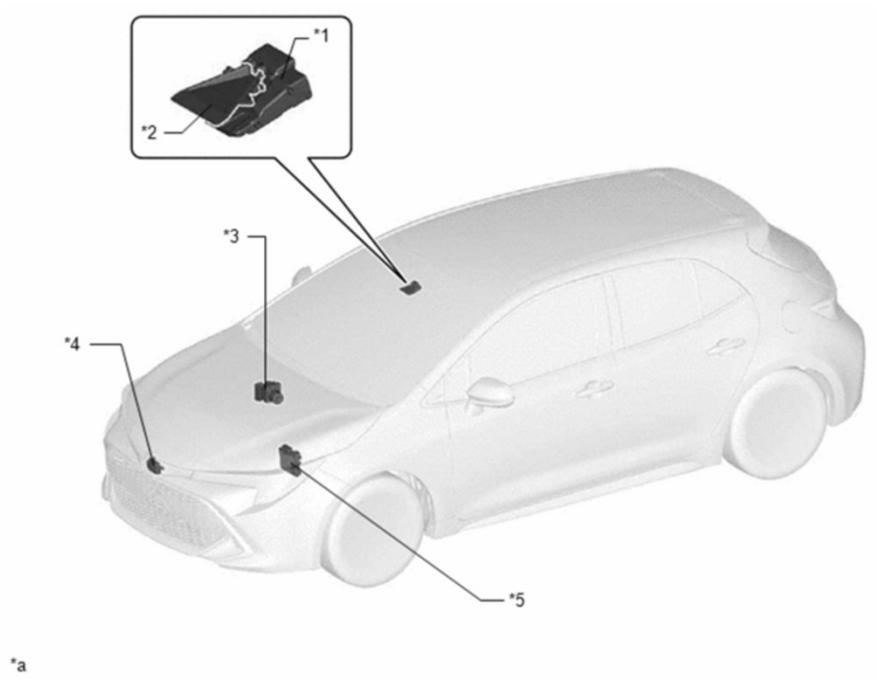
2019 Corolla Hatchback - TSS 2.0



*1	FORWARD RECOGNITION CAMERA	*2	CAMERA HEATER (FORWARD RECOGNITION HOOD WITH HEATER SUB-ASSEMBLY)
*3	BRAKE ACTUATOR ASSEMBLY • SKID CONTROL ECU	*4	MILLIMETER WAVE RADAR SENSOR ASSEMBLY
*5	ECM	-	-
*a	The illustration shown is an example only.	-	-

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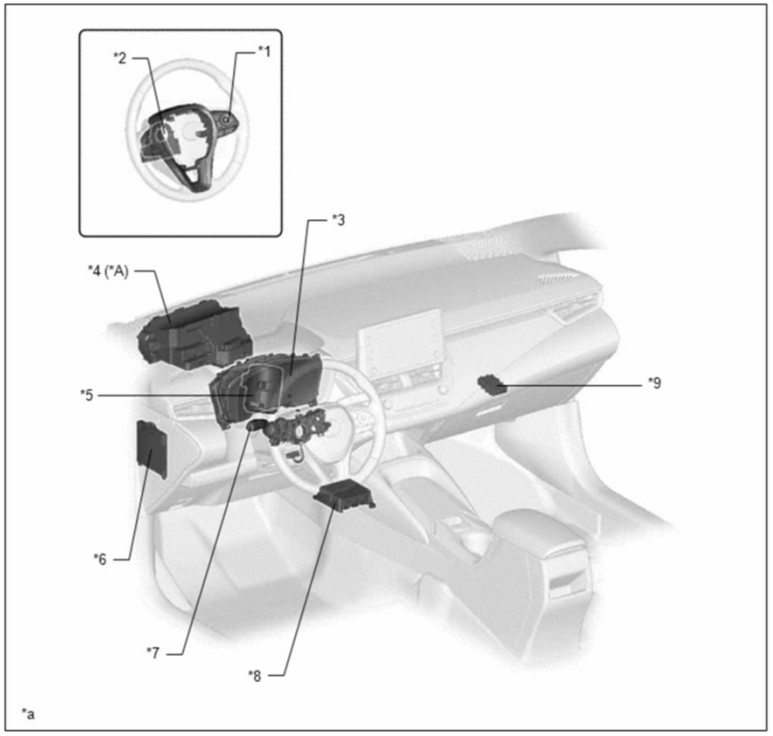
2019 Corolla Hatchback - TSS 2.0



*A	Models with Headup Display	-	-
*1	STEERING PAD SWITCH ASSEMBLY • LDA MAIN SWITCH • METER CONTROL SWITCH	*2	STEERING SENSOR
*3	COMBINATION METER ASSEMBLY • LDA INDICATOR • MULTI-INFORMATION DISPLAY • BUZZER	*4	METER MIRROR SUB-ASSEMBLY
*5	POWER STEERING ECU WITH MOTOR ASSEMBLY	*6	MAIN BODY ECU (MULTIPLEX NETWORK BODY ECU)
*7	TURN SIGNAL SWITCH	*8	AIRBAG SENSOR ASSEMBLY • YAWRATE SENSOR
*9	CENTRAL GATEWAY ECU (NETWORK GATEWAY ECU)	-	-
*a	The illustration shown is an example only.	-	-

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TSS 2.0 – Event Date / Time

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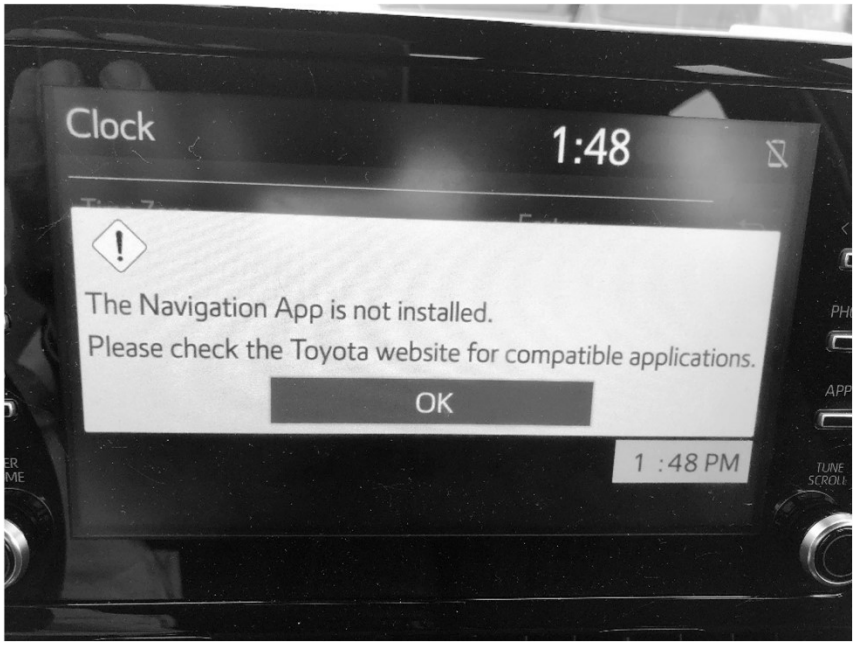
TSS 2.0 – Event Date / Time

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	ODO
108	00:00:09.8	-	2019-02-08-16:38:26.200	Accelerator pedal opening angle is medium or higher immediately after shifting to R	162
107	00:10:04.3	-	2019-02-08-16:02:30.100	Sudden braking history	162
107	00:10:04.4	-	2019-02-08-16:02:30.200	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	162
107	00:10:04.4	-	2019-02-08-16:02:30.200	ABS operation history	162
107	00:10:04.3	-	2019-02-08-16:02:30.100	PCS operation history (when acceleration above certain amount is detected)	162
107	00:06:41.1	-	2019-02-08-15:59:06.900	PCS operation history (pre-collision brake assist operation)	161
105	00:06:42.9	-	2019-02-07-23:28:42.500	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	142
104	00:25:42.8	-	2019-02-07-21:27:21.100	Accelerator high position in mid to high speed	114
104	00:05:30.8	-	2019-02-07-21:07:09.100	TRC operation history	100

Back



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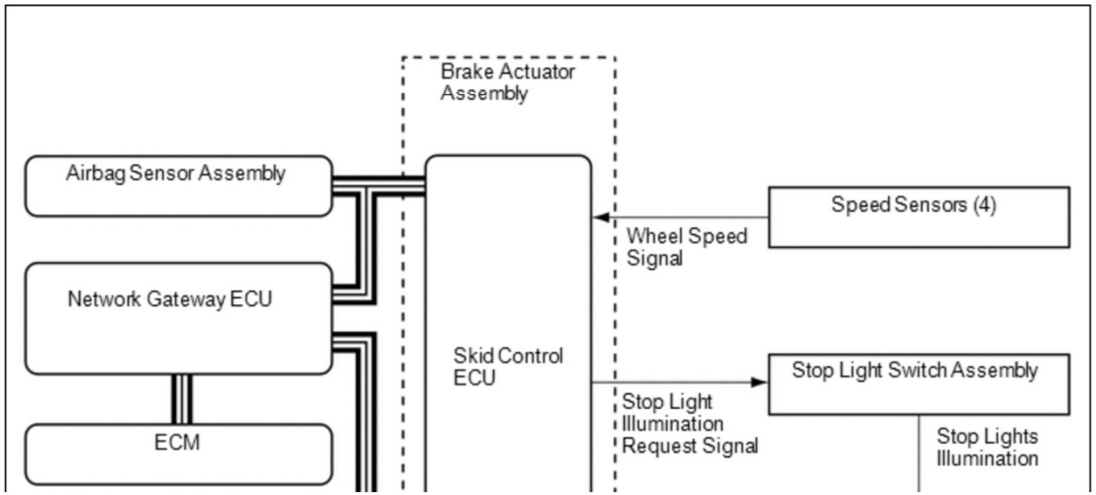
Secondary Collision Brake !!

SECONDARY COLLISION BRAKE

SYSTEM CONTROL

(a) The secondary collision brake system lessens damage from a secondary collision by controlling the automatic brakes and stop light illumination after a collision occurs.

(b) When the airbag sensor assembly detects a collision and operates the airbags, it sends a collision detection signal to the skid control ECU. When the skid control ECU receives a collision detection signal, it determines that operation can start, operates the brake actuator and performs braking. Simultaneously, the skid control ECU sends a stop light illumination request to the stop light switch assembly, illuminating the stop lights.



TSS 2.5 / 2.5+

- 2021 Camry and Highlander and Lexus IS to start
- Camera / Millimetre Wave Radar optimized improve response range and enhanced low light pedestrian and daytime bicycle recognition.
- Pre-Collision System adds new Features

TSS-C	*TSS-P*	*TSS 2.0*	*TSS 2.5+*
TSS-C features Pre-Collision System, Lane Departure Alert, and Automatic High Beam.	TSS-P enhances TSS-C, and adds Dynamic Radar Cruise Control to its suite of systems.	Builds on TSS-P, and consists of up to six active safety and driver assistance systems.	TSS 2.5+ adds new capabilities to the Pre-Collision and Dynamic Radar Cruise Control Systems, and enhances Lane Tracing Assist.

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Lexus Safety System+

LEXUS SAFETY SYSTEM+

Lexus Safety System+ is a comprehensive suite of advanced technologies and intelligent sensors that help keep you protected and create a more confident driving experience to focus on the road ahead[2].

LSS+[2] (RC, RC F, LC, GX, LX):
- Pre-Collision System with Daytime Pedestrian Detection
- Lane Departure Alert with Steering Assist
- Automatic High Beam
- All-Speed Dynamic Radar Cruise Control

LSS+ 2.0[2] (ES, UX, NX, RX, LS):
- Pre-Collision System with Low-light Pedestrian and Daytime Bicycle Detection
- Lane Tracing Assist
- Automatic High Beam
- All-Speed Dynamic Radar Cruise Control

LSS+ 2.5[2] (IS)
- Pre-Collision System with Pedestrian Detection, Intersection Support and Emergency Steering
- Lane Departure Alert with Steering Assist
- Automatic High Beam
- All-Speed Dynamic Radar Cruise Control with Curve Speed Management

LSS+ A[2] (Select LS Packages, LSh):
- Pre-collision System with Pedestrian Alert, Active Steering Assist and Front Cross Traffic Alert
- Lexus CoDrive with automatic high beam

AVAILABILITY BY MODEL

21 MODEL YEAR	LEXUS SAFETY SYSTEM+	LEXUS SAFETY SYSTEM+ 2.0	LEXUS SAFETY SYSTEM+ 2.5
UX		✓	
NX		✓	
RX/RXL		✓	
GX	✓		
LX	✓		
ES		✓	
IS			✓
LS		✓	
LC	✓		
RC	✓		
RC F	✓		

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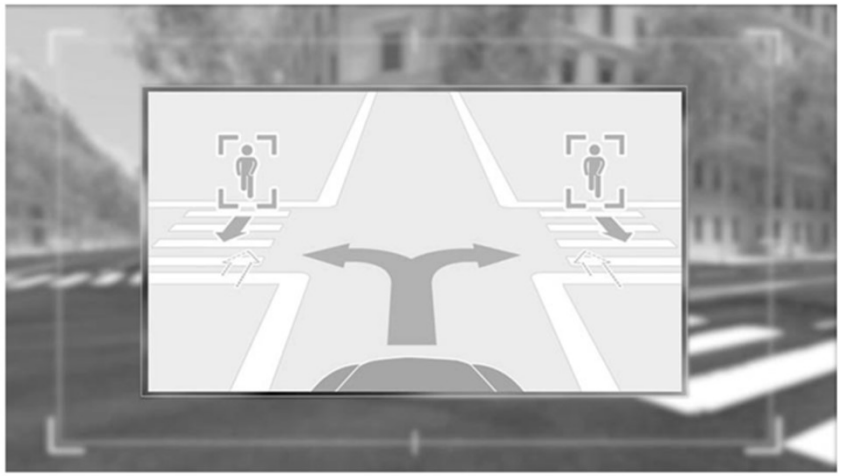
Intersection Support

- Left turn at an intersection, the system
- Capable of detecting an oncoming vehicle / activating automatic braking when required.
- Detects pedestrians crossing the road when turning left or right at an intersection / automatic braking when required.



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Emergency Steering Assist

- Updated in the TSS 2.5+ package
- Provides additional steering input to correct the vehicle's trajectory and keep within Lane
- More stability when emergency steering is active while avoiding a pedestrian, bicyclists, or another vehicle.

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Curve Speed Management

- When active, an icon will be displayed in the multi-information display and head-up display.
- The vehicle will automatically return to the preset speed after passing the curve.

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Toyota Safety Sense 2.5+ Overview

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TSS Version Comparison / Reference

	TSS 2.5+	TSS 2.5	TSS 2.0
Pre-Collision System			
Daylight/Lowlight Vehicle & Pedestrian Detection	✓	✓	✓
Daytime Cyclist Detection	✓	✓	✓
Emergency Steering Assist	Pedestrian Cyclist Vehicle	Pedestrian	
Front-to-Front Risk Detection from Oncoming Vehicles	✓		
Left Turn Oncoming Traffic Detection	✓	✓	
Left/Right Turn Oncoming Pedestrian Detection	✓	✓	
Dynamic Radar Cruise Control			
Turn Signal-Linked Control	✓ Enhanced	✓	✓
Curve Speed Management	✓		
Lane Keeping			
Lane Departure Alert w/ Steering Control	✓	✓	✓
Lane Tracing Assist	✓ Enhanced	✓ Enhanced	✓
Sway Warning System	✓	✓	✓
Lights			
Automatic High Beam	✓	✓	✓

PRE-COLLISION SYSTEM

TSS-C – PRE-COLLISION SYSTEM[2]

Pre-Collision System (PCS) provides collision avoidance or collision mitigation support in certain situations where a frontal collision is likely to occur, and in some cases, will automatically apply the brakes to help the driver.

Watch the TSS-C Pre-Collision System video

TSS-P – PRE-COLLISION SYSTEM WITH PEDESTRIAN DETECTION[3]

PCS has been enhanced to also help detect pedestrians in certain situations.

Watch the Pre-Collision System with Pedestrian Detection video

TSS 2.0 – PRE-COLLISION SYSTEM WITH DAYTIME/LOW-LIGHT VEHICLE AND PEDESTRIAN DETECTION, PLUS DAYTIME BICYCLE DETECTION[3]

PCS is also designed to help detect bicyclists in daytime situations, as well as vehicles and pedestrians in both daytime and low-light situations.

Watch the Pre-Collision System with Pedestrian and Bicycle Detection Video

TSS 2.5+ – PRE-COLLISION SYSTEM WITH PEDESTRIAN (DAYTIME & LOW-LIGHT) & DAYTIME BICYCLE DETECTION, INTERSECTION SUPPORT & EMERGENCY STEERING ASSIST[3]

PCS in TSS 2.5+ may warn you of oncoming pedestrians or vehicles when making a turn at an intersection and provide emergency braking in certain conditions. It also features Emergency Steering Assist, which is designed to stabilize the driver's emergency steering maneuver and keep the vehicle in its lane when avoiding a pedestrian, cyclist or another vehicle.

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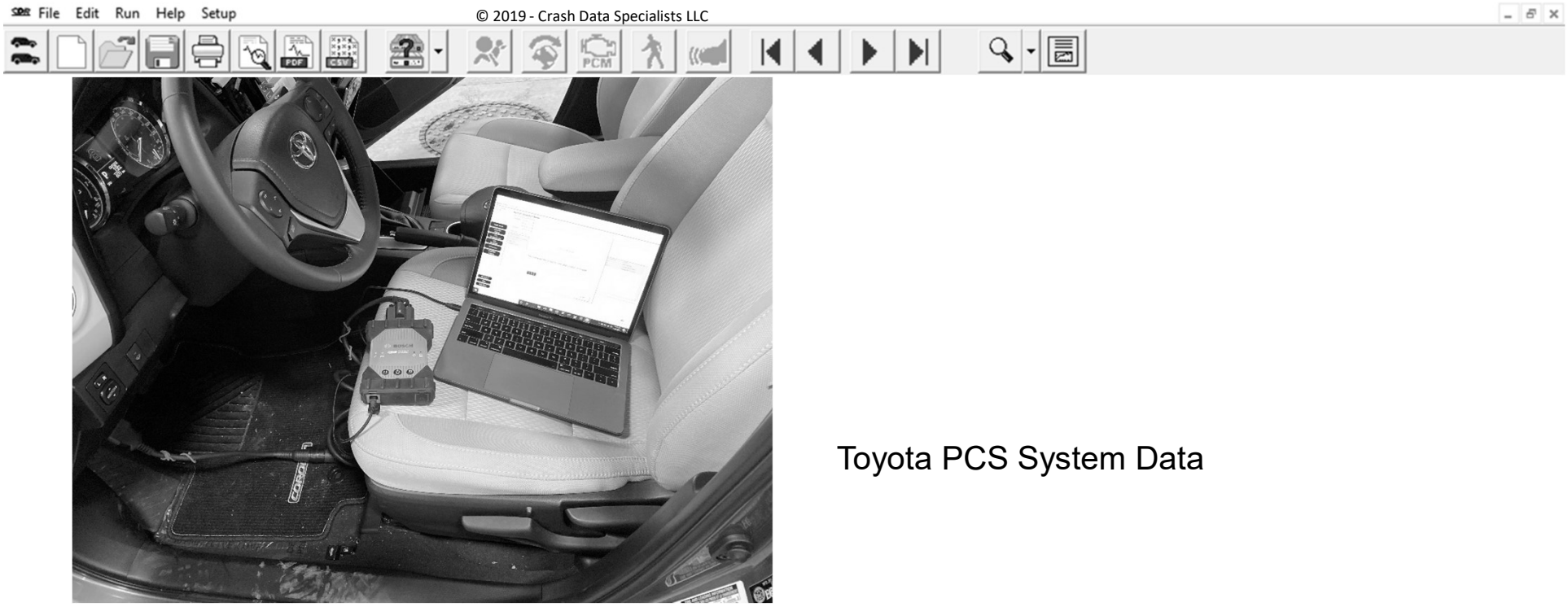

Toyota PCS System Data

Retrieving Toyota Vehicle Control History, Images and FFD records

What Data Can I get?

- Vehicle Control History
  - With Pre & Post Trigger data it may help you better understand pre- and post- collision vehicle inputs and behavior
- MAY contain images
  - Similar pre/post trigger timing as VCH
  - Low-Res black & white images
- TSS 2.0 data easily reconciled with corresponding CDR data
- VCH may exist when event delta-v < 8km/h threshold

TSS 1.0

- For Images to be recorded and linked the VEHICLE must brake
- A PCS warning buzzer is not sufficient for images to be linked with VCH and recorded
- If vehicle begins to brake and driver responds, then related images should be stored
- CHECK: perform CDR download FIRST. Look for deceleration with no driver input: i.e. no brake switch = on, no brake oil pressure, but speed drops more than idling down

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TSS 1.0 – Will I Have Images / PCS FFD?

CDR Pre-Crash Data

Pre-Crash Data, -5 to 0 seconds (Most Recent Event, TRG 1)

Time (sec)	-4.55	-4.05	-3.55	-3.05	-2.55	-2.05	-1.55	-1.05	-0.55	-0.05	0 (TRG)
Vehicle Speed (MPH [km/h])	19.9 [32]	19.9 [32]	19.9 [32]	20.5 [33]	20.5 [33]	18.6 [30]	9.3 [15]	5 [8]	0.6 [1]	0 [0]	0 [0]
Accelerator Pedal, % Full (%)	13.5	18.0	23.5	21.5	14.5	23.5	67.0	0.0	0.0	0.0	0.0
Percentage of Engine Throttle (%)	3.0	4.0	4.5	5.5	3.0	4.5	0.0	3.0	0.0	0.0	0.0
Engine RPM (RPM)	1,300	1,300	1,300	1,300	1,300	1,200	1,000	1,800	900	700	700
Service Brake, ON/OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	ON	ON	ON	ON
Brake Oil Pressure (Mpa)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.44	4.37	7.87	6.34
Longitudinal Acceleration , VSC Sensor (m/sec^2)	0.359	0.287	0.287	0.287	0.502	-4.450	-8.973	-1.005	-4.307	8.973	8.973
YawRate (deg/sec)	0.00	-0.49	0.00	0.00	-0.49	-0.49	1.46	-0.49	-0.49	0.00	3.90
Steering Input (degrees)	3	3	3	3	3	3	3	6	15	24	27
Shift Position	D	D	D	D	D	D	D	D	D	D	D

TSS 2.0

- Adds new Image capture categories:
  - Vehicle behavior related image data: 1 Set of images
  - PCS related image data: 3 Sets of images
  - LKA related image data: 1 Set of images
  - ICS related image data : 3 Sets of images
  - Collision related image data : 2 Sets of images

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Vehicle Behavior Related Image Data

- Judged by vertical/lateral acceleration of vehicle
- Threshold can be exceeded by full-braking on dry asphalt.
- Strong braking input by the driver may record image data.

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PCS Related Image Data

- Applying PCS-braking is the trigger.

File Edit Run Help Setup
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LKA Related Image Data

- Detected by the angular velocity of steering wheel
  - threshold is designed to capture lateral avoidance behavior into the next lane without braking against the object ahead
  - Time to Contact ~ 1.4s.
- Judgement will be carried out only at 30km/h or more
- Threshold defined in deg/s
- Focus on dynamic "quick" handling in avoidance behaviours
  - Will not trigger on static "tight" cornering

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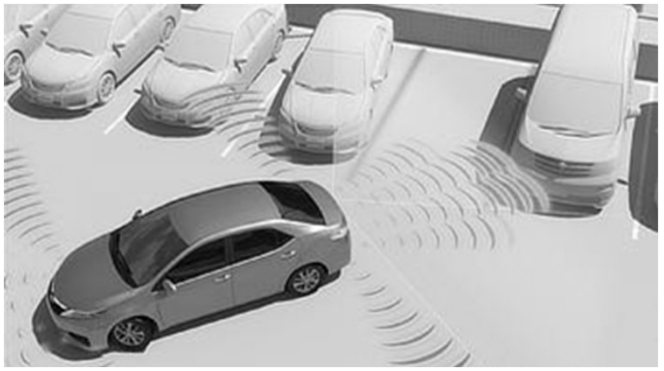
ICS Related Image Data

- ICS = Intelligent Clearance Sonar
- Vehicle applying ICS deceleration is the trigger
- All ICS brake regression events will be recorded



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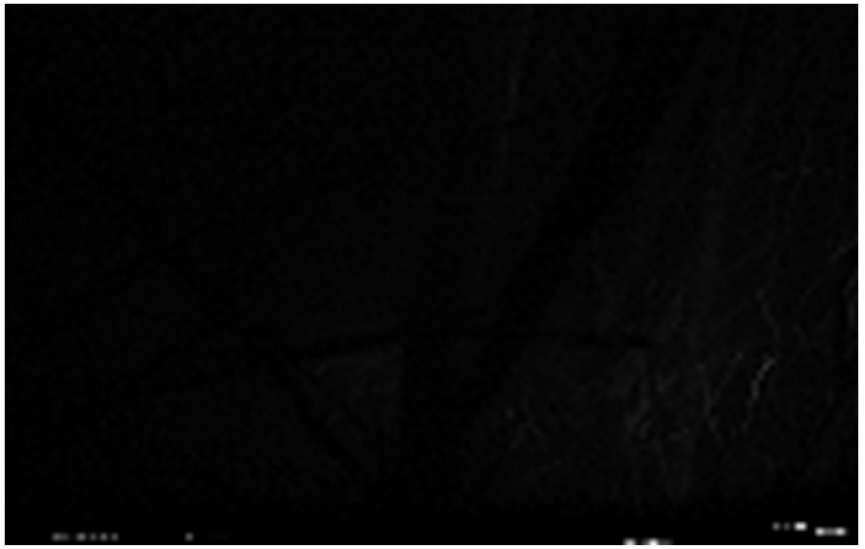
Collision Related Image Data

- Detected by vertical/lateral acceleration of vehicle
- Threshold “much” higher than 1g
- Threshold can’t be exceeded by hard braking
- Trigger suggests external forces were applied against the vehicle



Updated VCH Trigger List

- Long List! – Refer to www.techinfo.Toyota.com !!

TRIGGER ITEM		RECORD DATA				
		BEFORE DETECTION		AFTER DETECTION		SAMPLING CYCLE (SEC.)
ID NO.	ITEM	RECORDING TIME (SEC.)	NUMBER OF POINTS	RECORDING TIME (SEC.)	NUMBER OF POINTS	
20-3	ICS operation history (brake control)					
20-4	ICS operation history (stop)					
20-5	ICS operation history (regression)					
20-6	ICS operation history (temporarily not available)					
22	Lane change assist operation history	8	20	0	1	0.4
23	FHL operation history	5	25	3	16	0.2
24	SCB operation history	5	34	5	35	0.15
26-1	Stop assist history (shift to deceleration phase)					
26-2	Stop assist history (shift to stop phase)					

Trigger Parameters

General data* *(Recorded for all Triggers)		Powertrain system	Brake control system	Driver Assistance system
•Vehicle speed •Num. of ignition cycles •Time from IG-ON •Odometer •Accelerator pedal % •Shift position •Brake SW •Brake fluid pressure •Steering angle	•Fuel injection command •Longitudinal G sensor •Lateral G sensor •Yaw rate	•Engine RPM •Gear position •Turbo RPM •Ready-ON (HV) •% of Engine throttle •Engine load factor	•VSC/ABS/TRC activation •VSC/TRC switch •Wheel speed sensors	•Cruise control SW •Cruise control operation •Pre Collision System activation •Lane Keep Assist •Target distance •Target relative velocity •Drive mode

Vehicle Control History - Parameters

- Not all parameters are shown
- Refer to Vehicle Control History section of on-line Service Manual
- Techinfo.Toyota.com*
- *Subscription Required

ITEM NAME	ITEM EXPLANATION	DISPLAYED VALUE	REMARKS
Vehicle Speed	Displays the vehicle speed.	0 to 200 km/h	-
Accelerator Opening Ratio	Displays the accelerator position sensor opening angle.	0 to 100%	-
Engine RPM Data	Displays the engine speed.	0 to 12800 rpm	-
Cruise Control	Displays whether the cruise control is operating.	ON: Operating OFF: Not Operating / Unequipped Vehicles	Unequipped vehicles display OFF.
Shift Position Signal	Displays the shift lever position.	N / D / R / P / 5 / 4 / 3 / 2 / LO / B / SD / Sequential Mode / Not Specified	-
Brake SW	Displays the stop light switch status.	ON: Stop light illuminated OFF: Not Illuminated	-
Brake Oil Pressure	Displays the fluid pressure inside the master cylinder.	0 to 12.144 MPa	-
Yaw Rate Sensor Signal	Displays the angular velocity when turning.	-61 to 61 deg/s	Becomes a positive value when turning left.
Longitudinal G Sensor Signal	Displays the front and rear acceleration detected by the sensor.	-8.9725 to 8.9725 m/s^2	• Acceleration becomes a positive value • Displays "INVALID" if below the smallest value
Lateral G Sensor Signal	Displays the right and left acceleration detected by the sensor.	-8.9725 to 8.9725 m/s^2	Becomes a positive value when turning left.
Throttle Opening Ratio	Displays the engine throttle opening amount.	0 to 100%	Only displayed for diesel vehicles ("INVALID" is displayed for other vehicles)
Command Value of Fuel Injection Amount	Displays the fuel injection volume (ECU command value).	0 to 120 mm^3/st	Only displayed for diesel vehicles.
Steering Signal	Displays the steering angle.	-567 to 567 deg	Becomes a positive value when turning left.
HV Recognition Flag		HV: Hybrid vehicle	-

Vehicle Control History - Parameters

ITEM NAME	ITEM EXPLANATION	DISPLAYED VALUE	REMARKS
	Displays whether the vehicle is a hybrid vehicle.	Other: Gasoline or diesel vehicle	
Diesel Flag	Displays whether the vehicle is a diesel vehicle.	Diesel: Diesel vehicle Other: Gasoline or hybrid vehicle	-
Sensor Type Identification (Security)	Displays whether a clinometer sensor is installed.	INSTALLED / NOT INSTALLED	Displays a fixed value for each vehicle.
Sensor Type Identification (G Sensor Layout)	Displays the acceleration sensor position angle for the vehicle forward direction.	45° / 90°	Displays a fixed value for each vehicle.
Sensor Type Identification (G Sensor)	Displays the number of acceleration sensors.	1 / 2	Displays a fixed value for each vehicle.
Sensor Type Identification (Yaw Rate Sensor)	Displays the number of yaw rate sensors.	1 / 2	Displays a fixed value for each vehicle.
Yaw G Sensor Installation Information	Displays normal direction or reverse direction (180°opposite) for the sensor position angle (acceleration sensor position angle).	NORMAL DIRECTION / REVERSE DIRECTION	Displays a fixed value for each vehicle.
Shift Gear	Displays the shift step (meter display value) when in sequential mode.	OFF: Sequential mode off GEARS 1 to 8: Shift step on meter display	-

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Vehicle Control History - Parameters

Engine Load Factor	Displays the engine load.	0 to 100%	Hybrid and diesel vehicles display "INVALID"
READY Signal	Displays whether driving is possible.	ON / OFF	Conventional and diesel vehicles display "INVALID".
Gear Position Signal	Displays the current gear step.	1 / 2 / 3 / 4 / 5 / 6 / 7 / 8	• Hybrid vehicles display "INVALID" • CVT normally displays "1"
Turbine RPM/Input Shaft RPM	Displays the torque converter output axis speed (input shaft speed).	0 to 12650 rpm	-
FR Wheel Speed	Displays the front right wheel speed.	-67.67 to 260km/h	-
FL Wheel Speed	Displays the front left wheel speed.	-67.67 to 260km/h	-
RR Wheel Speed	Displays the rear right wheel speed.	-67.67 to 260km/h	-
RL Wheel Speed	Displays the rear left wheel speed.	-67.67 to 260km/h	-
ACC Target Distance	Displays the vehicle-to-vehicle distance to the preceding vehicle targeted by ACC (vehicle-to-vehicle distance mode).	0 to 251 m	-

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Vehicle Control History - Parameters

ITEM NAME	ITEM EXPLANATION	DISPLAYED VALUE	REMARKS
ACC Target Condition		PRECEDING VEHICLE EXISTS / NO PRECEDING VEHICLE / FIXED SPEED MODE / RADAR MALFUNCTION	
ACC Relative Speed	Displays the relative vehicle speed of the preceding vehicle targeted by ACC (vehicle-to-vehicle distance mode).	-128 to 127km/h	-
PCS Reception Data Invalid Flag	Displays the VALID/INVALID status of PCS related data.	VALID / INVALID	-
ALM Request Flag	Displays whether an operation request was output to the warning function from PCS.	ON: Operation request OFF: No operation request	-
ABK Request Flag	Displays whether an operation request was output to the warning brake function from PCS.	ON: Operation request OFF: No operation request	Only displayed for vehicles with a driver monitor camera
PBA Request Flag	Displays whether an operation request was output according to the pre-collision brake assist function from PCS.	ON: Operation request OFF: No operation request	-

Vehicle Control History - Parameters

Target Distance	Displays the PCS control target vehicle-to-vehicle distance.	0 to 163.83 m	If a target is recognized, but PCS control is determined unnecessary, the largest value is displayed.
Target Relative Velocity	Displays the PCS control target vehicle relative speed.	-51.175 to 0 m/s	If a target is recognized, but PCS control is determined unnecessary, the largest value is displayed.
Target Lateral Position	Displays the PCS control target side position.	-5.12 to 5.08 m	• If a target is recognized, but PCS control is determined unnecessary, the largest value is displayed. • If the target is on the right side of the vehicle a positive value is displayed, and if the target is on the left side a negative value is displayed.
Target Number	Displays the target number determined by PCS control for targets recognized by the sensors (maximum 8 targets).	1 to 8	When the target number is 1 and the related record data (control target vehicle-to-vehicle distance, relative speed and side position) all display the largest values, PCS control has been determined unnecessary.

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TSS 2.0

- New for 2019+
  - Rav4 / Corolla HB / Lexus ES350
- Capture of images on Triggers:
  - PCS
    - No longer requires car braking to store images
  - LKA
  - EDR
  - ICS – Intelligent Clearance Sonar
  - “Vehicle Behaviour”
- Low-Res Camera images before / after trigger point

Introducing Toyota Safety Sense™ (TSS), a new series of multi-feature active safety packages incl
TSS is designed to support the driver's awareness, decision making and vehicle operation over a w
together in an integrated system, these features help address three key areas of accident protectio
keep drivers within their lane and enhancing road safety during nighttime driving.

The TSS systems are in addition to the Star Safety System technologies.

Toyota Safety Sense P

TSS-P is an active safety package that combines the Pre-Collision System with Pedestrian Detection (PCS w/PD), Lane Departure Alert with Steering Assist (LDA w/SA)[1], Automatic High Beam (AHB) and Dynamic Radar Cruise Control (DRCC). TSS-P combines a camera and millimetre-wave radar technology.

Toyota Safety Sense C

TSS-C is an active safety package that
Automatic High Beam (AHB). TSS-C co

FEATURE

OPERATION

Pre-Collision System (PCS)

TSS-C: Provides collision avoidance or damage mitigation support in speed ranges in which frontal collisions are likely to occur.

Pre-Collision System with Pedestrian Detection (PCS w/PD)

TSS-P: Provides collision avoidance or damage mitigation support in speed ranges in which frontal collisions are likely to occur. In certain conditions, the Pre-Collision System with Pedestrian Detection (PCS w/PD) may also help to detect pedestrians and help prevent or mitigate a collision.

Step 1: Detection of leading vehicle/ pedestrian (TSS-P only).

Step 2: Audio and visual alert "Brake". Notification may vary.

BRAKE

Step 3: If driver applies brakes: Pre-Collision Brake Assist may provide additional braking force during emergency braking to help avoid or mitigate damage from a collision.

Step 4: If driver does not apply brakes: Automatic Braking may apply the brakes to help avoid or mitigate damage from a collision.

ALARM BRAKING STOP
IF DRIVER DOES NOT REACT
IF DRIVER REACTS
ALARM DRIVER BRAKES STOP

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Image Recording

Image Data Specifications

TRIGGER ITEM		RECORD DATA				
		BEFORE DETECTION		AFTER DETECTION		SAMPLING CYCLE (SEC.)
ID NO.	ITEM	RECORDING TIME (SEC.)	NUMBER OF POINTS	RECORDING TIME (SEC.)	NUMBER OF POINTS	
15	Sudden braking history	6	10	6	10	0.6
16	Sudden turning history	6	10	6	10	0.6
18-3	PCS operation history (pre-collision brake assist operation)	6	10	6	10	0.6
18-4	PCS operation history (prior brake operation)					
18-5	PCS operation history (pre-collision brake operation)					
18-11	PCS operation history (when acceleration above certain amount is detected)					
18-14	Pedestrian protection system over constant pressure sensor signal input					
19	Torque sensor signal above certain amount is detected during LKA/LDA operation	6	10	6	10	0.6
20-2	ICS operation history (max output limit)	6	10	6	10	0.6
20-3	ICS operation history (brake control)					

File Edit Run Help Setup

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PCS Data Viewer

PCS Data Viewer - 58ABZ1B11KU007324_20190405105359.vdas

File Help

File Information when GTS read out PCS Data

VIN	58ABZ1B11KU007324		
Key Cycle	135	Dealer Code	CDS LTD INC
ODO	163km	Retrieval Date	20190405

Airbag	
Parts No.	06880
Software	8917F06451

VCH or FFD Data Set in PCS Data

VCH FFD

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	ODO	Number of Overwrite of the Recording Area	Image FFD	Image FFD Trigger
108	00:00:09.8	-	2019-02-08-16:38:26.200	Accelerator pedal opening angle is medium or higher immediately after shifting to R	162	1	-	-
107	00:10:04.3	-	2019-02-08-16:02:30.100	Sudden braking history	162	1	*	Sudden braking history
107	00:10:04.4	-	2019-02-08-16:02:30.200	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	162	1	*	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)
107	00:10:04.4	-	2019-02-08-16:02:30.200	ABS operation history	162	15	-	-
107	00:10:04.3	-	2019-02-08-16:02:30.100	PCS operation history (when acceleration above certain amount is detected)	162	1	*	PCS operation history (when acceleration above certain amount is detected)
107	00:06:41.1	-	2019-02-08-15:59:06.900	PCS operation history (pre-collision brake assist operation)	161	1	*	PCS operation history (pre-collision brake assist operation)
105	00:06:42.9	-	2019-02-07-23:28:42.500	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	142	1	-	-
104	00:25:42.8	-	2019-02-07-21:27:21.100	Accelerator high position in mid to high speed	114	1	-	-
104	00:05:30.8	-	2019-02-07-21:07:09.100	TRC operation history	100	15	-	-

Show Detail

Back

Exit

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TSS 2.0 – Event Date / Time

File Edit Run Help Setup

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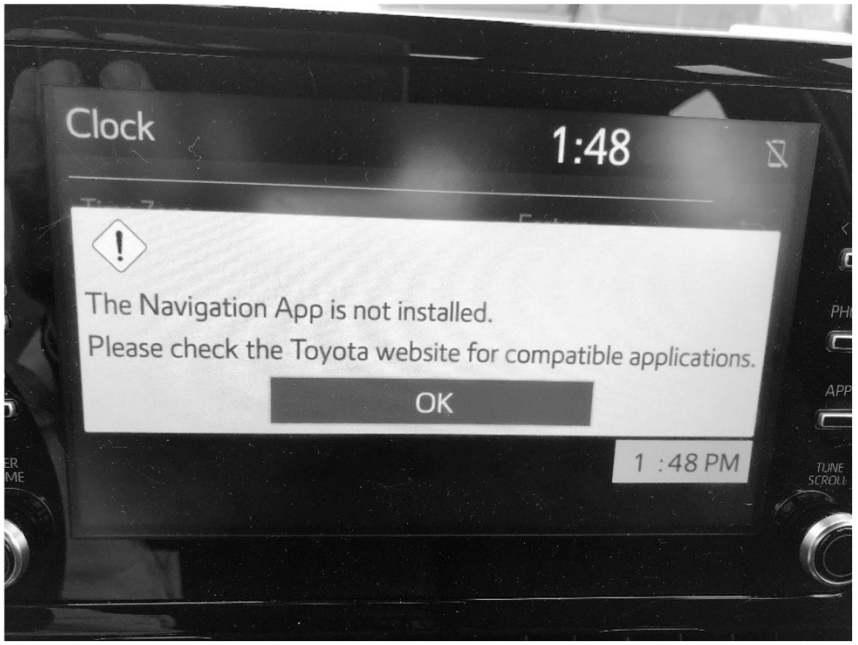
TSS 2.0 – Event Date / Time

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	ODO
108	00:00:09.8	-	2019-02-08-16:38:26.200	Accelerator pedal opening angle is medium or higher immediately after shifting to R	162
107	00:10:04.3	-	2019-02-08-16:02:30.100	Sudden braking history	162
107	00:10:04.4	-	2019-02-08-16:02:30.200	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	162
107	00:10:04.4	-	2019-02-08-16:02:30.200	ABS operation history	162
107	00:10:04.3	-	2019-02-08-16:02:30.100	PCS operation history (when acceleration above certain amount is detected)	162
107	00:06:41.1	-	2019-02-08-15:59:06.900	PCS operation history (pre-collision brake assist operation)	161
105	00:06:42.9	-	2019-02-07-23:28:42.500	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	142
104	00:25:42.8	-	2019-02-07-21:27:21.100	Accelerator high position in mid to high speed	114
104	00:05:30.8	-	2019-02-07-21:07:09.100	TRC operation history	100

Back



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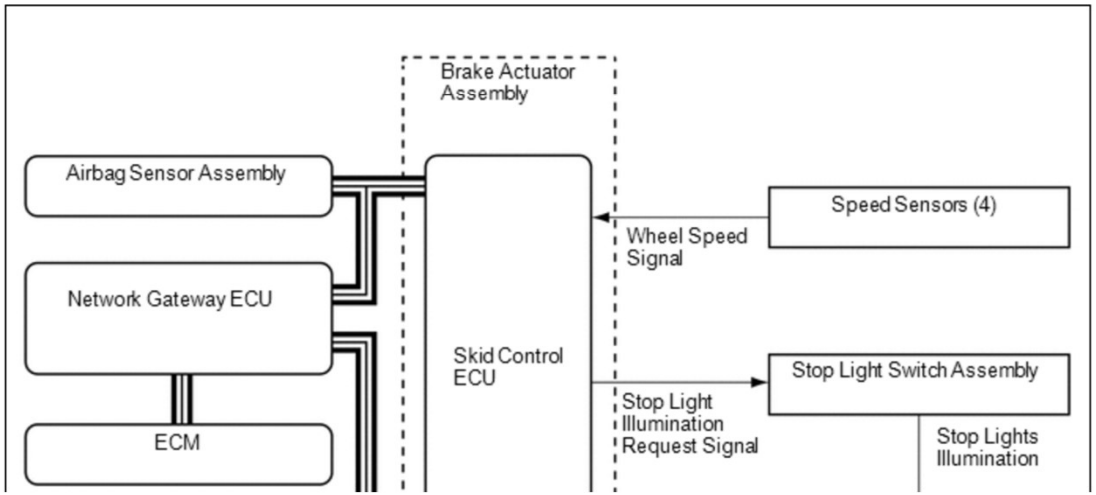
Secondary Collision Brake !!

SECONDARY COLLISION BRAKE

SYSTEM CONTROL

(a) The secondary collision brake system lessens damage from a secondary collision by controlling the automatic brakes and stop light illumination after a collision occurs.

(b) When the airbag sensor assembly detects a collision and operates the airbags, it sends a collision detection signal to the skid control ECU. When the skid control ECU receives a collision detection signal, it determines that operation can start, operates the brake actuator and performs braking. Simultaneously, the skid control ECU sends a stop light illumination request to the stop light switch assembly, illuminating the stop lights.



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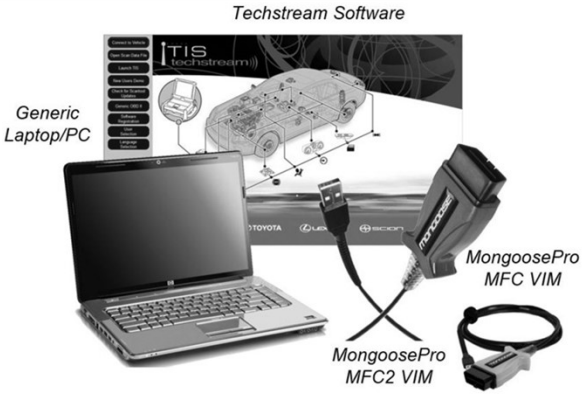
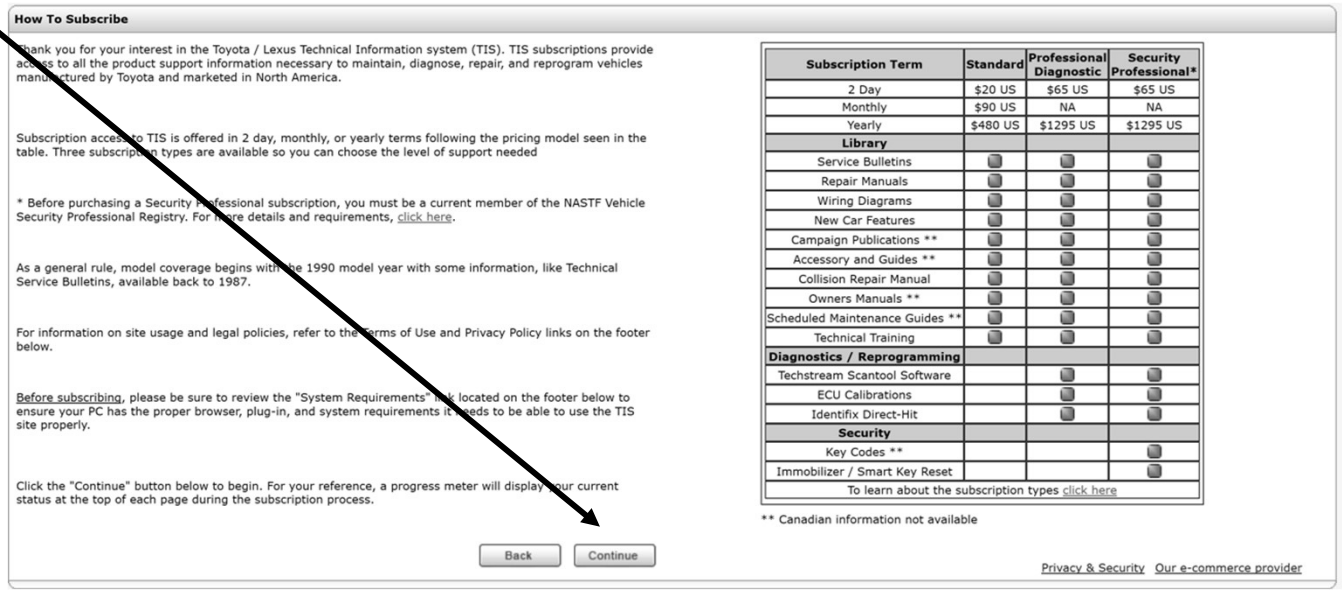
Toyota TechStream System

- Dealer Tool
  - PC Based – Toughbook with VIM
- Aftermarket Tool
  - VIM from Drew Technologies
  - Mongoose Pro MFC VIM or
  - Mongoose Pro MFC2 VIM
  - CDR900
- Software Subscription from techinfo.toyota.com



Toyota TechStream Setup

- Register for an account, go to:
- https://techinfo.snapon.com/TIS/Register.aspx
- Click “Continue”



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Click the "Continue" button below to begin. For your reference, a progress meter will display your current status at the top of each page during the subscription process.

Back | Continue

Subscription Term	Standard	Professional Diagnostic	Security Professional*
2 Day	$20 US	$65 US	$65 US
Monthly	$90 US	NA	NA
Yearly	$480 US	$1295 US	$1295 US
Library			
Service Bulletins	■	■	■
Repair Manuals	■	■	■
Wiring Diagrams	■	■	■
New Car Features	■	■	■
Campaign Publications **	■	■	■
Accessory and Guides **	■	■	■
Collision Repair Manual	■	■	■
Owners Manuals **	■	■	■
Scheduled Maintenance Guides **	■	■	■
Technical Training	■	■	■
Diagnostics / Reprogramming			
Techstream Scantool Software		■	■
ECU Calibrations		■	■
Identifix Direct-Hit		■	■
Security			
Key Codes **			■
Immobilizer / Smart Key Reset			■
To learn about the subscription types click here			

** Canadian information not available

Privacy & Security Our e-commerce provider

www.techinfo.Toyota.com

- Subscription basis

Subscription Type	Standard	Professional Diagnostic	Security Professional*
2 Day	**$20**	**$65**	**$65**
Monthly	**$90**	**NA**	**NA**
Yearly	**$480**	**$1295**	**$1295**

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This means a two-day subscription may actually provide greater than 48 hours of usable access.

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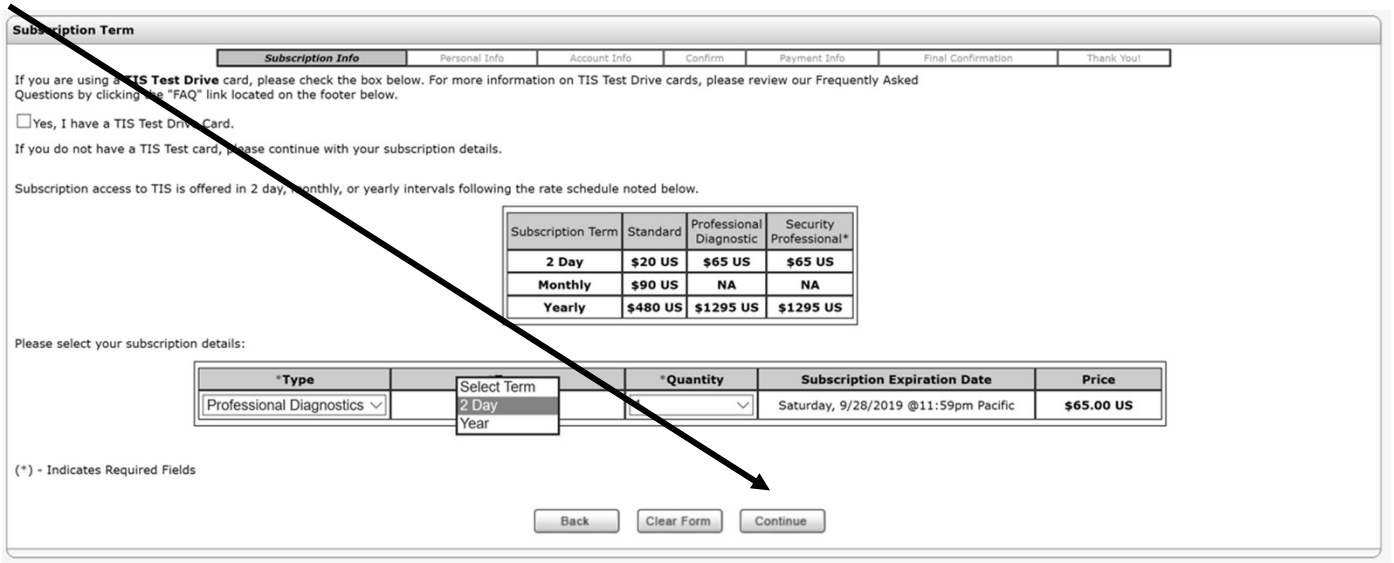
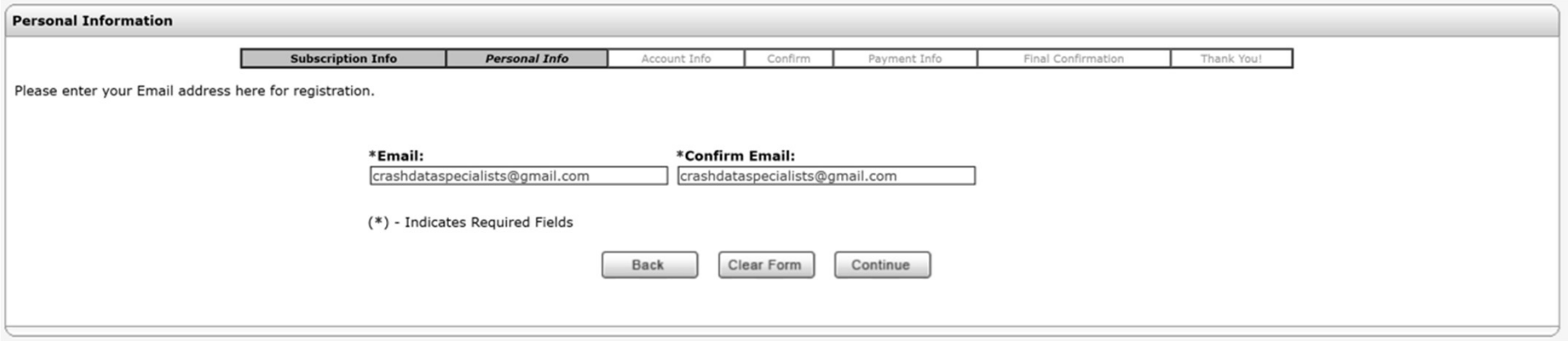
Toyota TechStream Setup

- Select: Professional Diagnostics / 2 Day or Annual / Quantity 1
- Click Continue



Toyota TechStream Setup

- Enter email to use for account. This will be your login name for using the website, downloading and installing the software



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Toyota TechStream Setup

- Complete mandatory fields with an * to complete account creation

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Toyota TechStream Setup

- Select and input a Password to complete account creation
  - Note Password Requirements!

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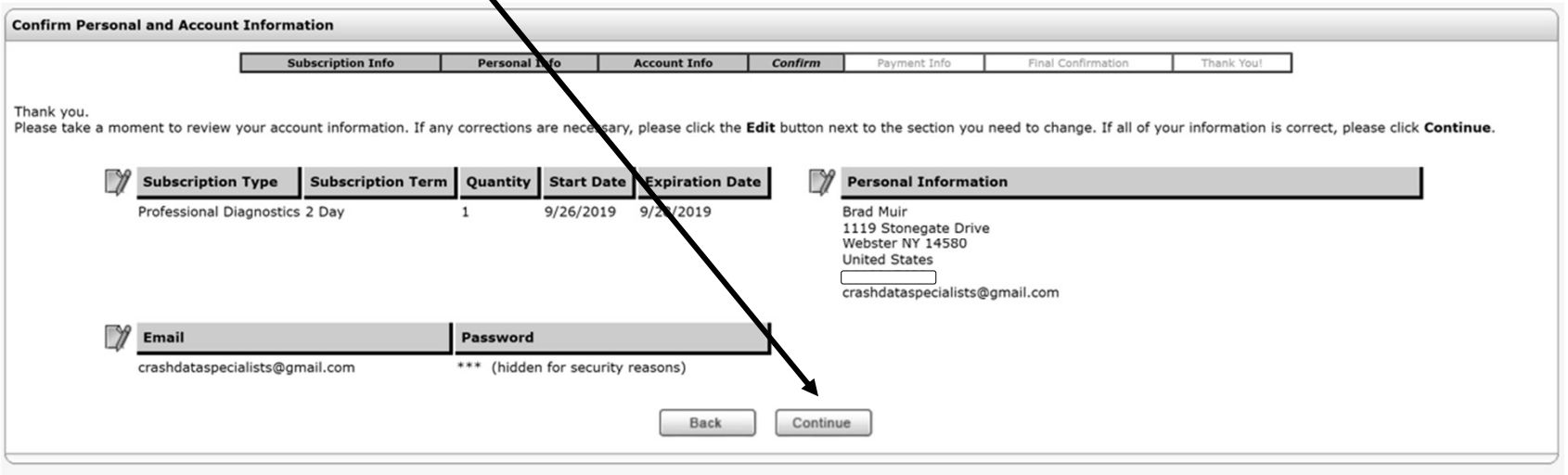
Toyota TechStream Setup

- Confirm summary of account information and subscription and Click Continue



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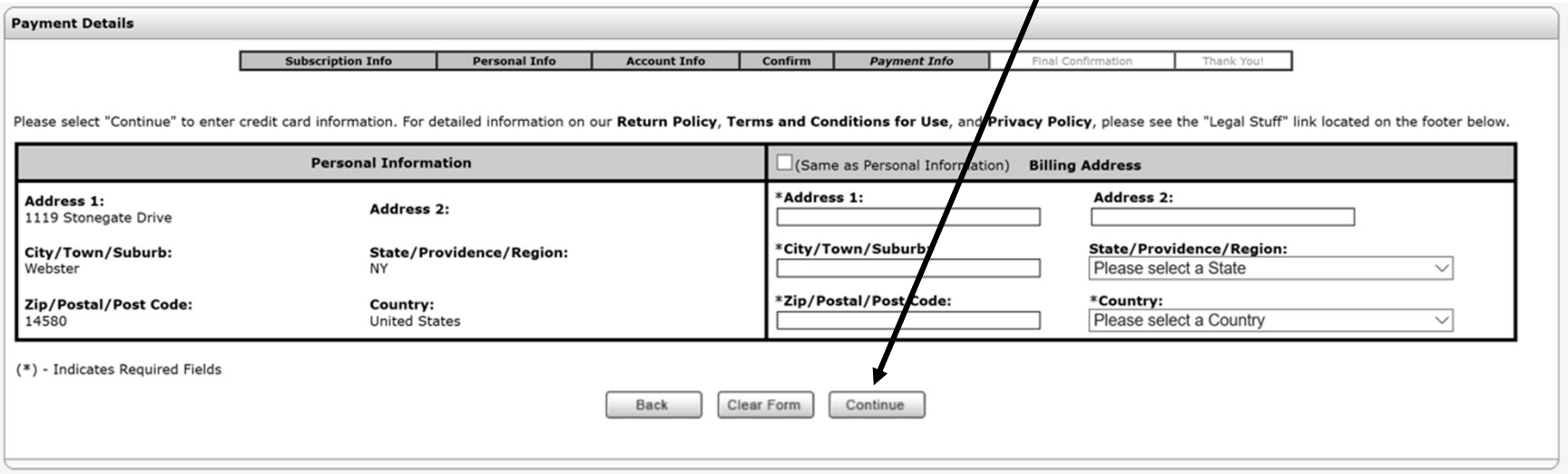
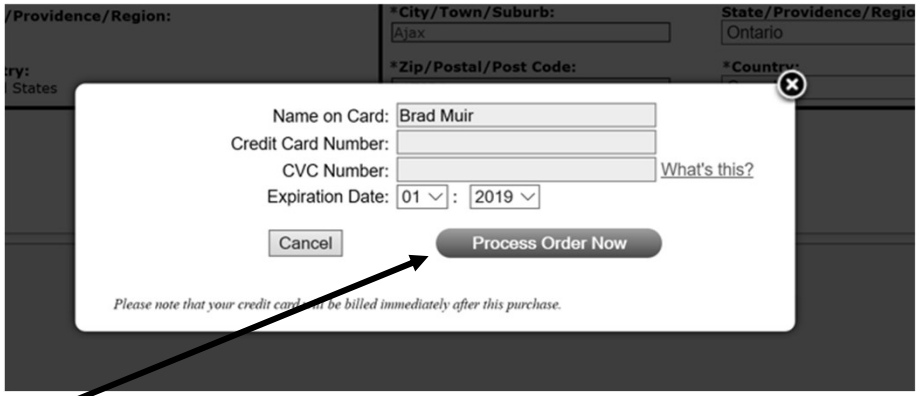
Toyota TechStream Setup

- For Credit Card payment, input the billing address or select “Same as personal information”, Click Continue



Toyota TechStream Setup

- A window will pop up for credit card information
- Payment **must be** made via Credit Card
- Visa / Mastercard and American Express accepted
- Click: “Process Order Now”



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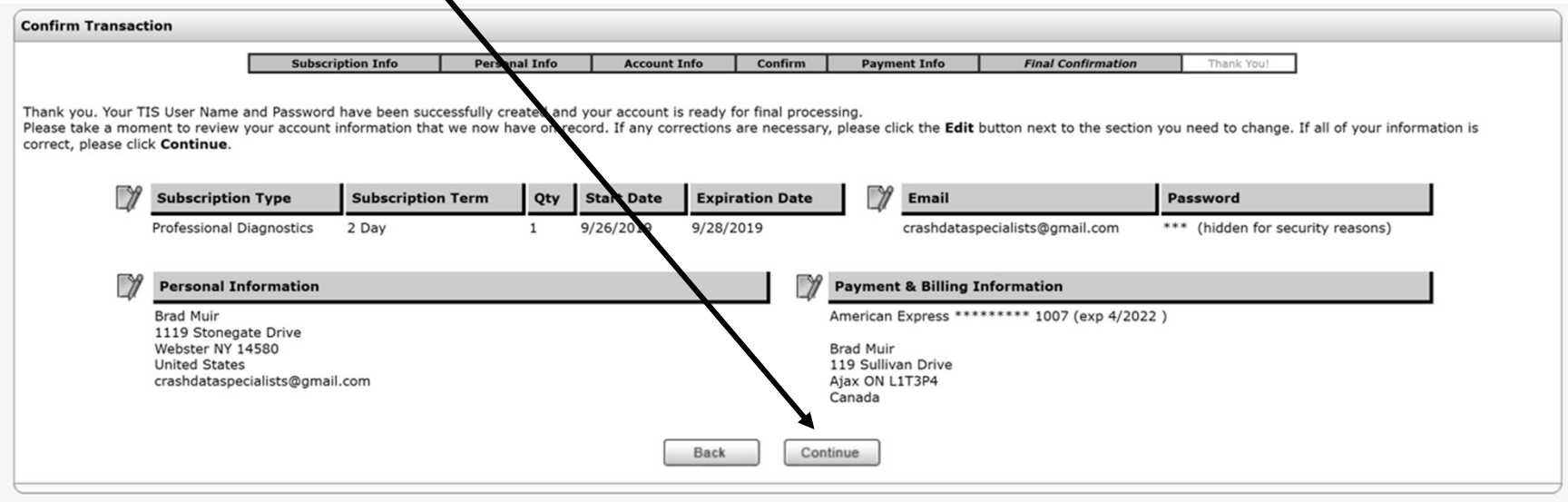
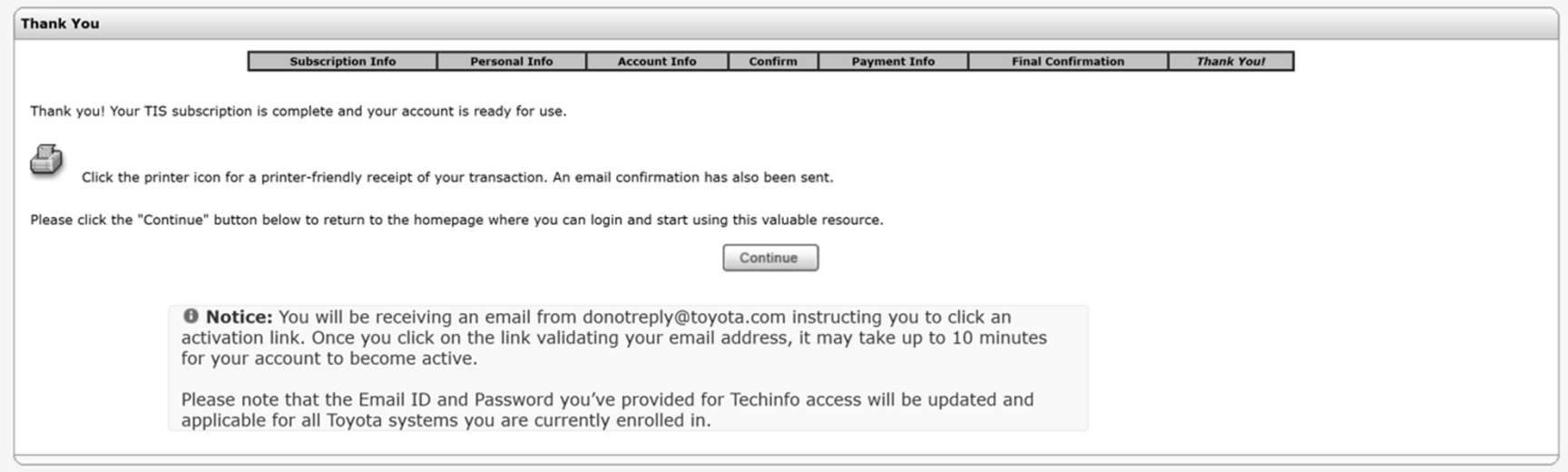
Toyota TechStream Setup

- Confirm or change relevant information
- Click Continue



Toyota TechStream Setup

- Follow instructions to Print Receipt
- Indicates an email with an Activation Link will be sent



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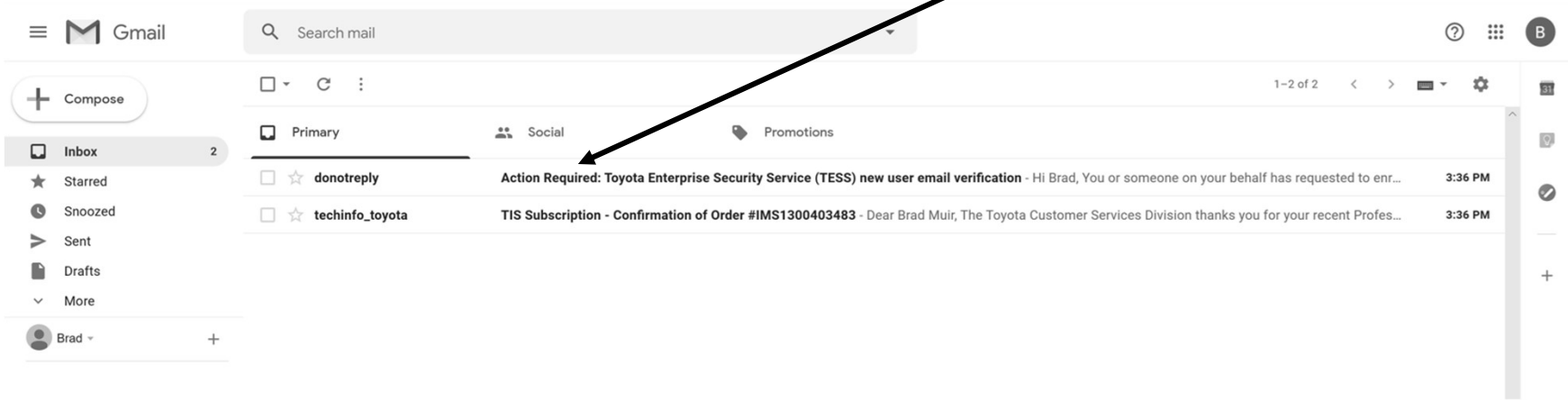
Toyota TechStream Setup

- Check your email and open the one with “Action Required” in subject line



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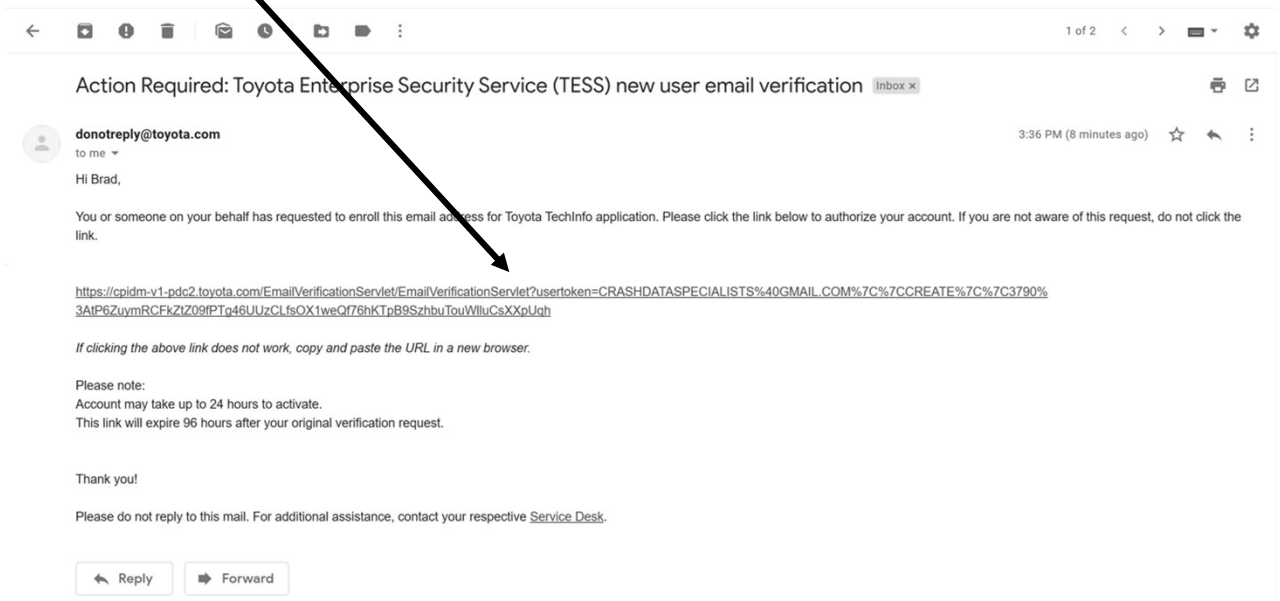
Toyota TechStream Setup

- Click the **link** in the email to Activate your account



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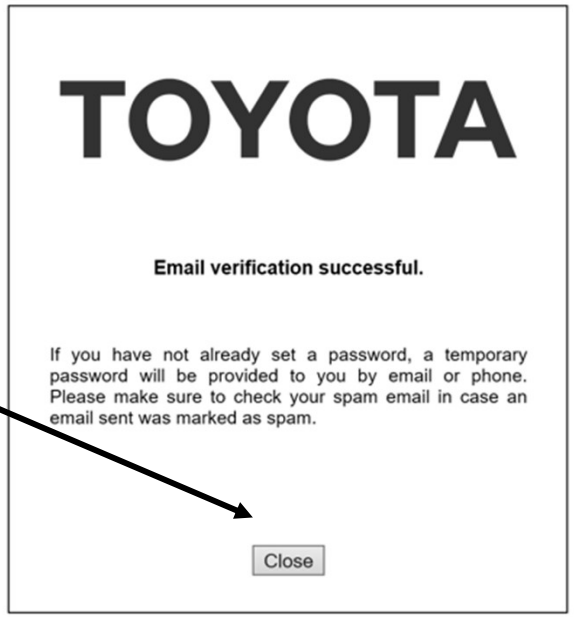
Toyota TechStream Setup

- Upon successful Activation this pop-up appears
- Click Close



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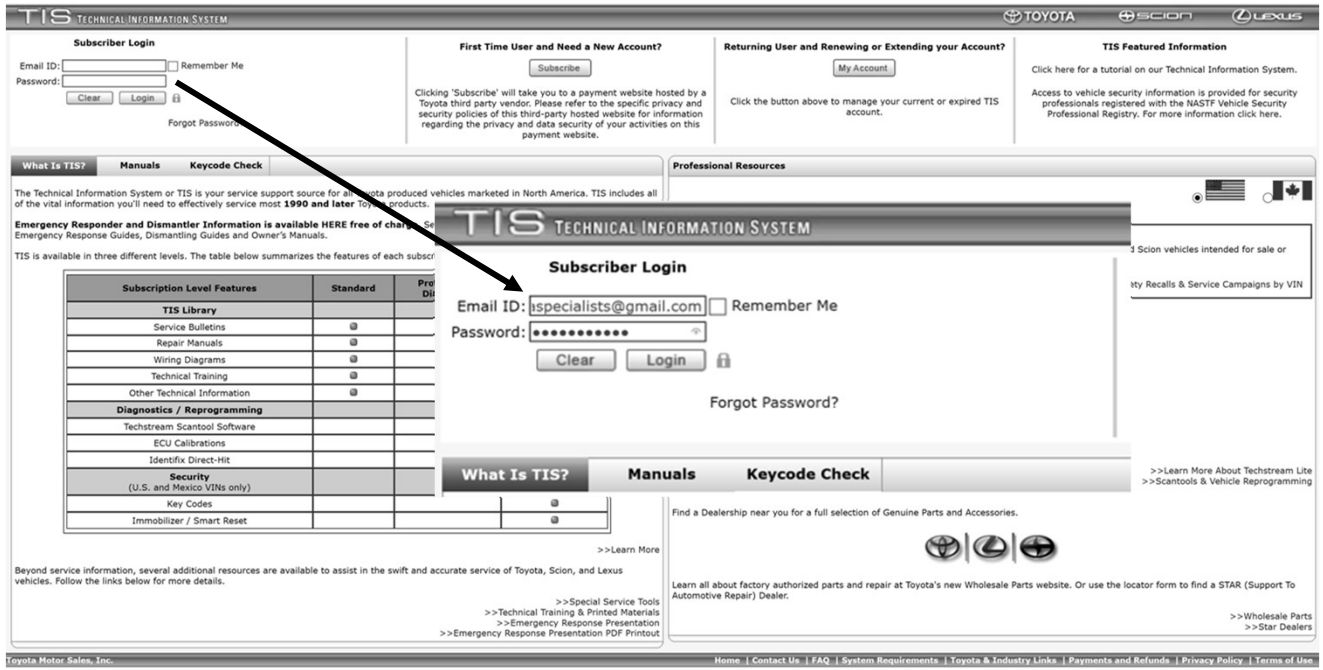
Toyota TechStream Setup

- Return to techinfo.toyota.com and log in



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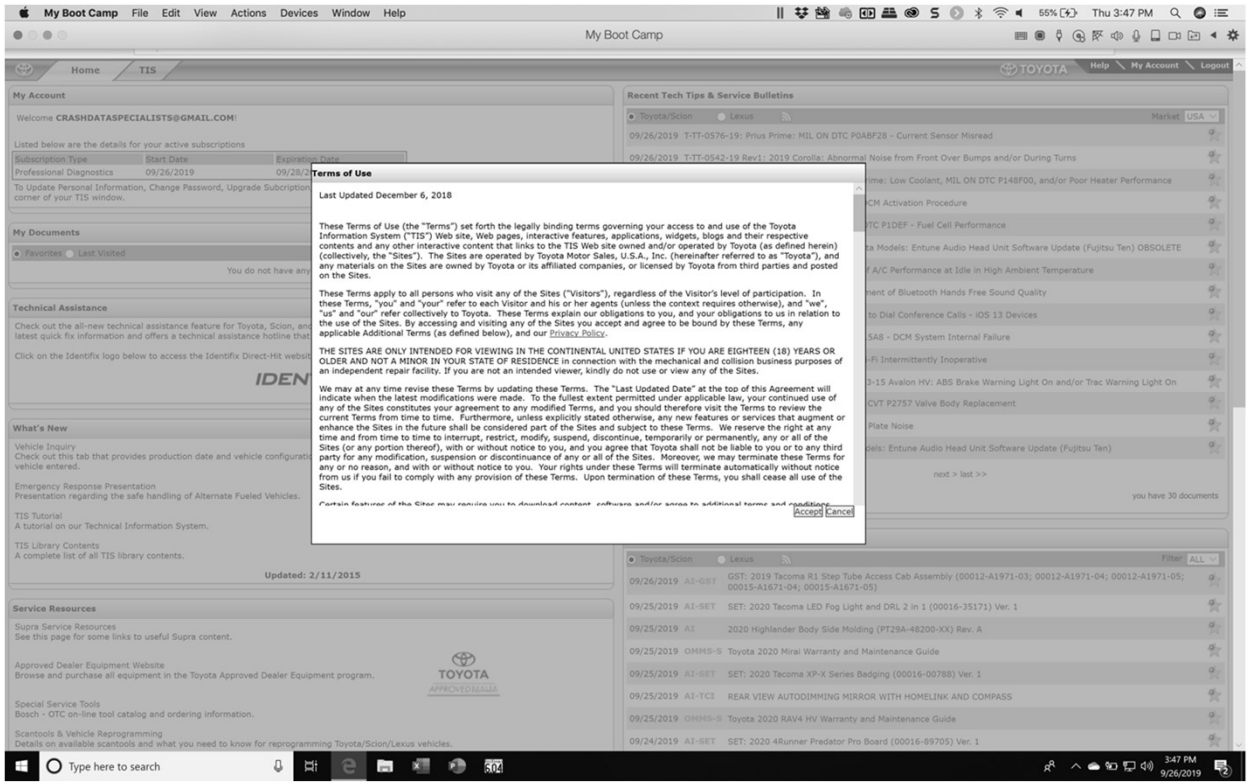
Toyota TechStream Setup

- Agree to the terms of use



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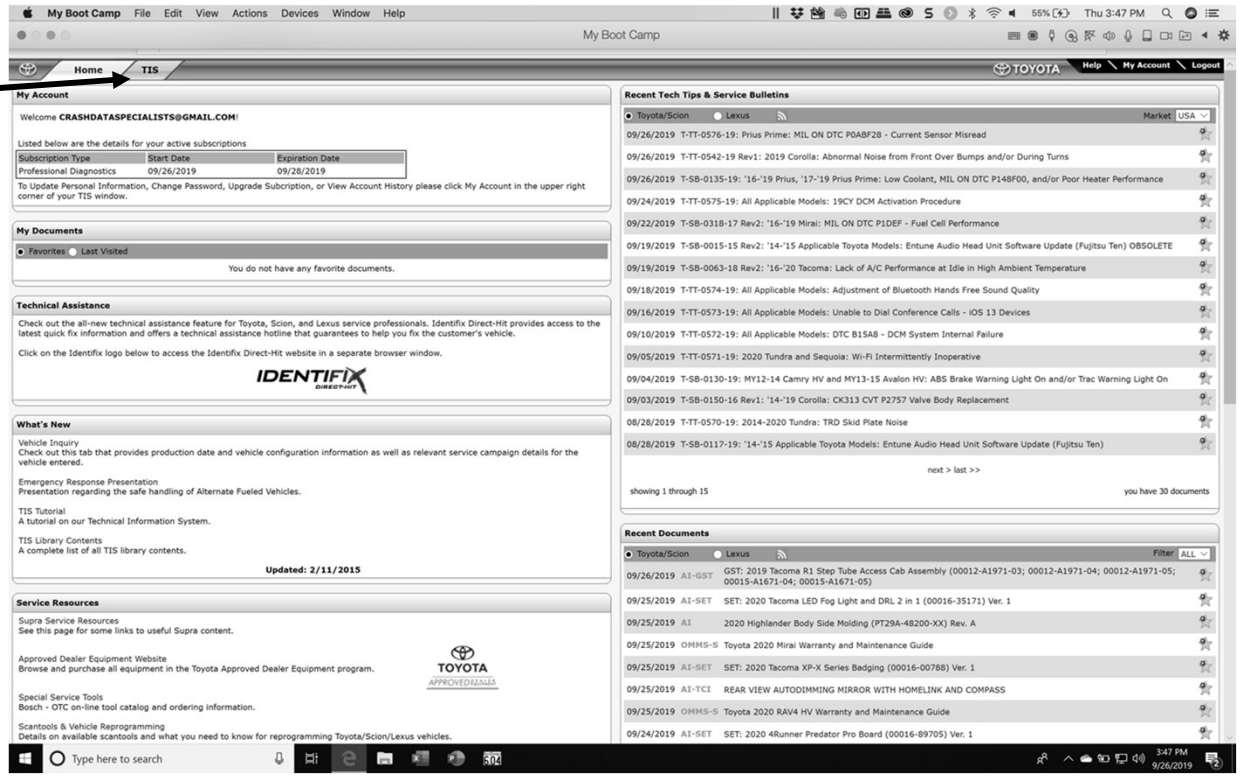
Toyota TechStream Setup

- Click “TIS” tab



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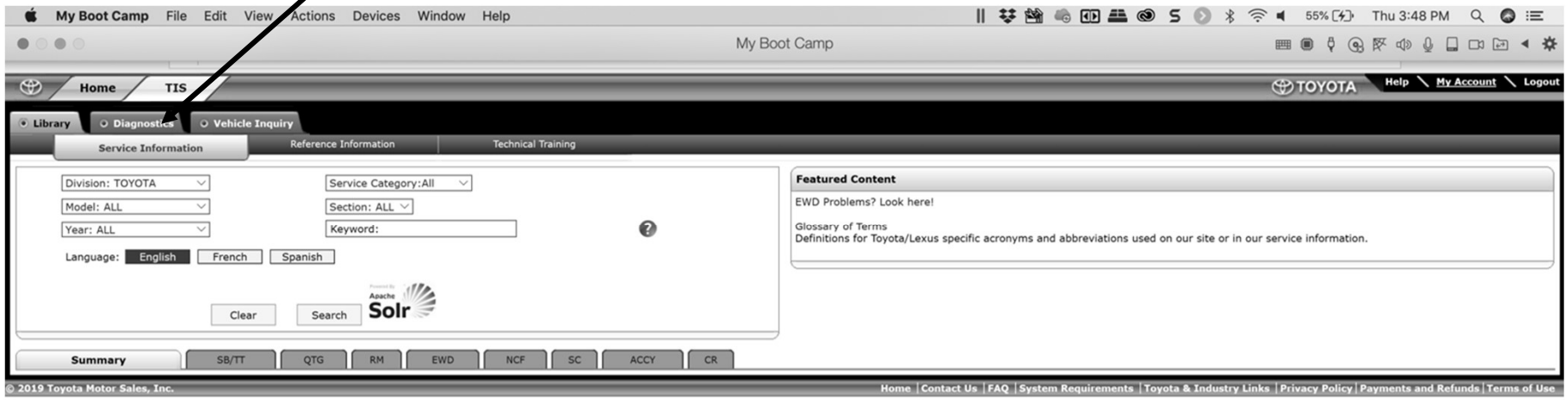
Toyota TechStream Setup

- Click the “Diagnostics” Tab



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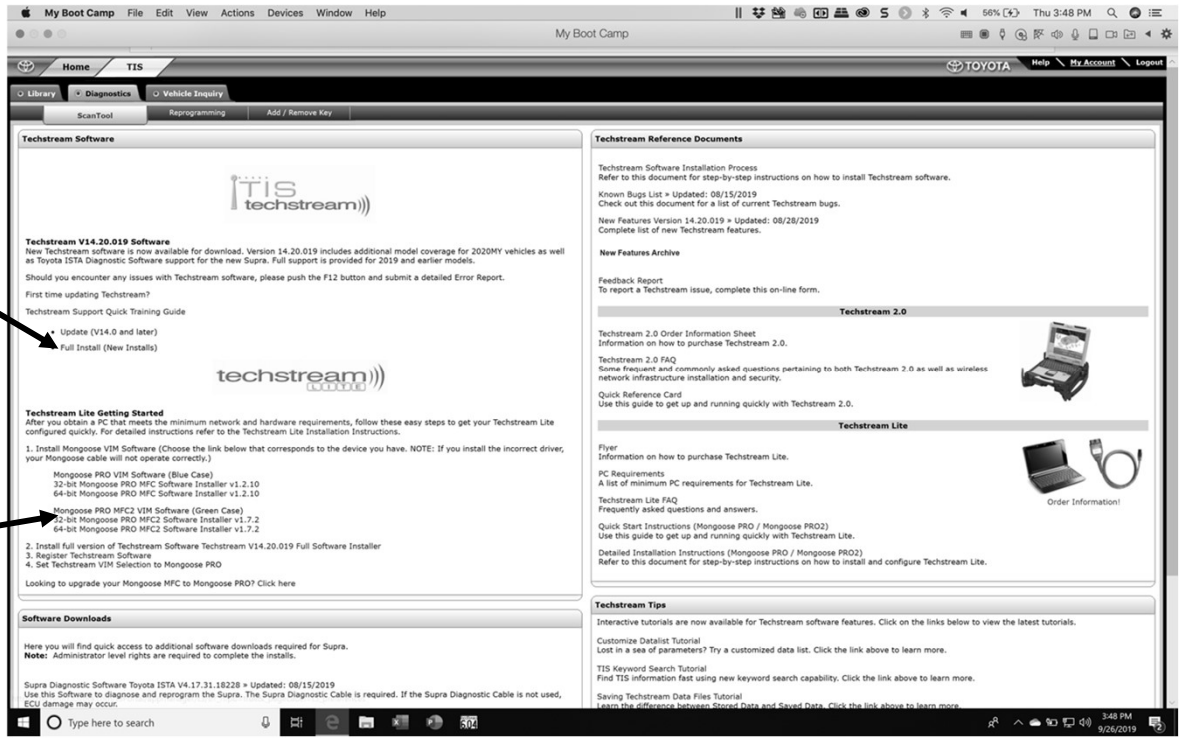
Toyota TechStream Setup

- Click to download the current “Full” version
- If using the Drew Tech Mongoose VIM the drivers are available here



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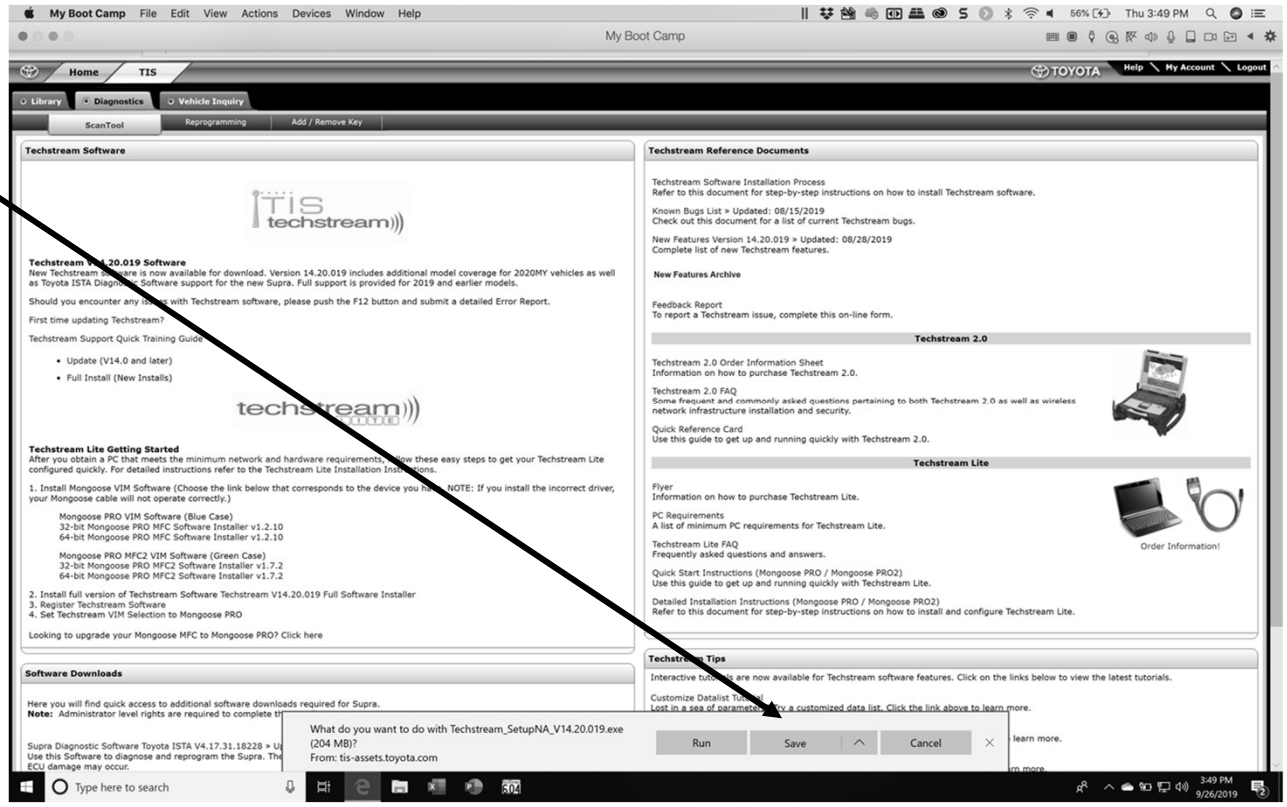
Toyota TechStream Setup

- Click “Save” as this will retain a copy on your computer in the “Downloads” directory
- This will allow re-install if there is an issue later



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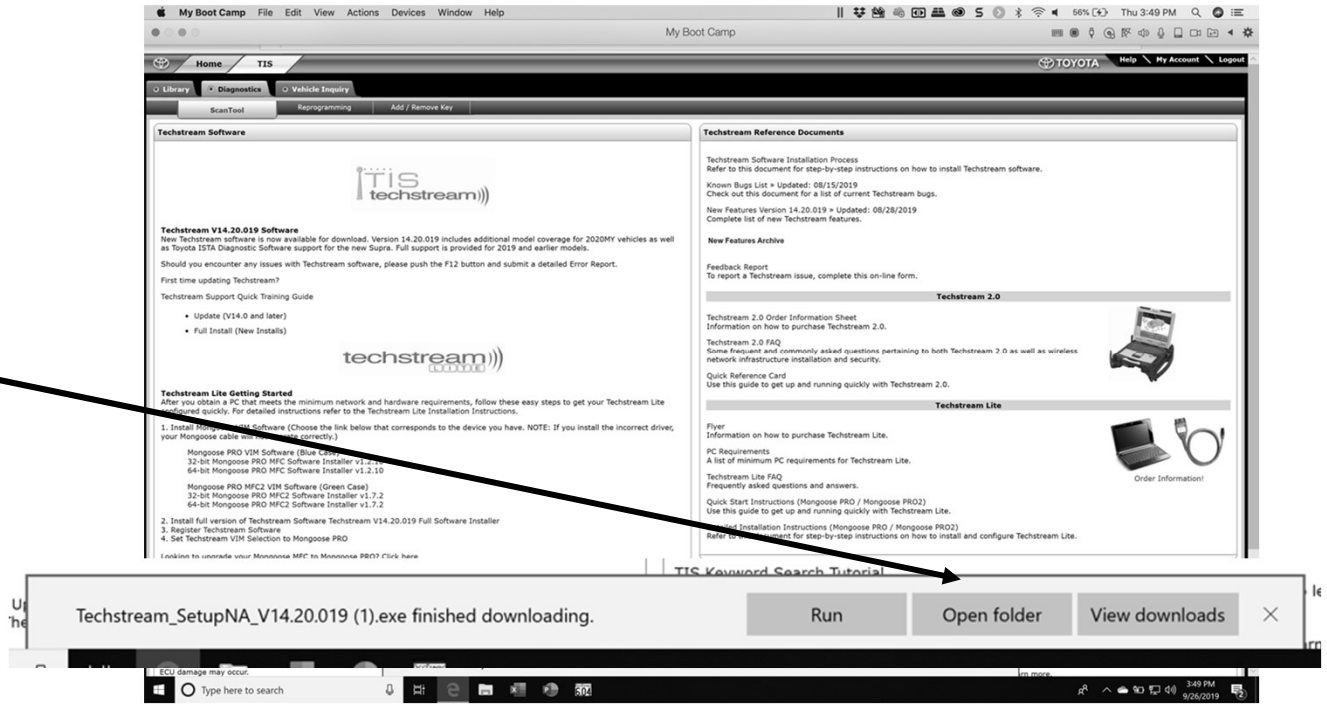
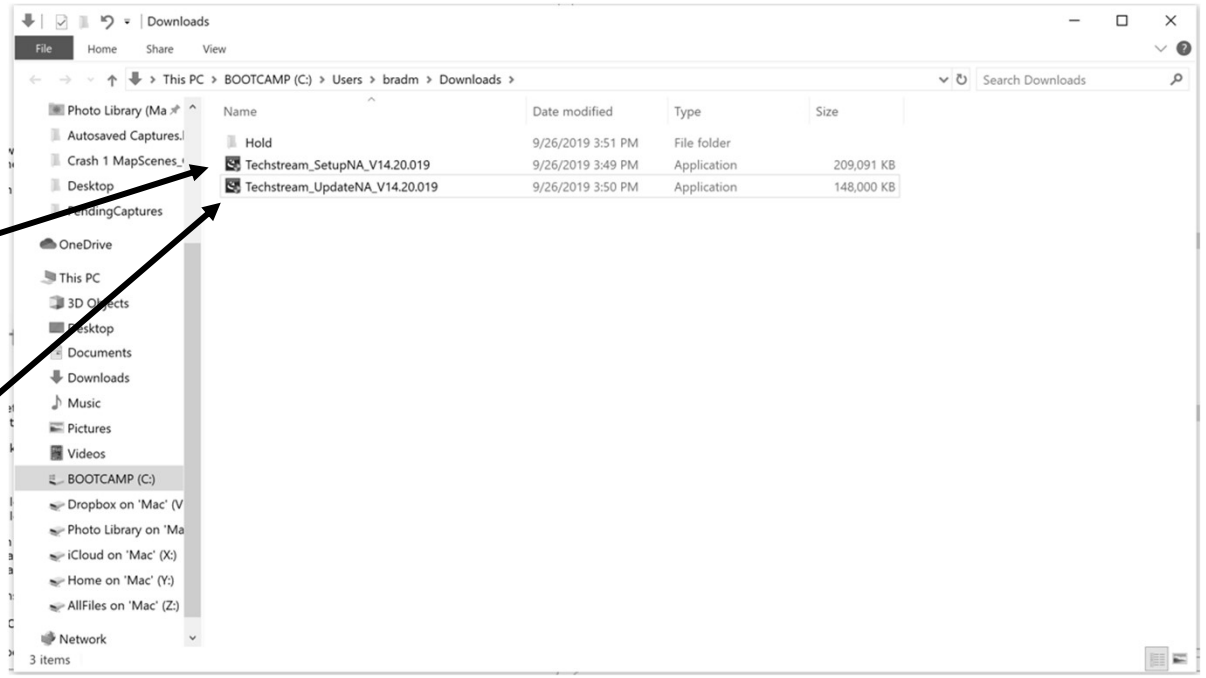
Toyota TechStream Setup

- After download the bottom footer message changes.
- Click “Open Folder”



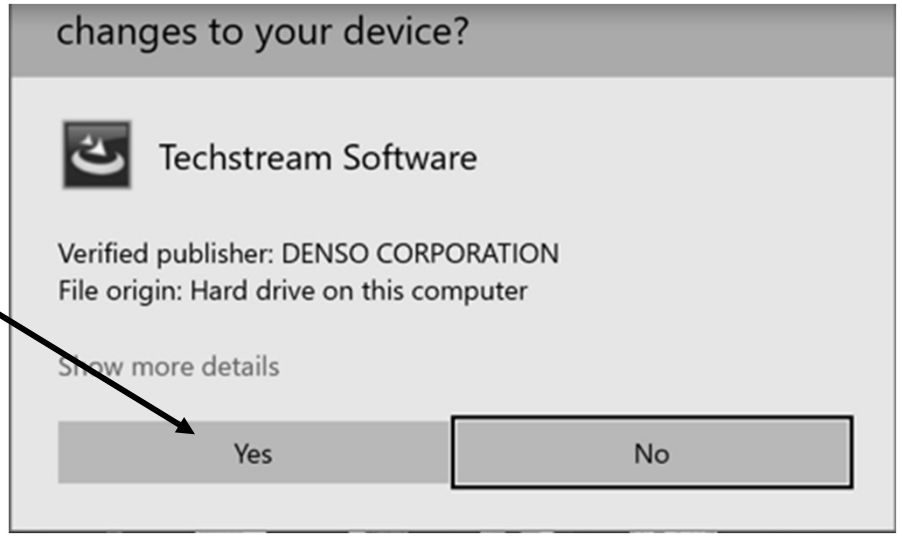
Toyota TechStream Setup

- An Explorer window opens.
- Locate the “Full” install you downloaded
- Double-Click to launch Install
- IF Installing an Update you downloaded, double-click that file to launch Install



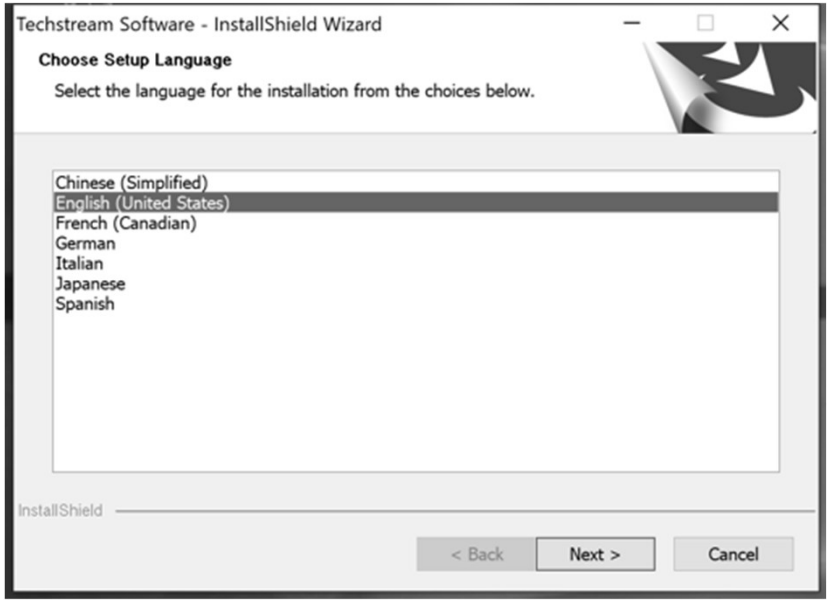
Toyota TechStream Setup

- Click “Yes” to continue…..
- In some environments you may want to “right-click” and run the setup program as an administrator.
- Only do this if there is a problem in the installation or launching of the software

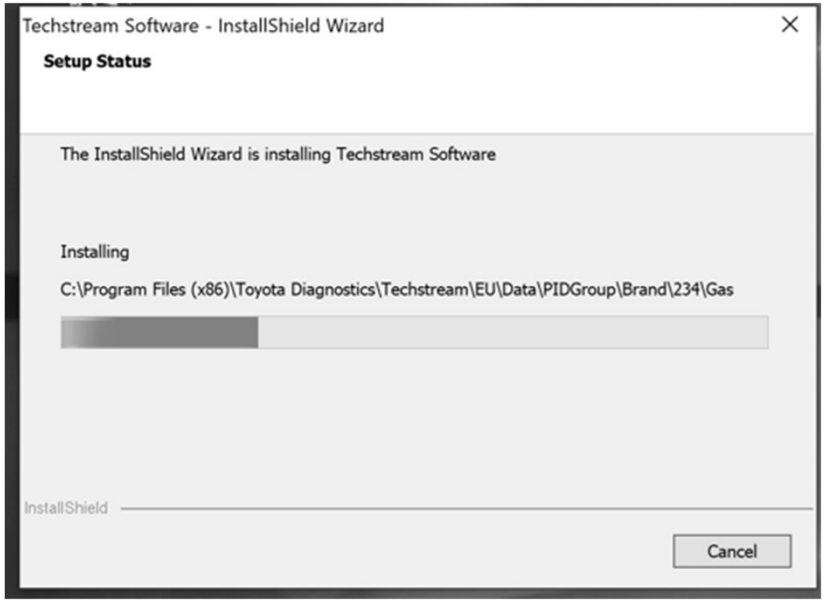


Toyota TechStream Setup

Select “English (United States)”, click “Next”



The software will not install. This may take a while!



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Toyota TechStream Setup

After successful installation two new shortcuts will appear on your desktop.

You can start the application from this shortcut

A new “TechStream Diagnostics” folder appears in your Start menu

Up can ;launch the application from this menu

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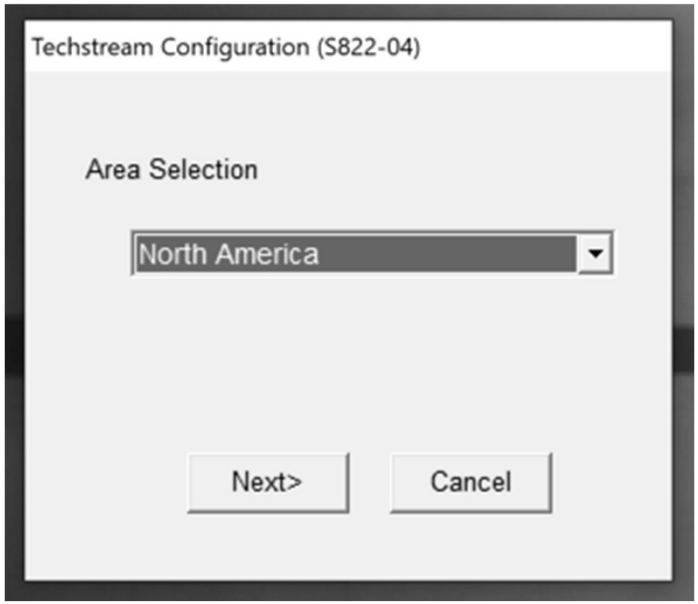
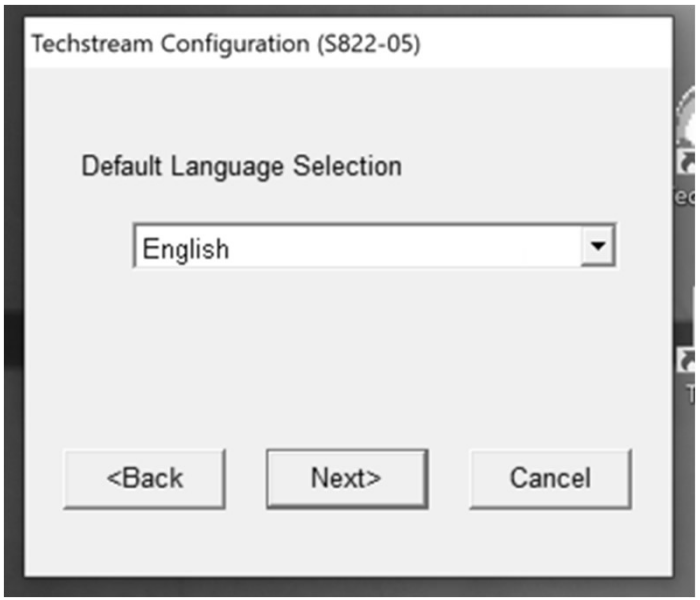
Toyota TechStream Setup

Select “North America” as the Region



Select “English” as the default language



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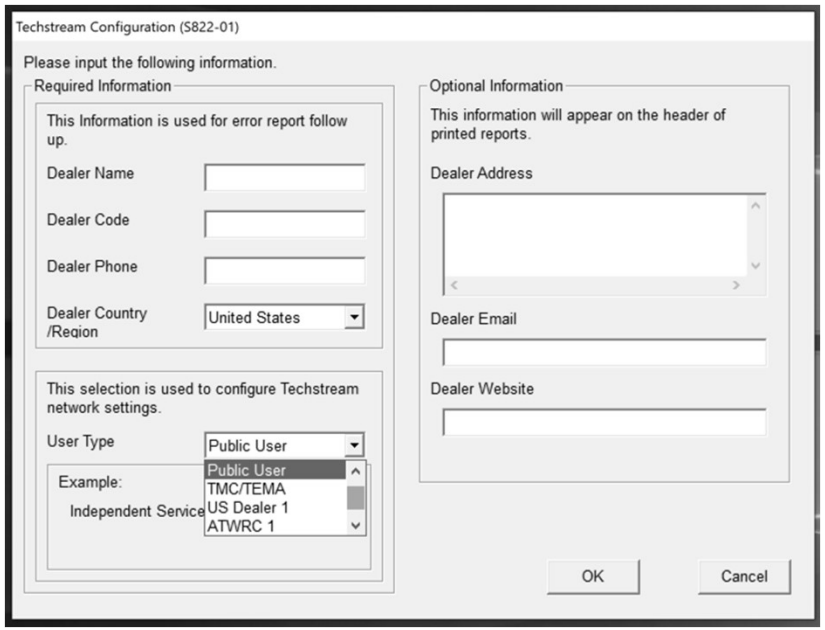
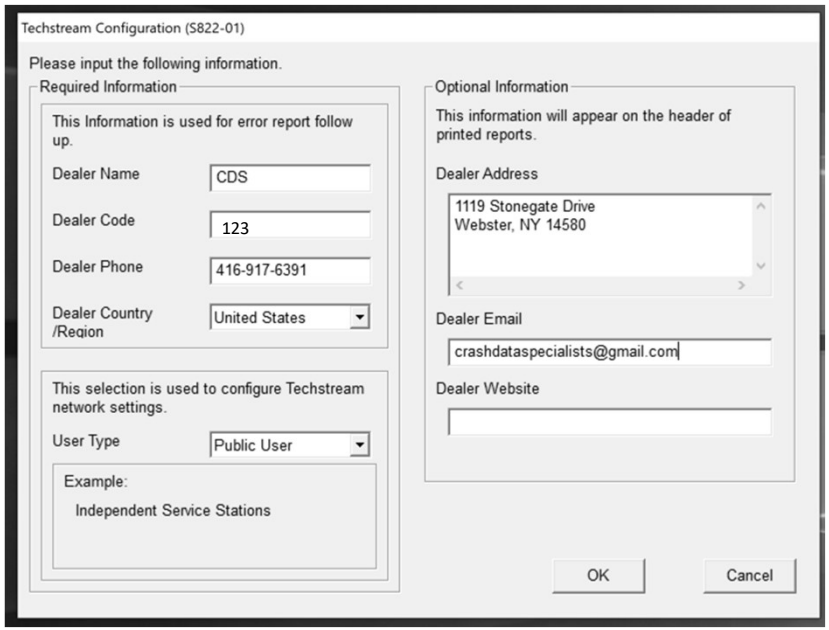
Toyota TechStream Setup

Select "Public User"

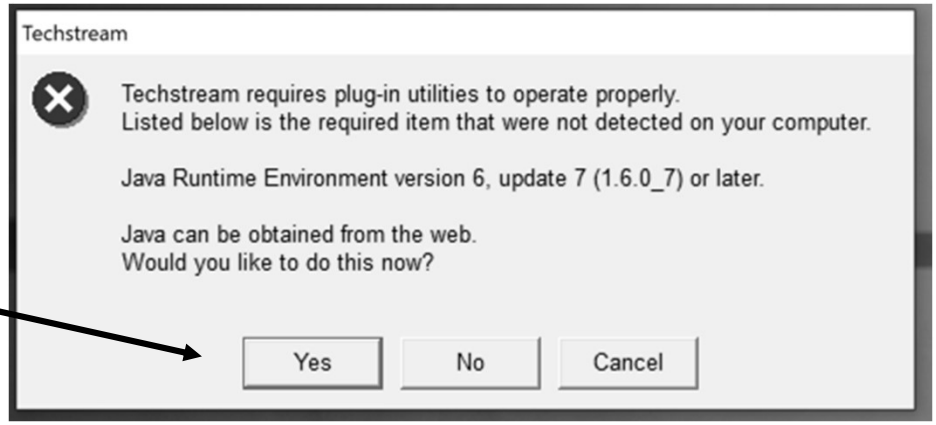


Complete all fields populated here:
If any but "dealer website" is blank in will not continue



Toyota TechStream Setup

- TechStream requires a Java Runtime environment to run properly.
- You must select “Yes” here unless you are certain you meet the requirements
- This will launch an Internet Explorer Window and prompt you to download and install the latest Java



Rick Ruth note: Mine did not require this

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Toyota TechStream Setup

- Click “Java Download”

Rick Ruth note: Mine did not require this

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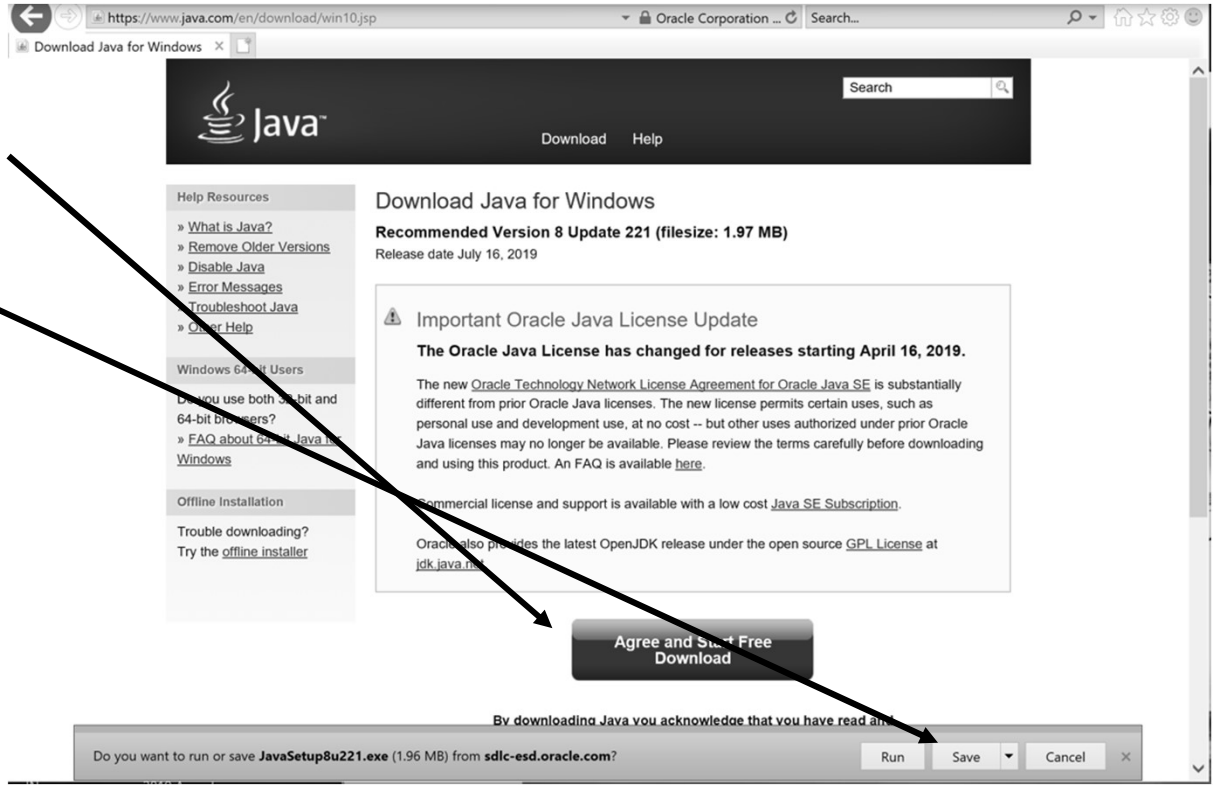
Toyota TechStream Setup

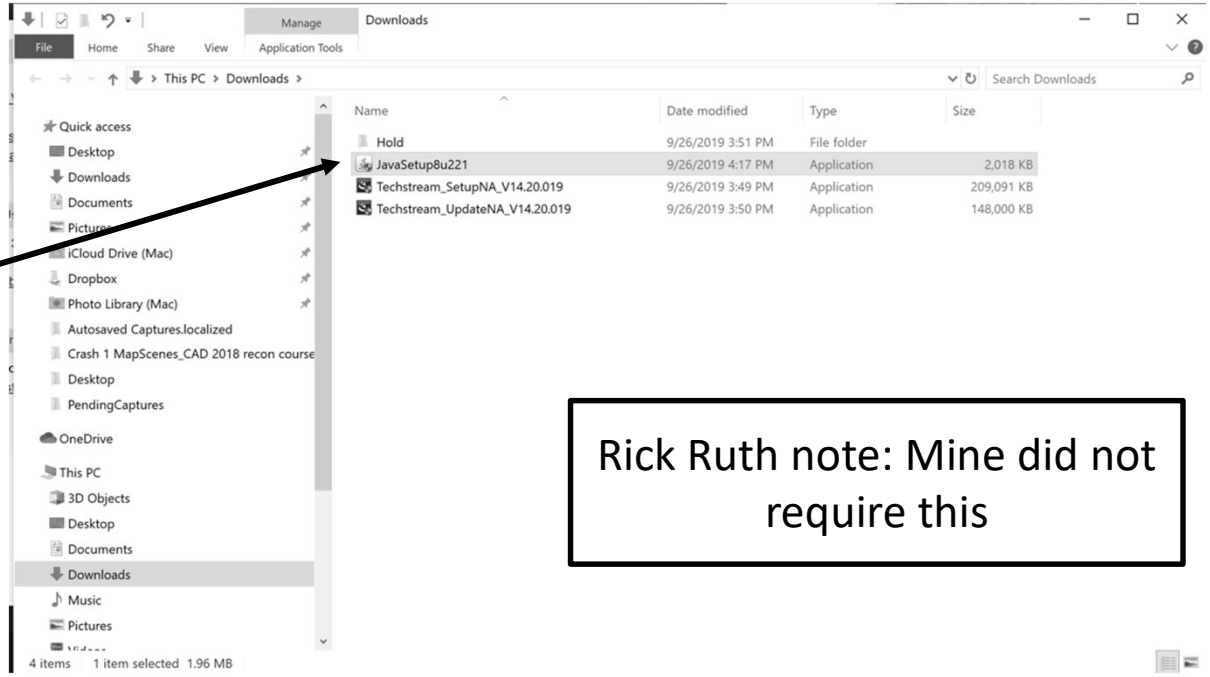
- Click: “Agree and Start Free Download”
- At the bottom footer prompt click: “Save”
  - This will keep a copy in your Downloads directory for later use if you must re-install the software!

Rick Ruth note: Mine did not require this



Toyota TechStream Setup

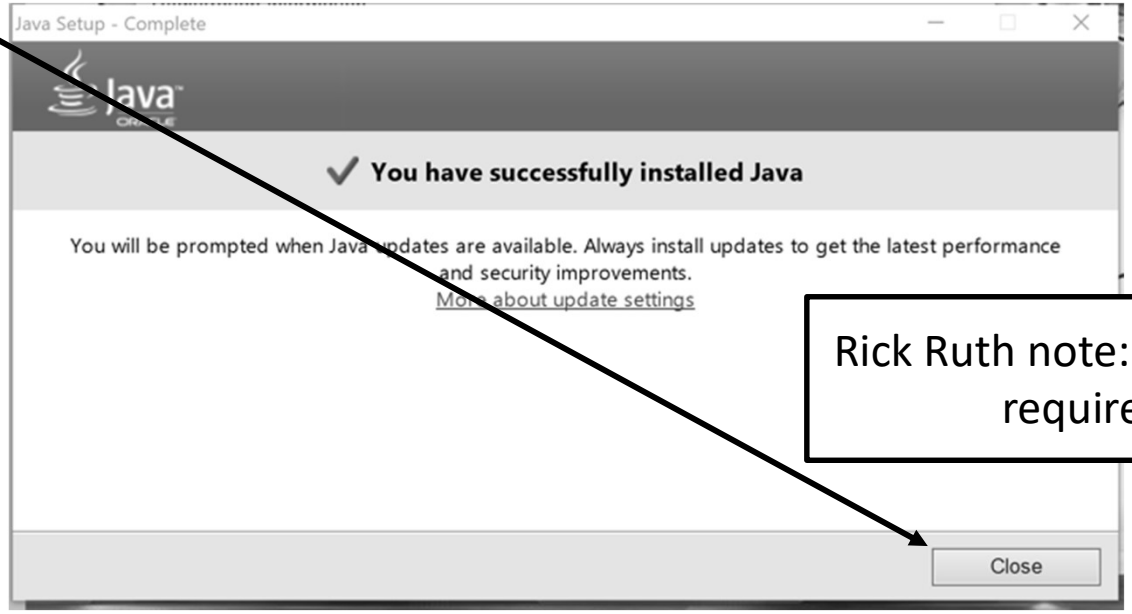
- Locate the Java setup program in your Downloads directory and double-click to launch the install



Rick Ruth note: Mine did not require this

Toyota TechStream Setup

- Upon successful completion this dialogue box appears.
- Click “Close”



Rick Ruth note: Mine did not require this

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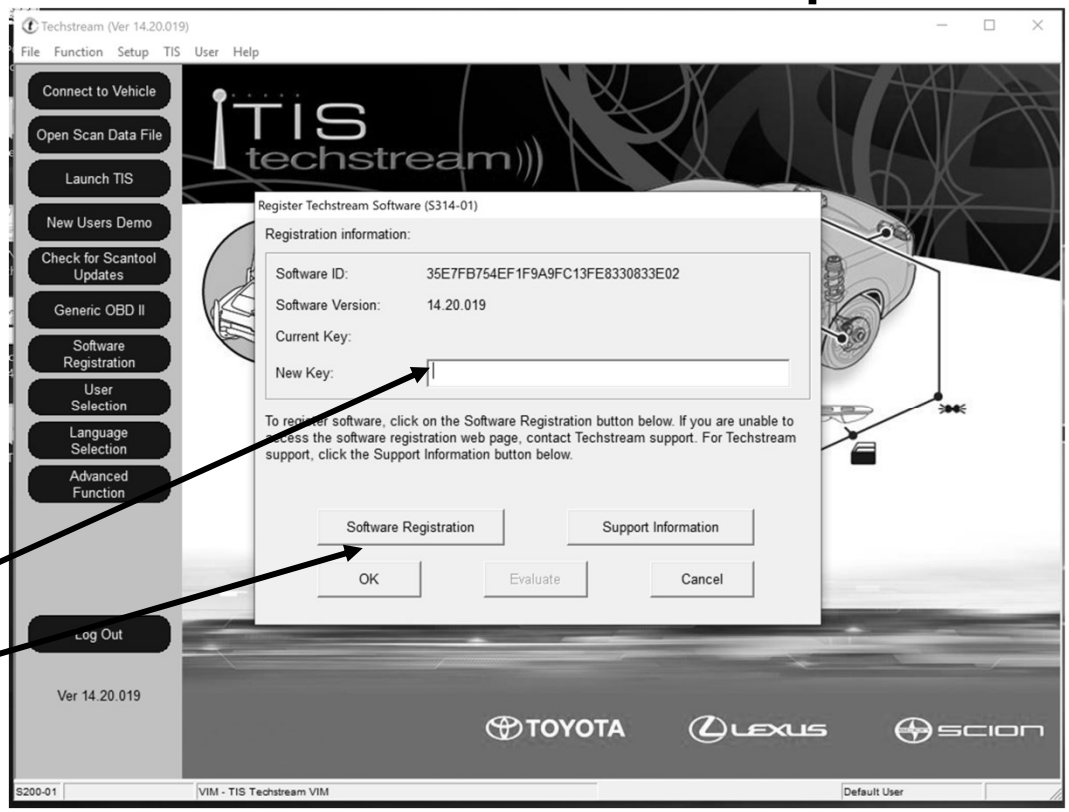
Toyota TechStream Setup

- Launch TechStream either from the Desktop short-cut or from the Start Menu Folder
- You are prompted for a Registration Key
- Click “Software Registration”



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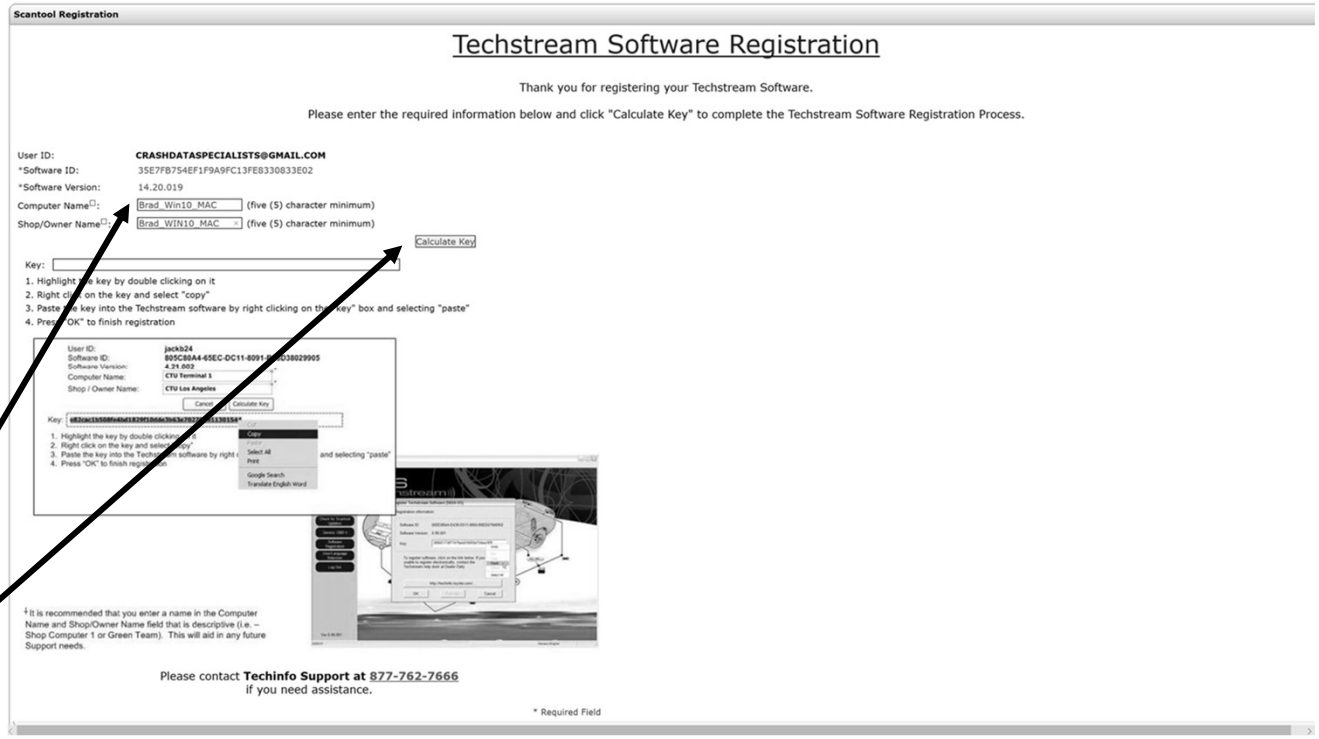
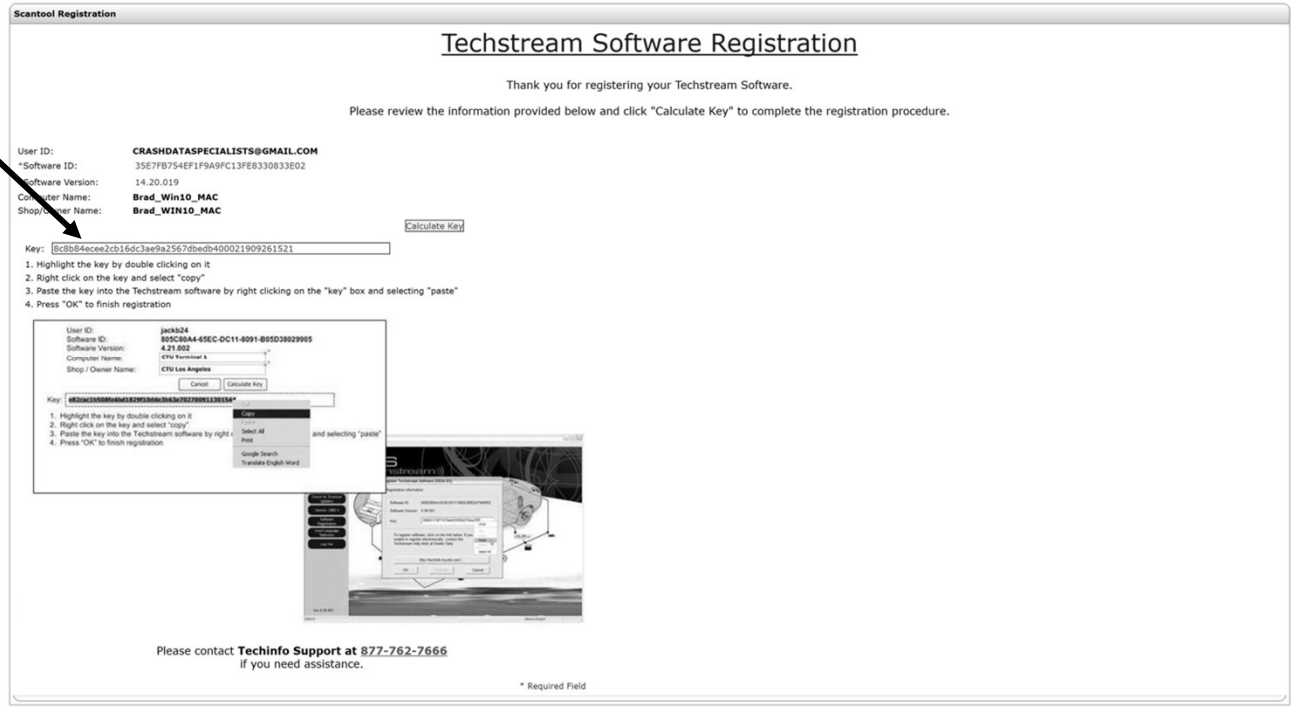
Toyota TechStream Setup

- You should still be logged in to your TechStream account
- If not, you will be prompted to log in
- Input a name under “Computer Name” and “Shop Owner”
- Click “Calculate Key”



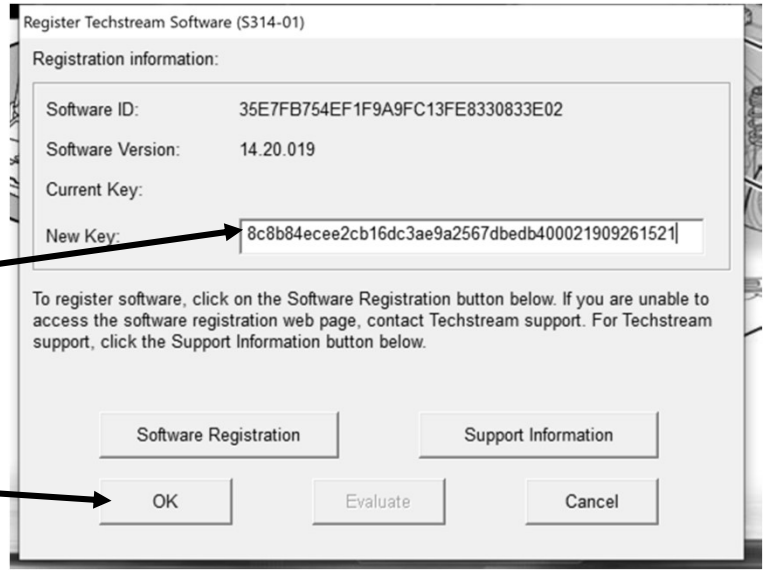
Toyota TechStream Setup

- “Double-click” the key number, right-click and click “Copy”
- Alternatively, click, drag to select the key number.



Toyota TechStream Setup

- “Paste” this into the “New Key” on the registration screen
- Click anywhere within the box and right-click, then click “Paste:
- Click OK to complete registration



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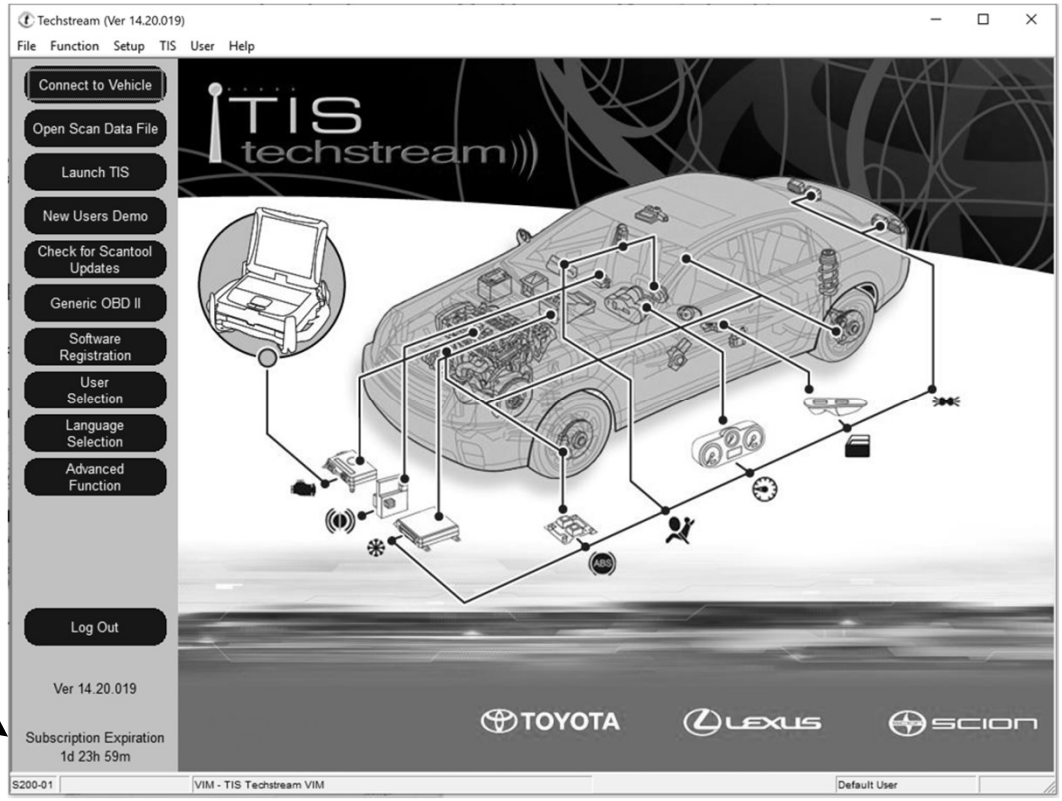
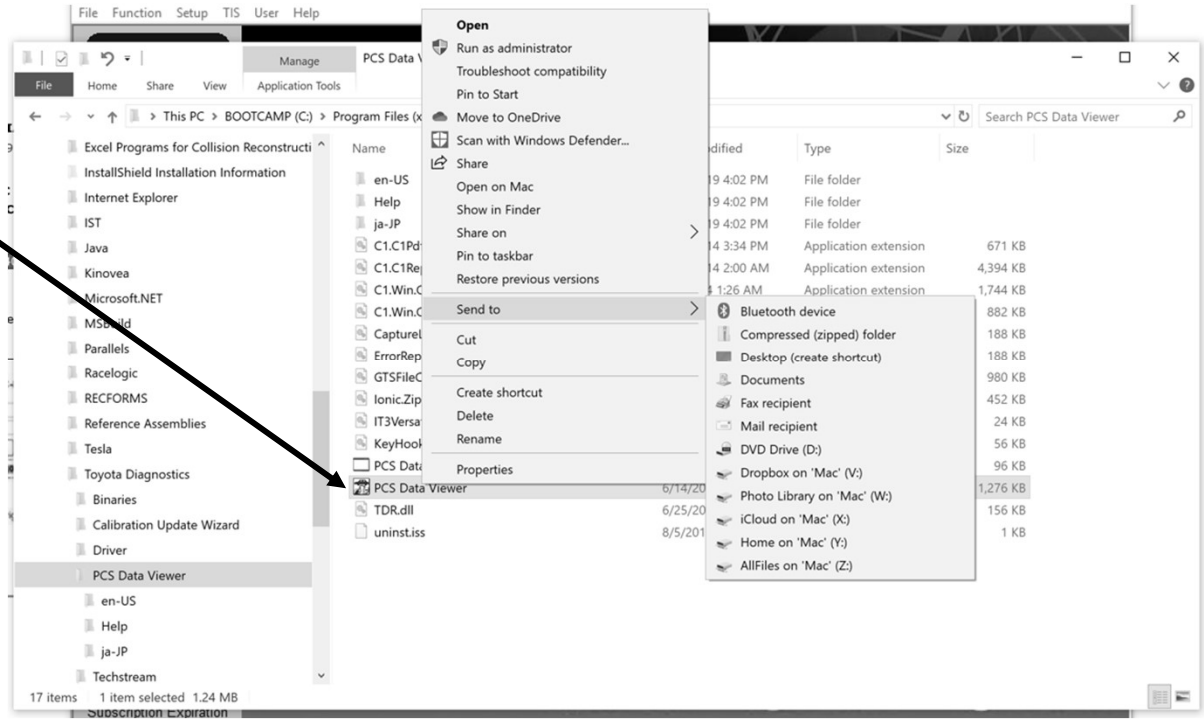
Toyota TechStream Setup

- You should return to this screen:
- Check your “Subscription Expiration”
- It should indicate either ~ 24 hours or 365 days depending on the type of subscription



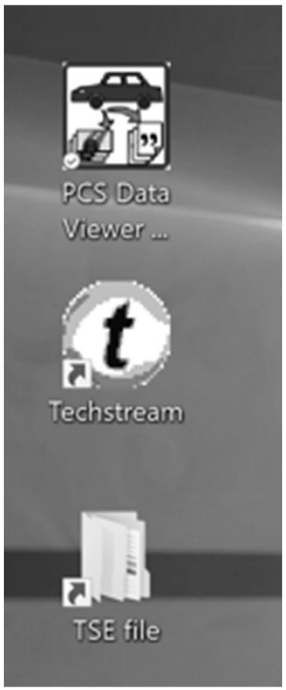
Toyota TechStream Setup

- To access VDAS files and Camera Images retrieved in the sections that follow, you will need to launch the “PCS Data Viewer” application
- This application is installed with TechStream, and once TechStream is activated, it can be launched
- Navigate from a Windows Explorer window to here:
- Right-click “PCS Data Viewer” and click “Send To” and chose “Desktop – Create Shortcut”



Toyota TechStream Setup

- Check your Windows Desktop
- You should see these 3 Icons:
  - PCS Data Viewer – launches the tool to view the VDAS file and create output that includes Vehicle Control History data and any related images (Depends on TSS 1.0 or 2.0 – version varies depending on make/model/year of vehicle
  - TechStream
  - TSE File – directory for information stored using TechStream



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Toyota TechStream – Vehicle Inquiry

- Your subscription includes access to Service manuals, Owners Manuals, and Vehicle “as-built” information
- Click “Vehicle Inquiry” Tab and input VIN of subject Vehicle
- Click “Lookup”

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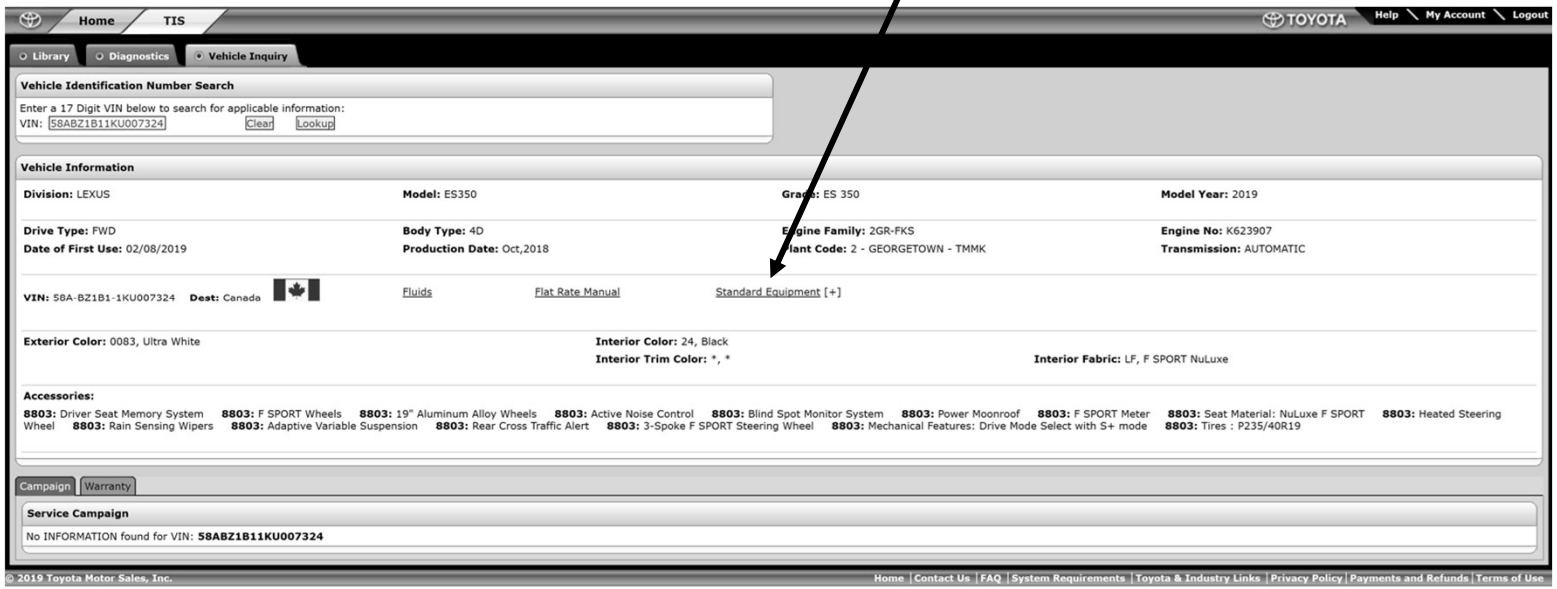
Toyota TechStream – Vehicle Inquiry

- "As-Built" record is shown. Click the "+" beside "Standard Equipment" to expand the displayed data



Toyota TechStream – Vehicle Inquiry

- A listing of the vehicle as-built equipment will be shown, including any optional equipment installed at the factory.
- This info can also be printed.

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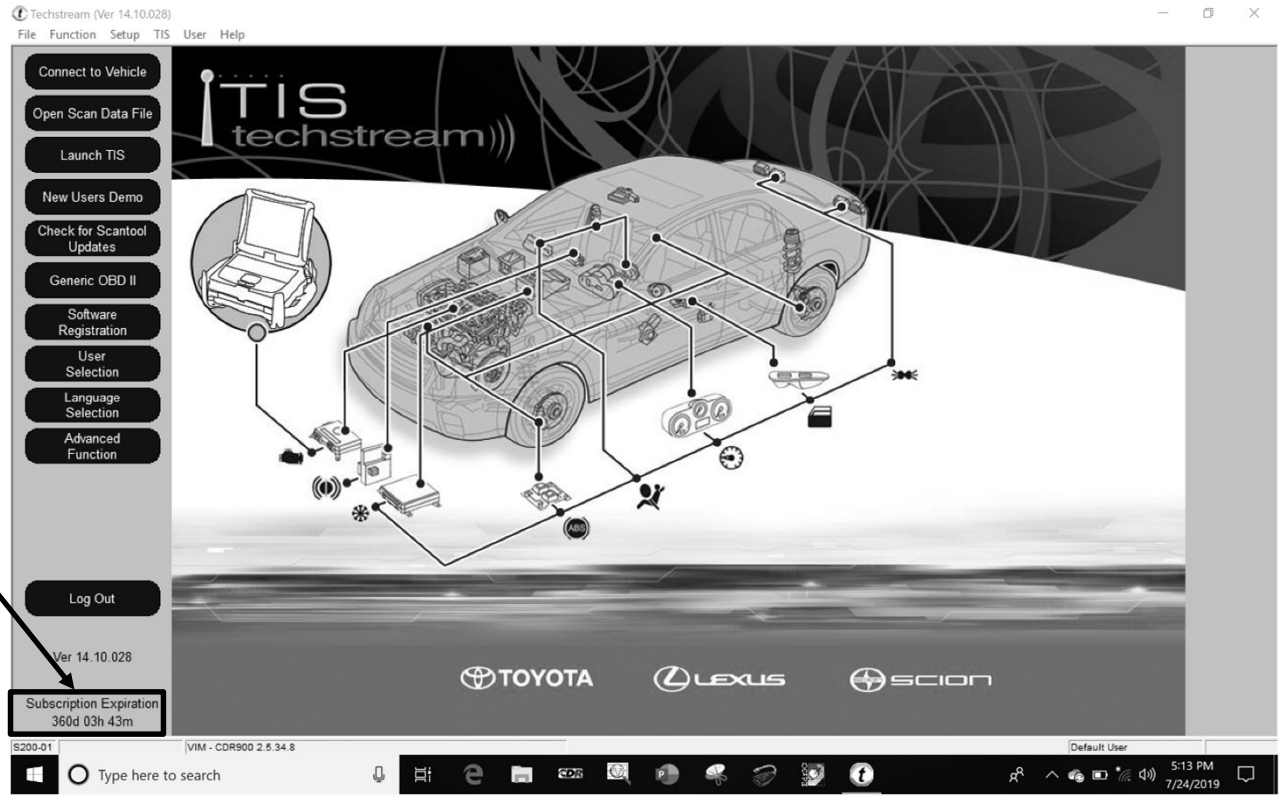
Toyota TechStream Operation

- Launch TechStream from the desktop icon
- Subscription Expiration
- Check VIM Setup



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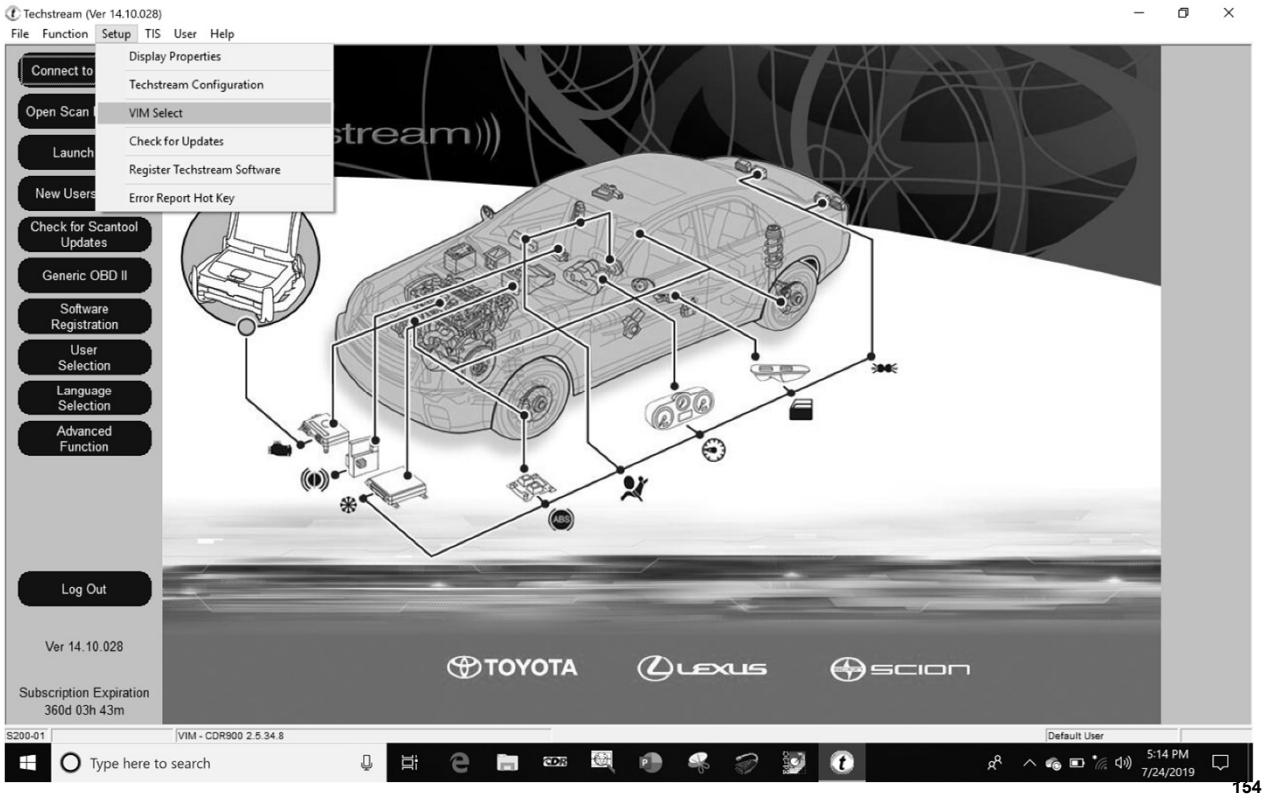
Toyota TechStream VIM Setup

- Click “Setup”
- Click “VIM Select”



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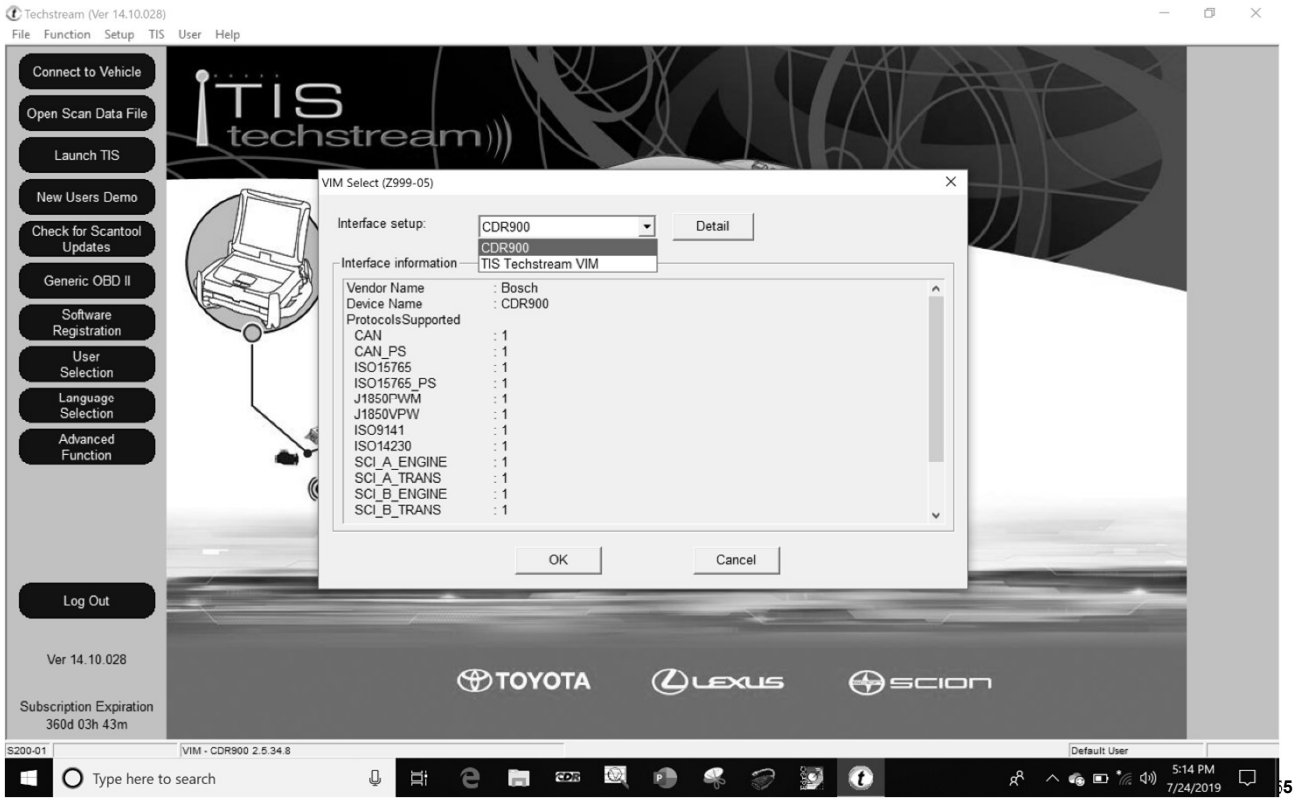
Toyota TechStream VIM Setup

- Available “VIM”
  - V ehicle
  - I nterface
  - M odule
- Several may be in list
- Choose “CDR900”
- Click OK



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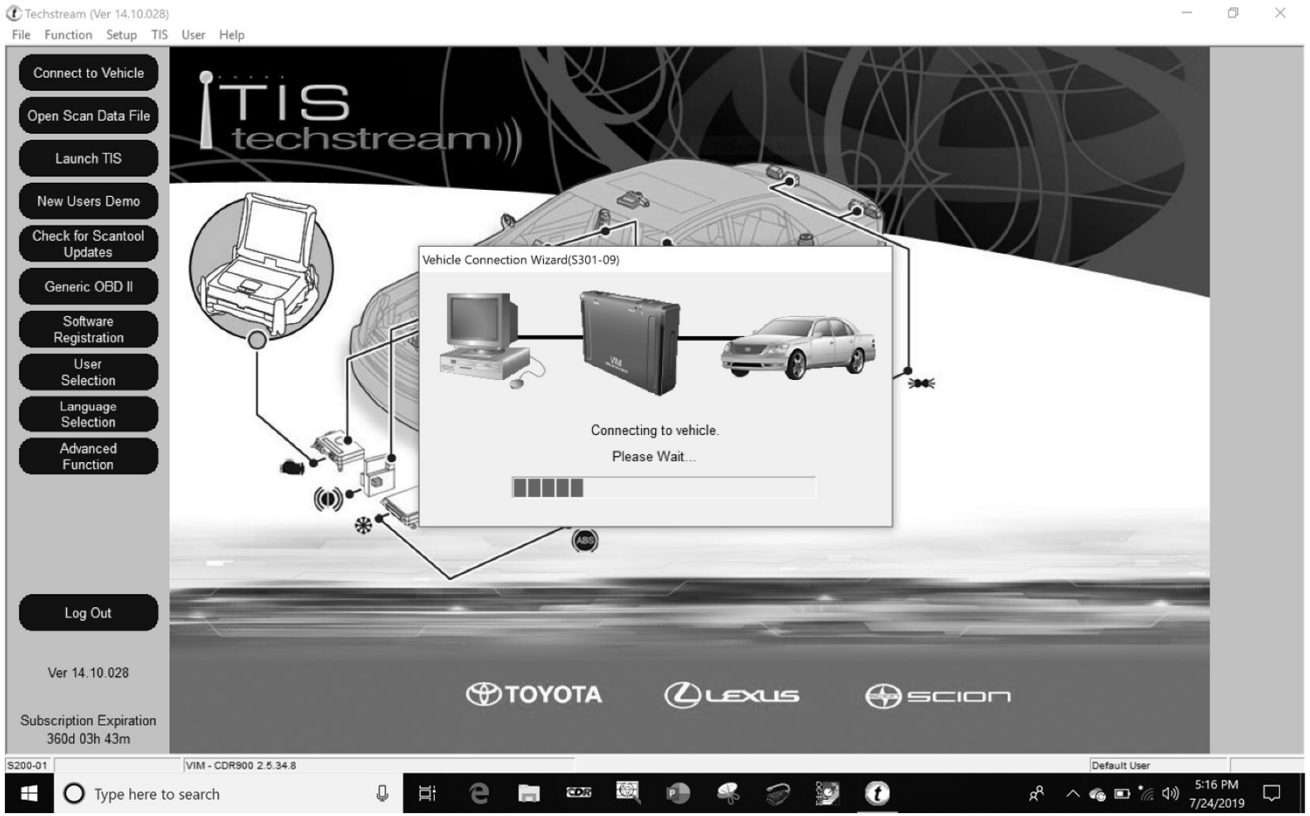
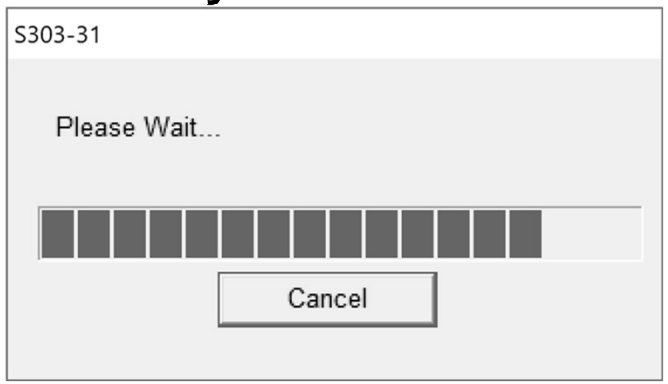
Toyota TechStream Operation

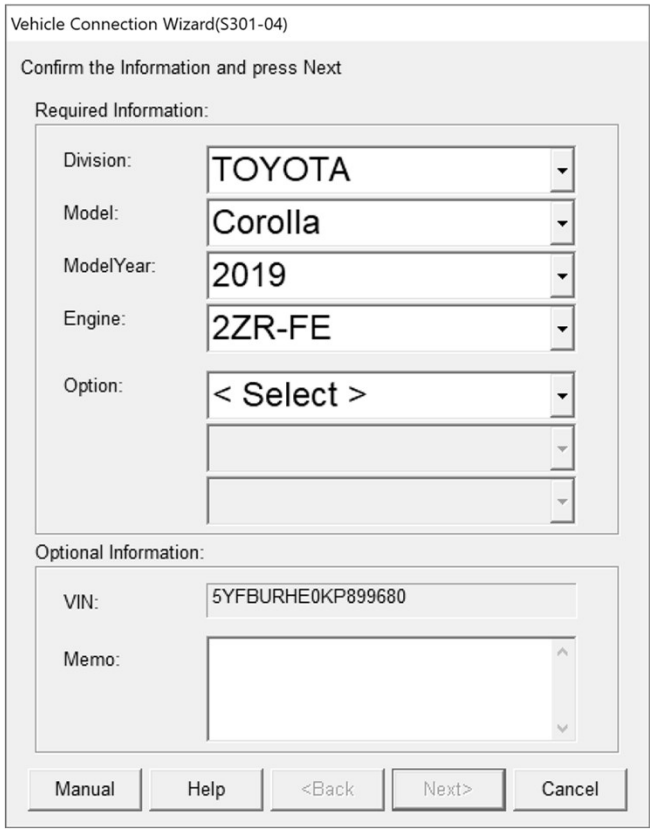
- Click “Connect to Vehicle”
- Make sure Ignition Key is ON !!



Toyota TechStream Operation

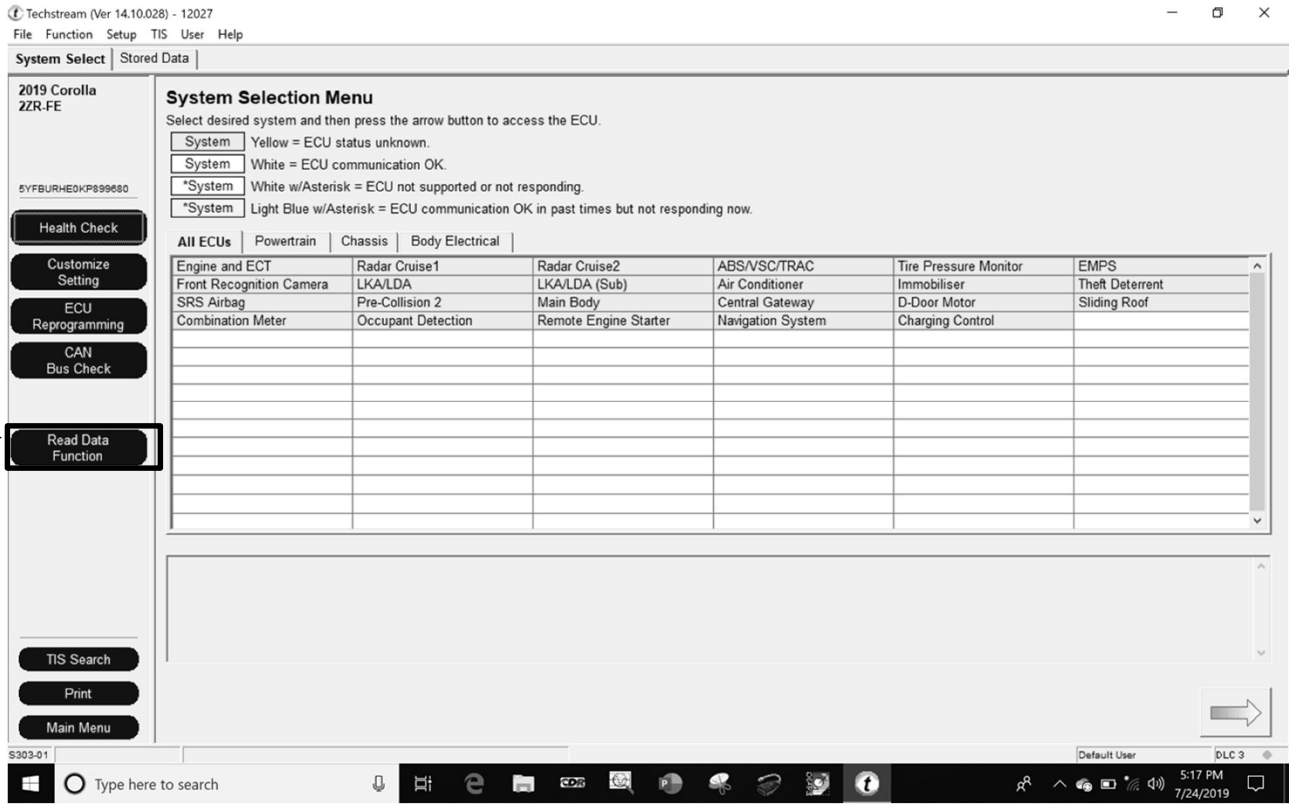
- TechStream connects to the vehicle and should auto detect
- VIN auto populated
- Additional Options may need to be user selected:
  - Smartkey
  - Radar Cruise
- Memo area





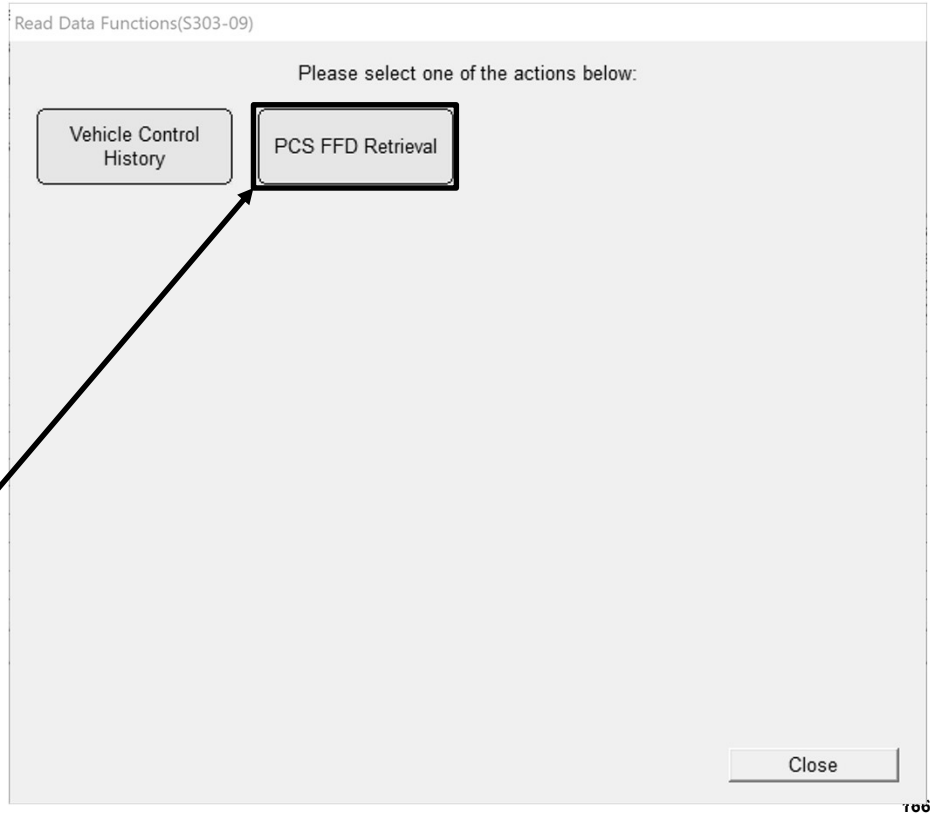
Toyota TechStream Operation

- Yellow ECU's are possible equipment
  - If you select "Health Check", after a Bus Scan, available ECU's show as White
- Click "Read Data Function"



Toyota TechStream Operation

- Two Paths to the Data
  - Vehicle Control History
    - Only retrieves VCH
  - PCS FFD Retrieval
    - Retrieves VCH and Images (If present)
- Click “PCS FFD Retrieval”
- (Vehicle Control History option results in one file with ALL data and no images)



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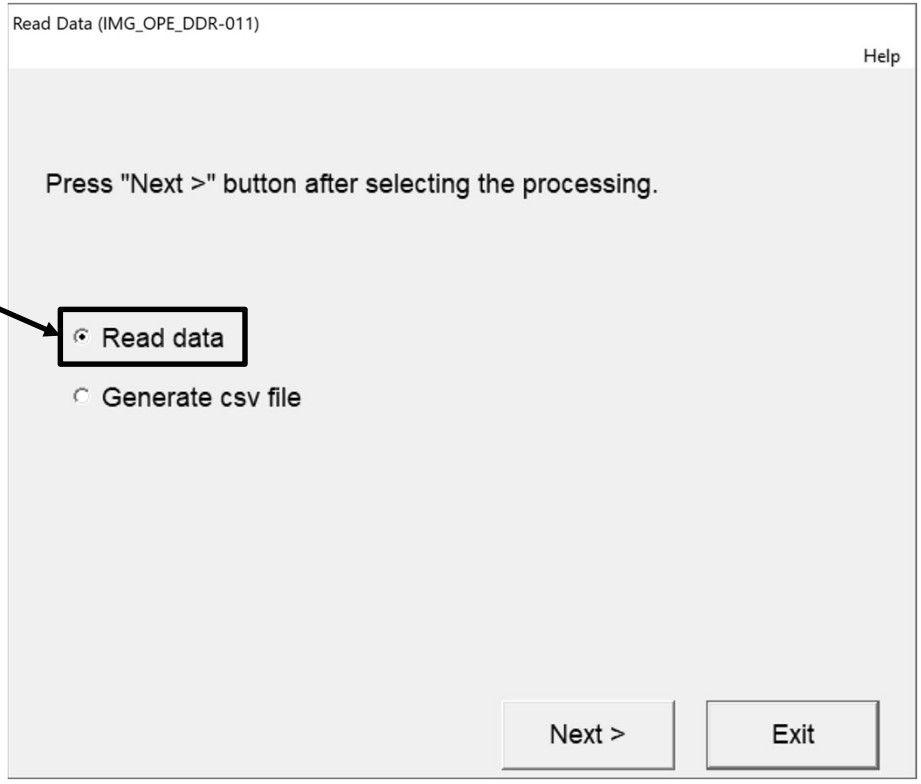
Toyota TechStream Operation

- Privacy Warning
  - Make sure to click the checkbox!
  - Click Next!

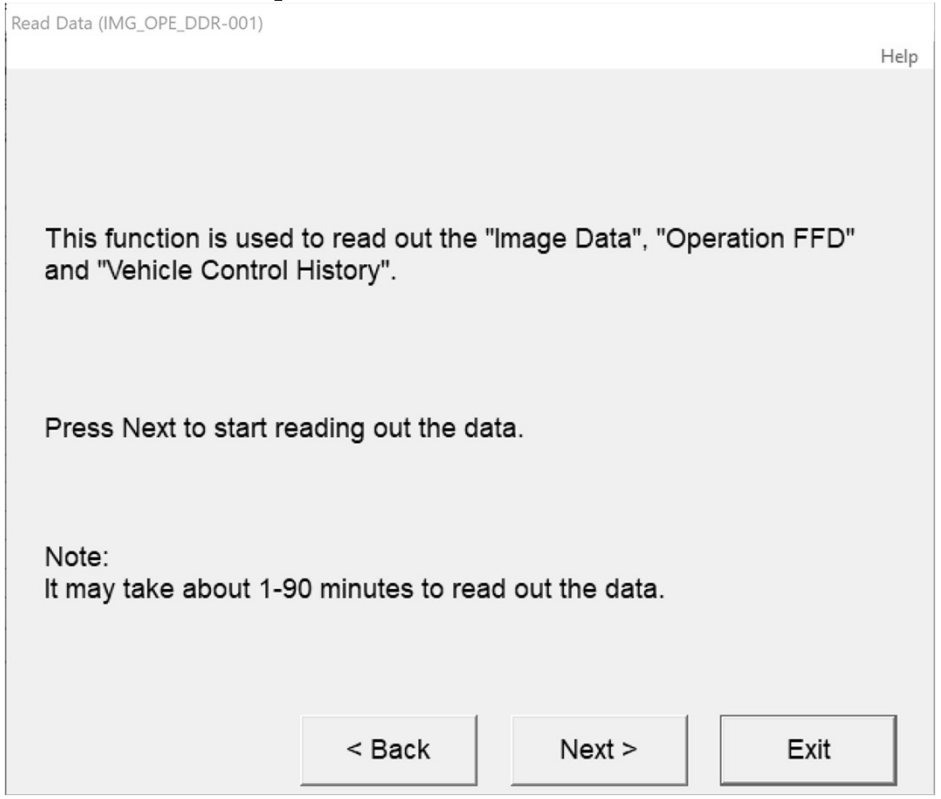
Toyota TechStream Operation

- Two Paths
  - Read Data
  - Generate csv file
- Click “Read data”
- Click Next
- (“Generate csv file” will not retrieve image data)



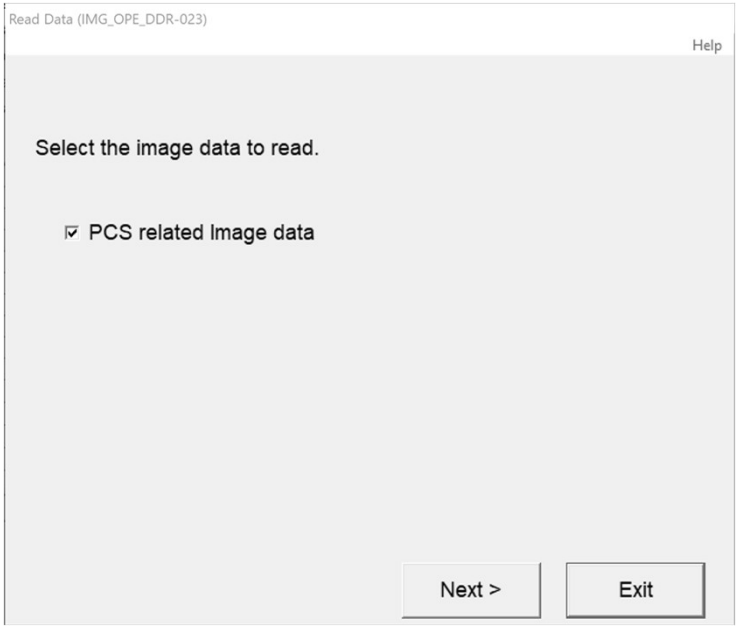
Toyota TechStream Operation

- Depending on the number of VCH records, retrieval may take some time
- Following screen gives time estimate
- Ensure sufficient battery or booster power for retrieval

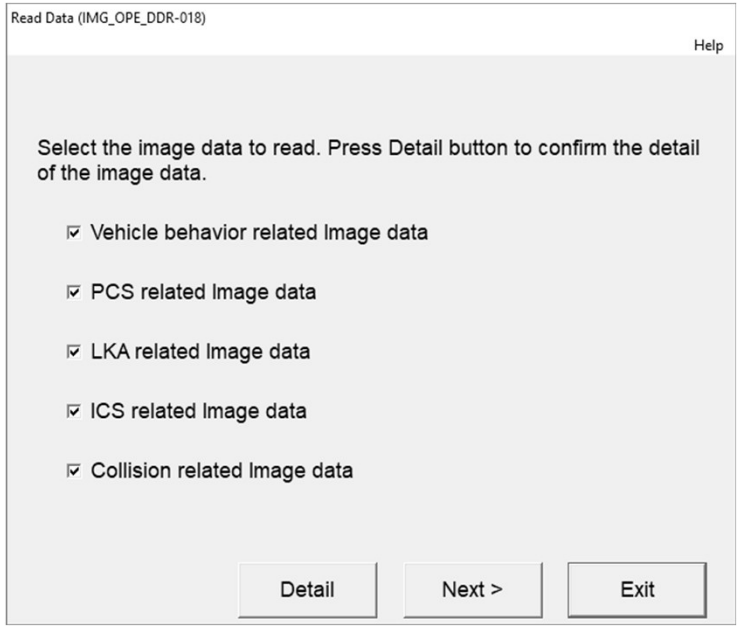


Toyota TechStream – TTS 1.0 vs. TSS 2.0

TSS 1.0 – Only Option is to read related image and related VCH data

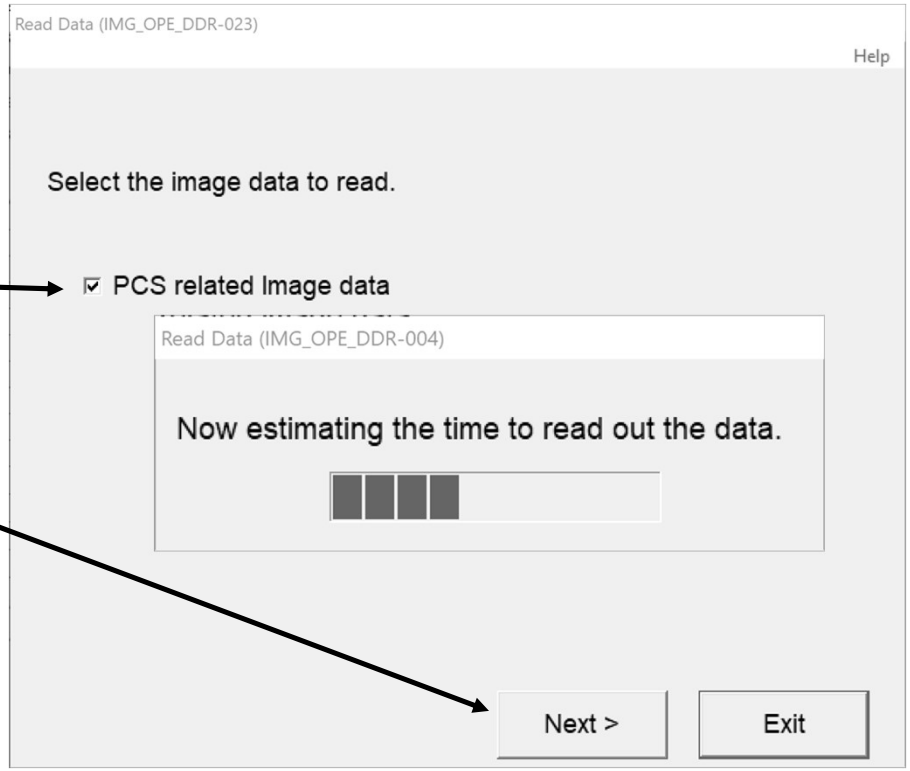


TSS 2.0 – 5 areas that may have images stored. Choose 1 or all to retrieve image and related VCH data



Toyota TechStream TSS 1.0

- Only choice is ”PCS related Image data”
- Select this box and click “Next”
- The time to retrieve the data will now be estimated
- Varies on number of events stored



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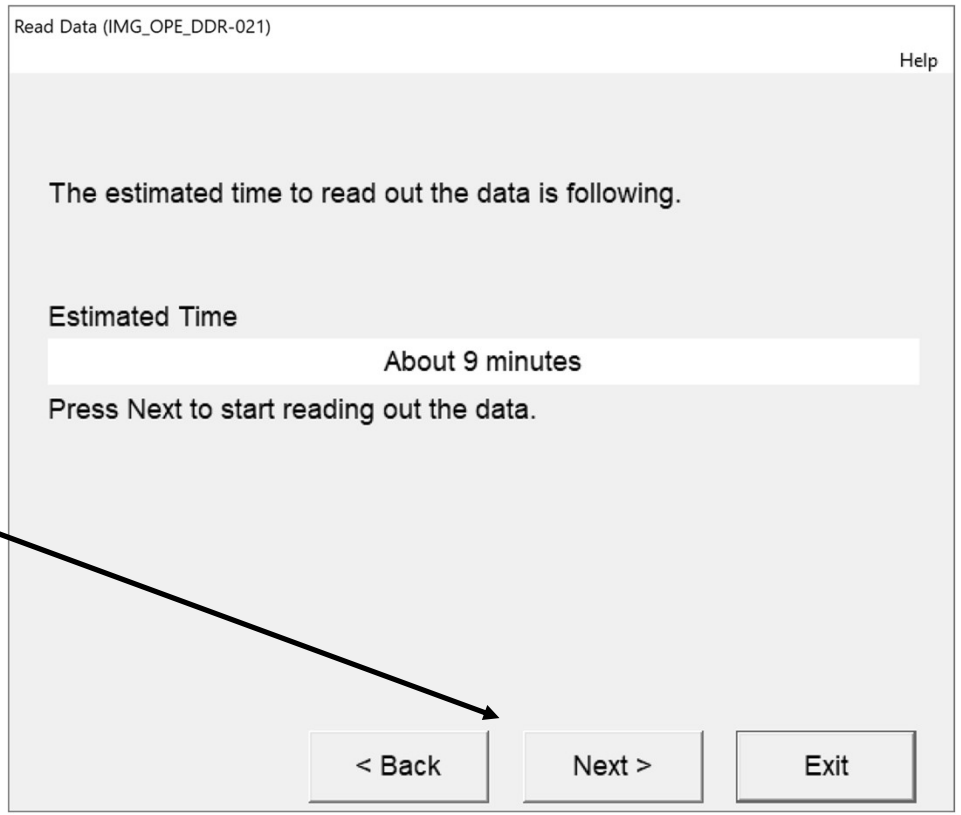
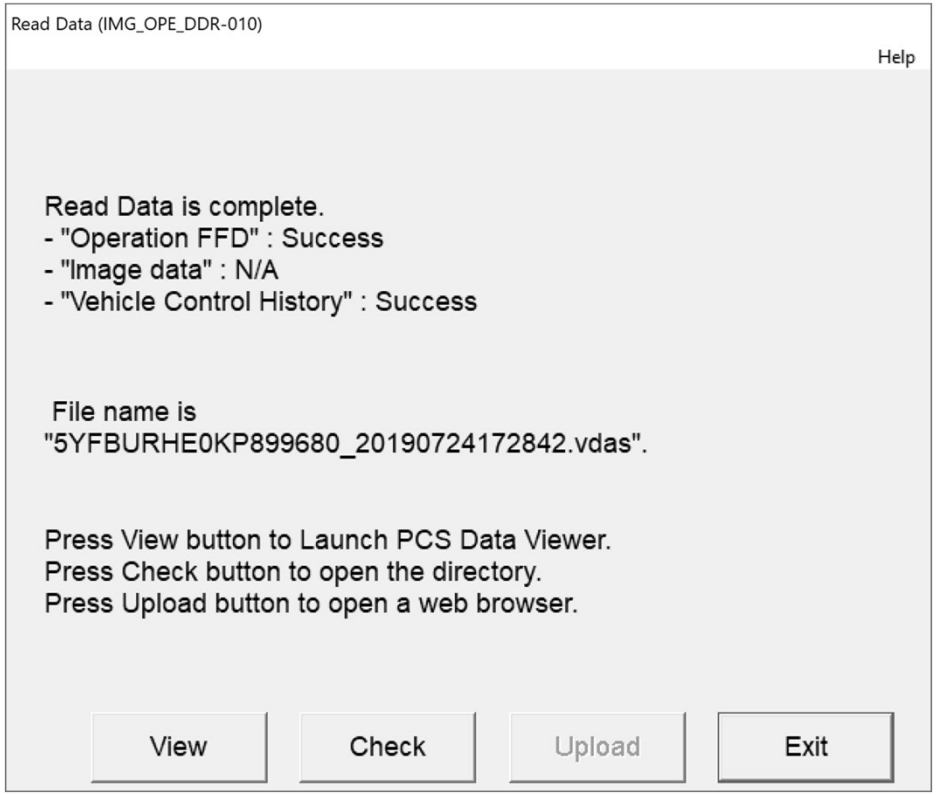
Toyota TechStream TSS 1.0

- Ensure sufficient battery power or booster battery available
- Click “Next” to begin reading the data



Toyota TechStream TSS 1.0

- Once complete note the summary data displayed
- If “Image data” is N/A then no images were stored relating to a PCS event
- If “Vehicle Control History” is ”Success” then the VCH data has been retrieved



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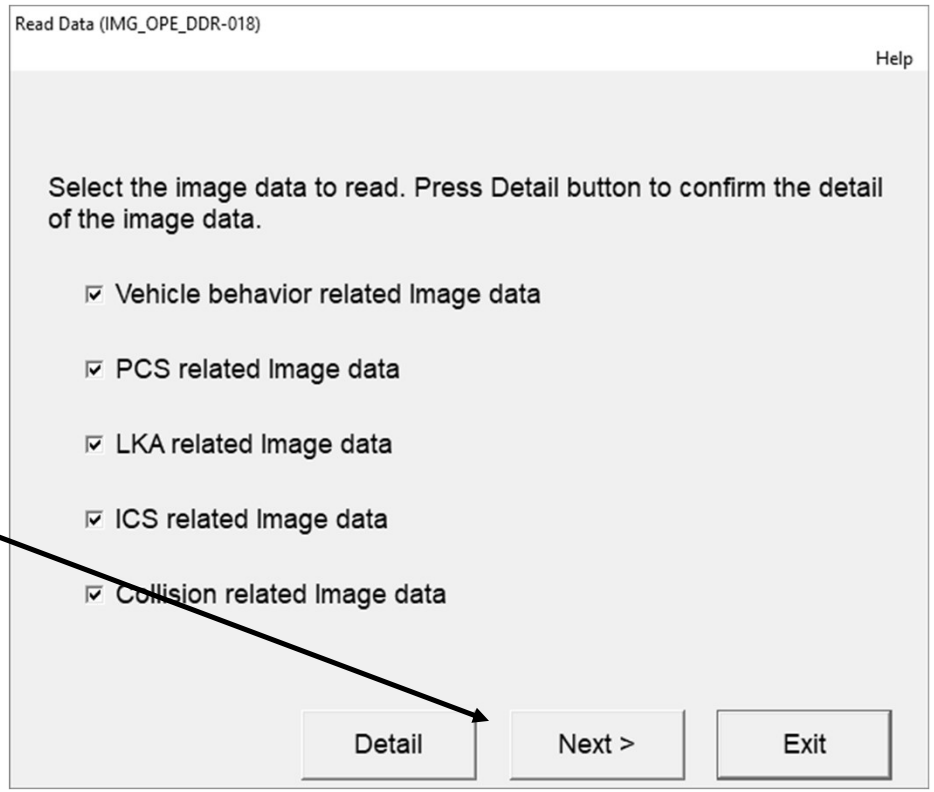
Toyota TechStream TSS 2.0

- Check Boxes for Image data you want
- Select this box and click “Next”
- The time to retrieve the data will now be estimated
- Varies on number of events stored



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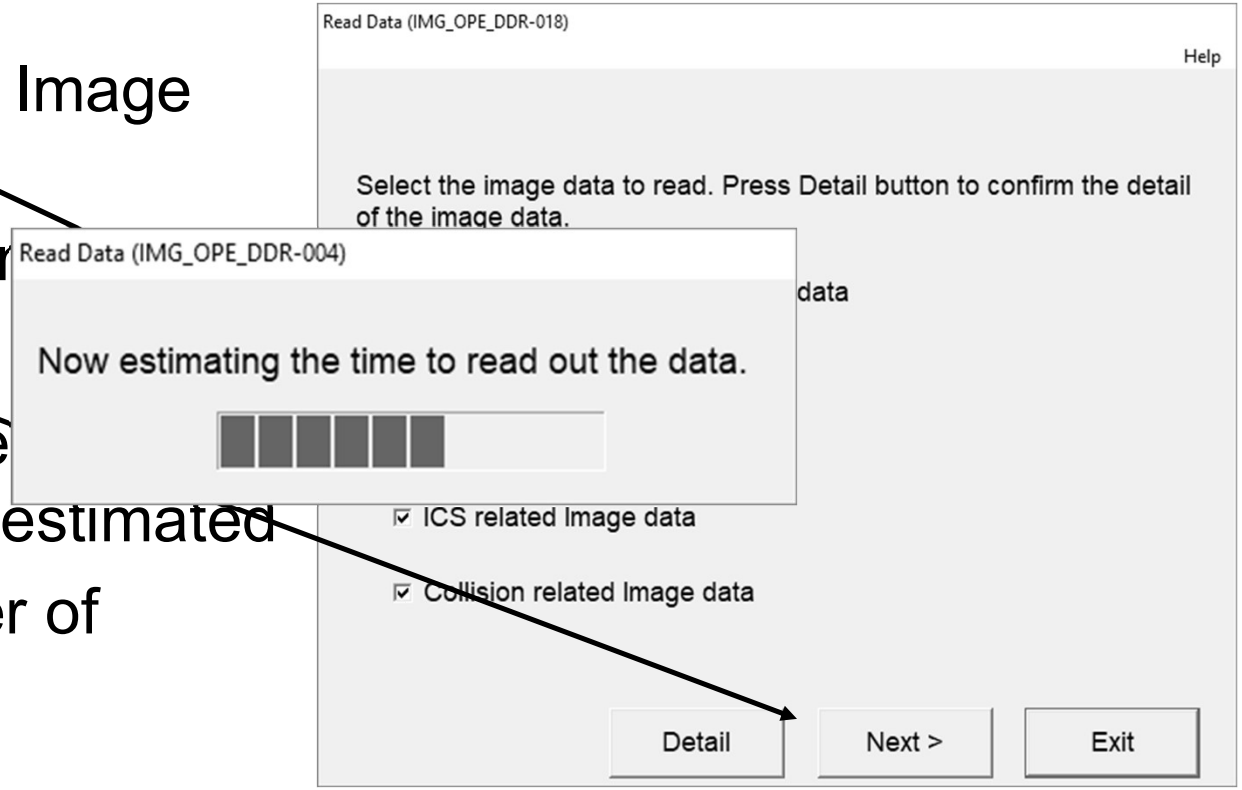
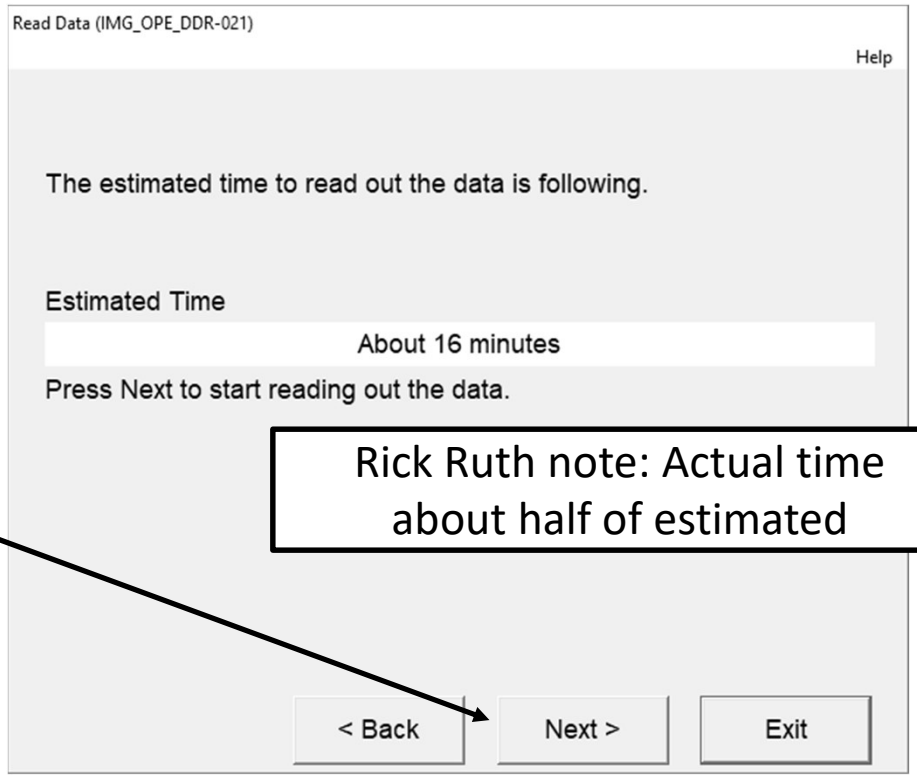
Toyota TechStream TSS 2.0

- Check Boxes for Image data you want
- Select this box ar click “Next”
- The time to retrie data will now be estimated
- Varies on number of events stored



Toyota TechStream TSS 2.0

- Ensure sufficient battery power or booster battery available
- Click “Next” to begin reading the data



Rick Ruth note: Actual time about half of estimated

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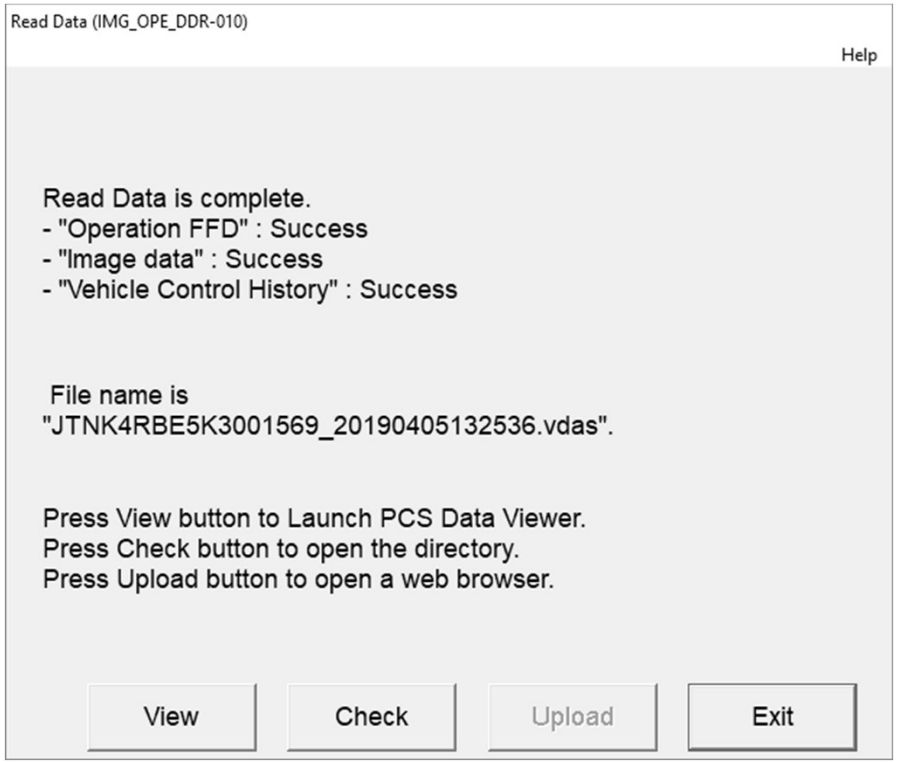
Toyota TechStream TSS 2.0

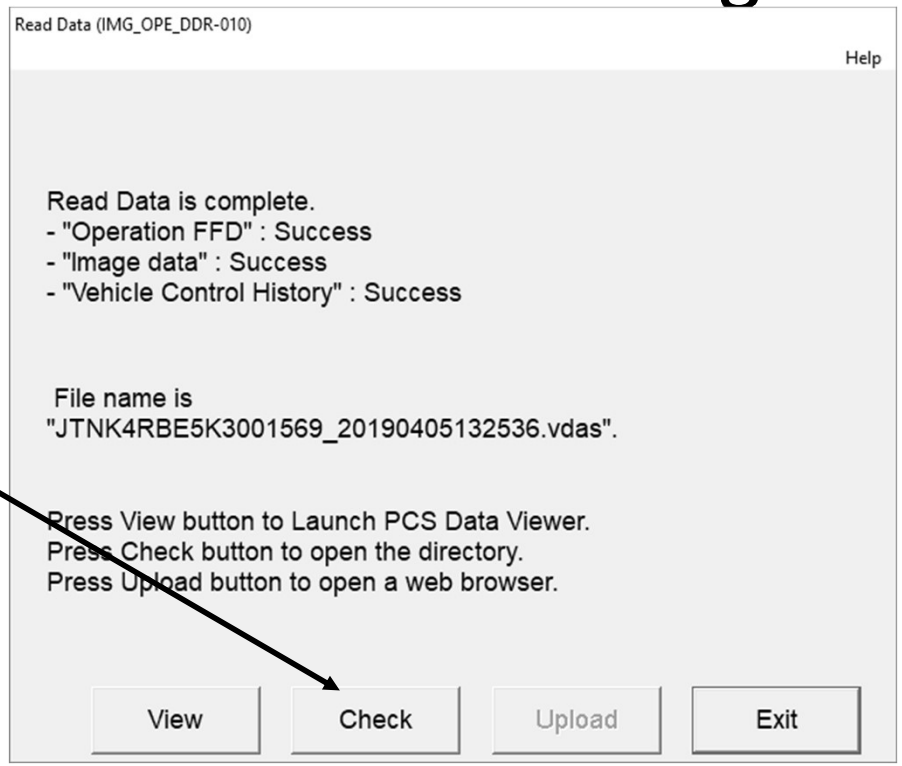
- Once complete, note the summary data displayed
- If “Image data” is N/A then no images were stored relating to a any events
- If “Vehicle Control History” is ”Success” then the VCH data has been retrieved

Rick Ruth note: “Check” shows it has already been saved



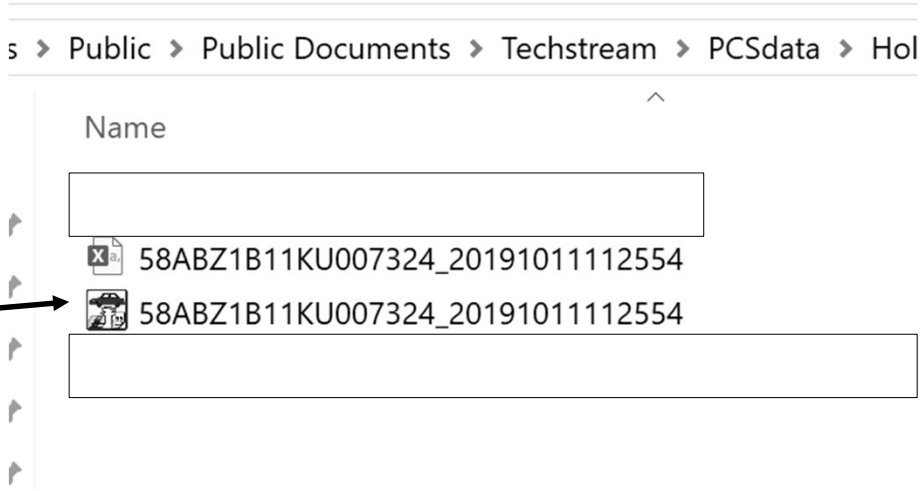
Toyota TechStream VDAS File Viewing

- Once complete, click “Check”
- This takes you to a File Explorer view of the location the .VDAS files are stored



Toyota TechStream – VDAS Viewer

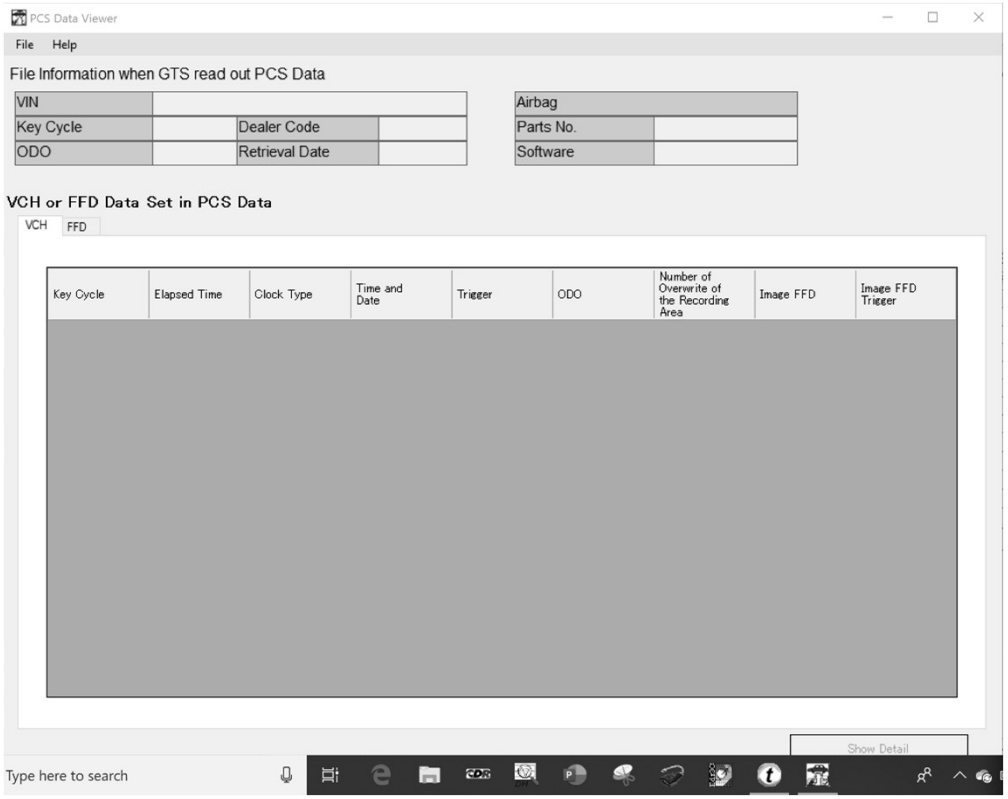
- After clicking “Check” a File Explorer view opens to the location the VDAS files are saved in.
- Double-clicking the .VDAS file will open it in the PCS Data Viewer



Rick Ruth note: I had to navigate to the file location, the initial prompt was not to it

Toyota TechStream – PCS Data Viewer

- Launch the PCS Data Viewer application
- Load the VDAS file saved in previous steps



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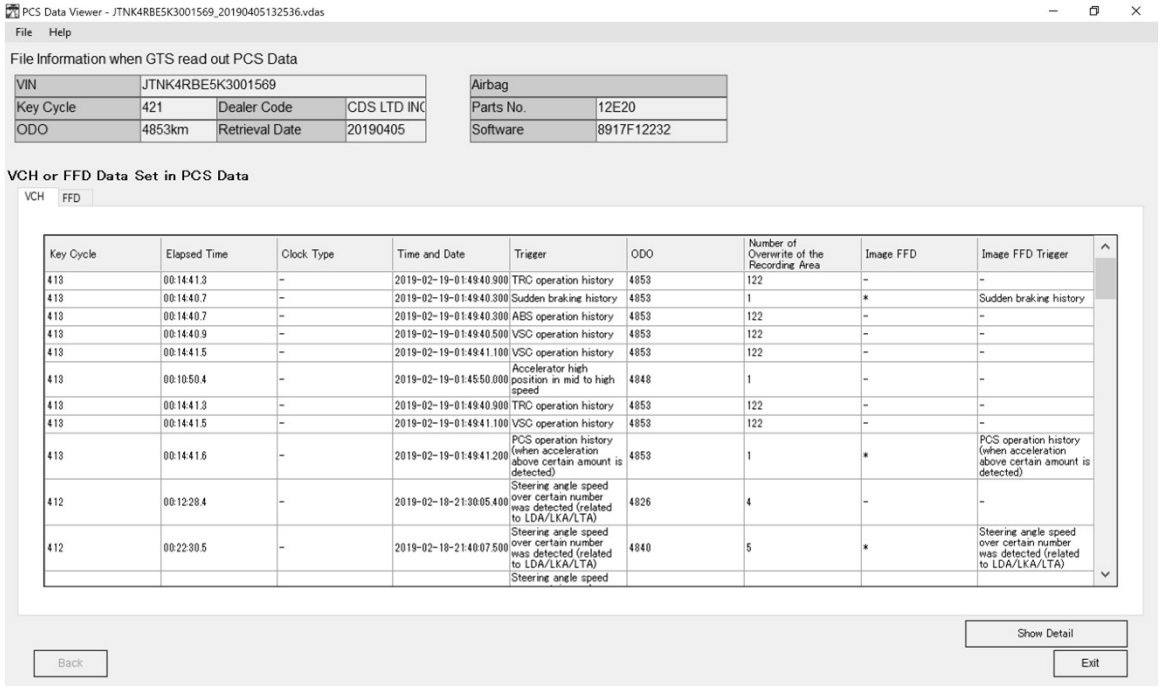
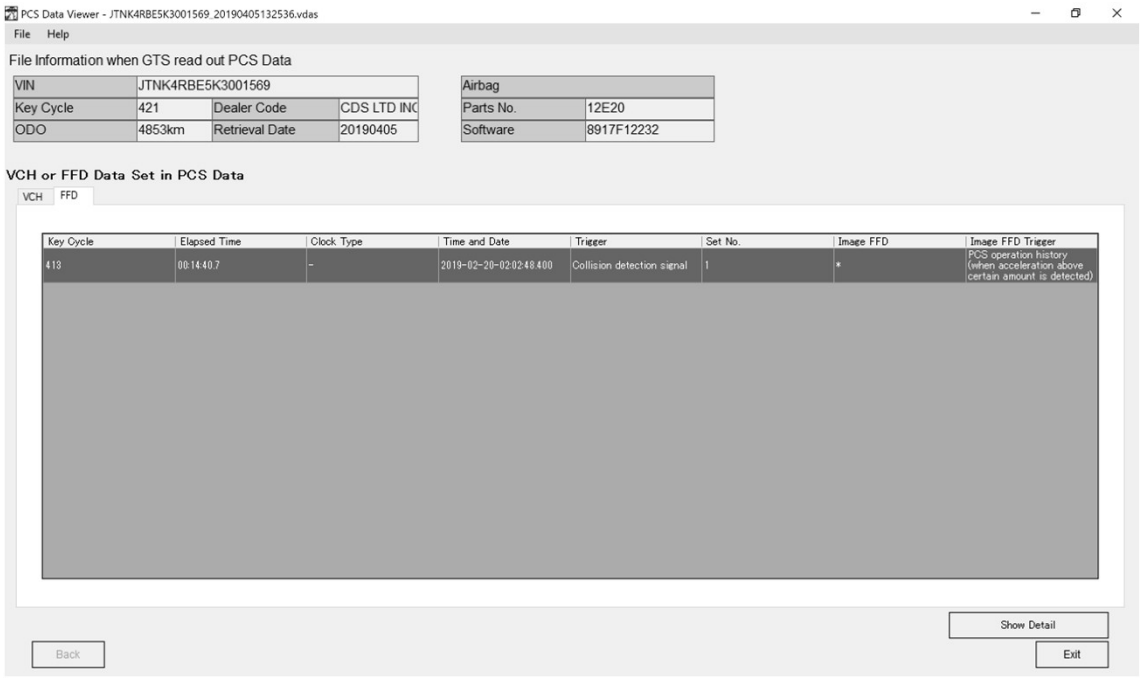
Toyota TechStream – PCS Data Viewer

- Under the VCH Tab all VCH events are listed
- Double-Click “Key Cycle” to sort them from highest to lowest



Toyota TechStream – PCS Data Viewer

- Under the FFD Tab “Freeze Frame Data” records are listed
- Double-Click “Key Cycle” to sort them from highest to lowest
- Events with an * in the Image FFD column will have associated images



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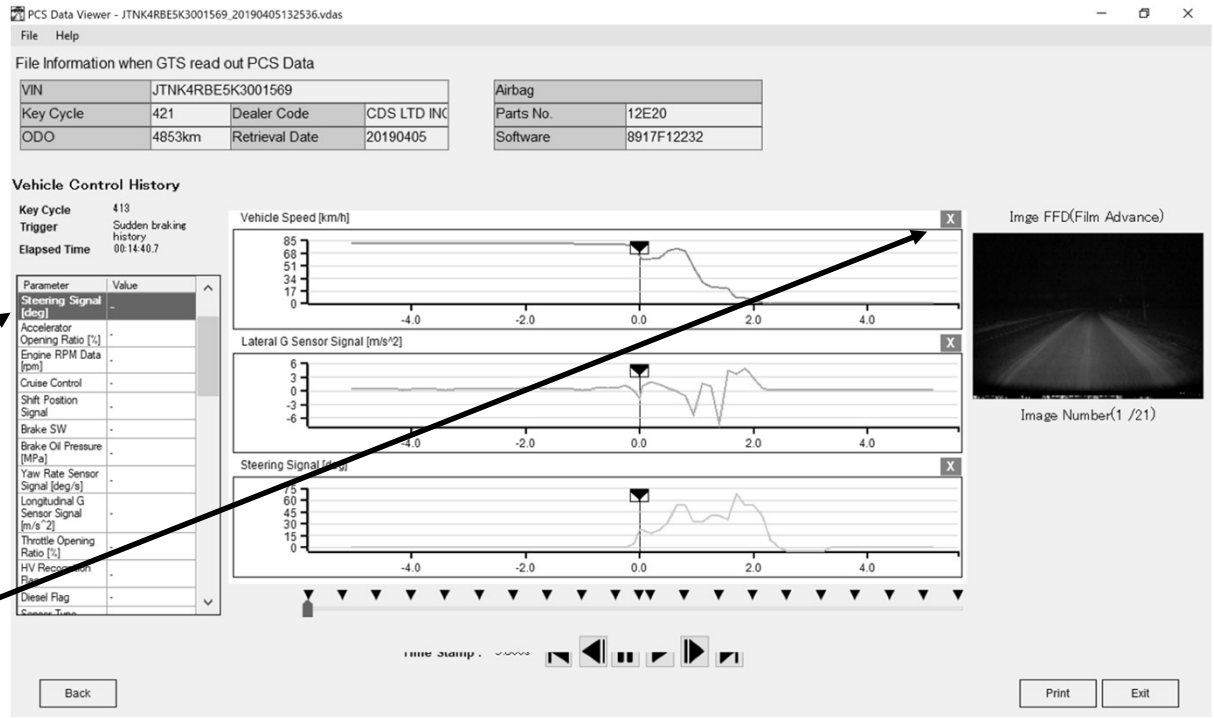
Toyota TechStream – PCS Data Viewer

- VCH and FFD views of the same event contain both over-lapping and different data
- Double click an item to view playback
- Click the red “X” to remove



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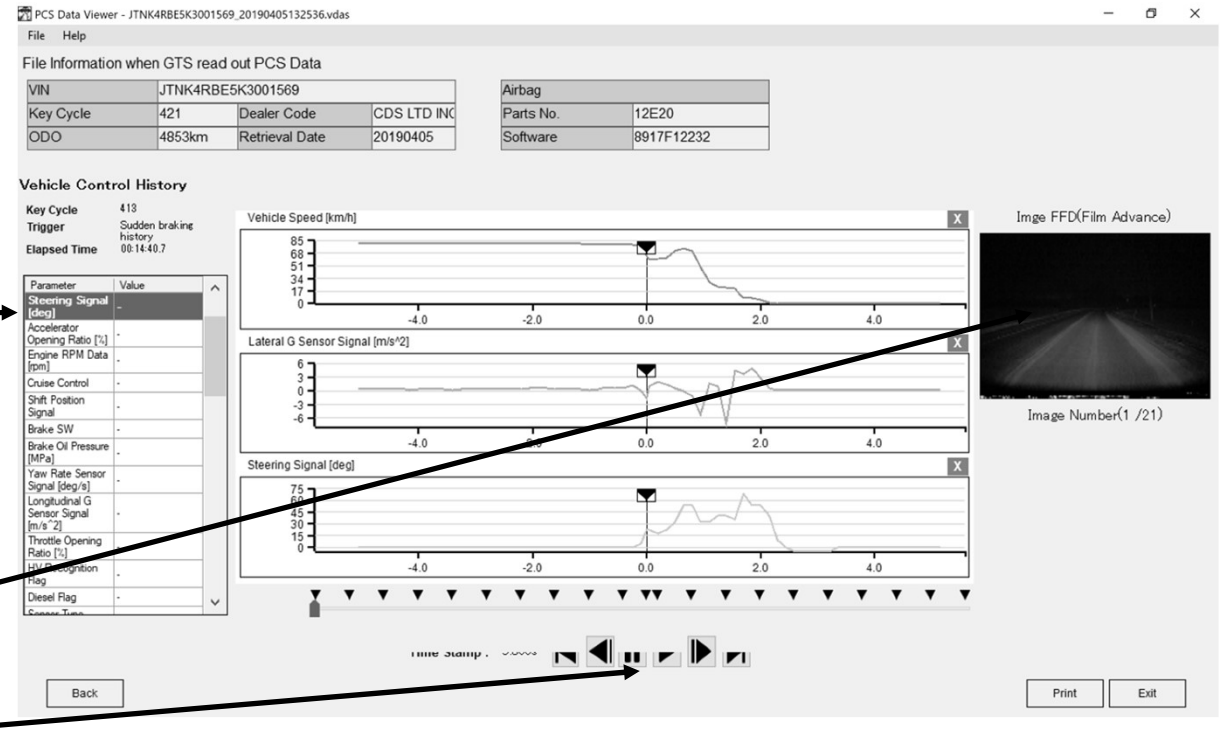
Toyota TechStream – PCS Data Viewer

- If either a VCH or FFD event with an * under Images is selected, and a parameter is selected for playback, associated images will be loaded for synched playback
- Playback viewer controls



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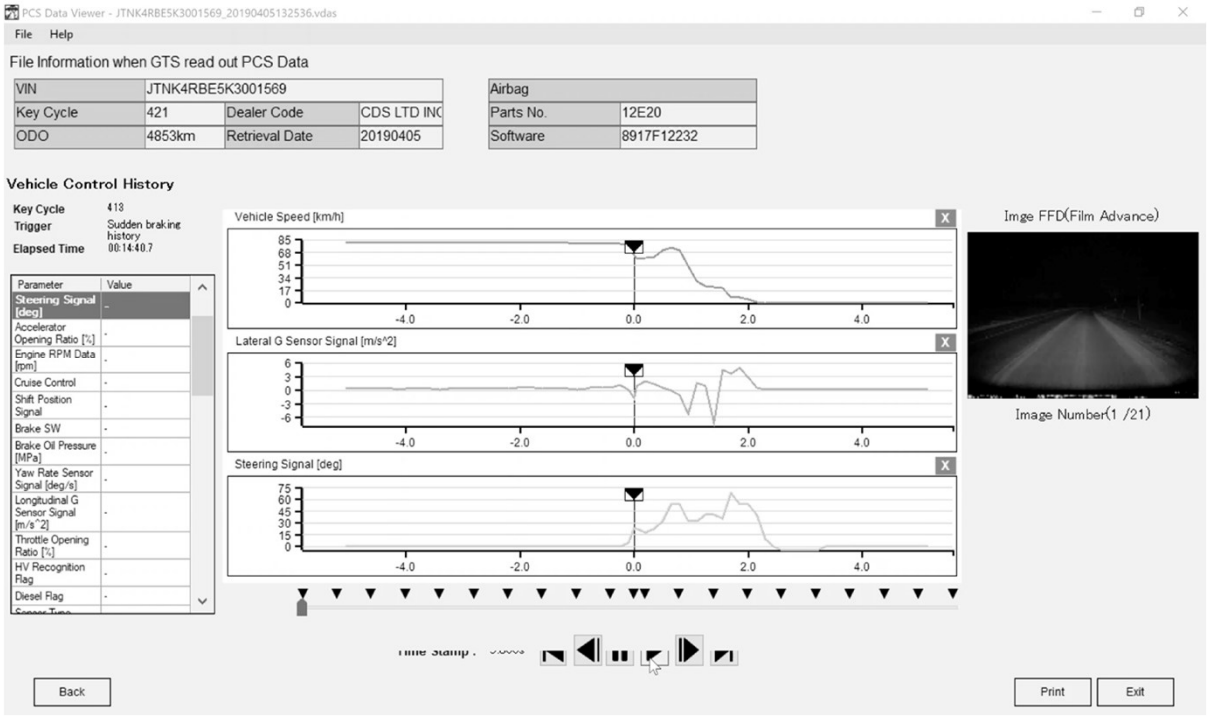
Toyota TechStream – PCS Data Viewer



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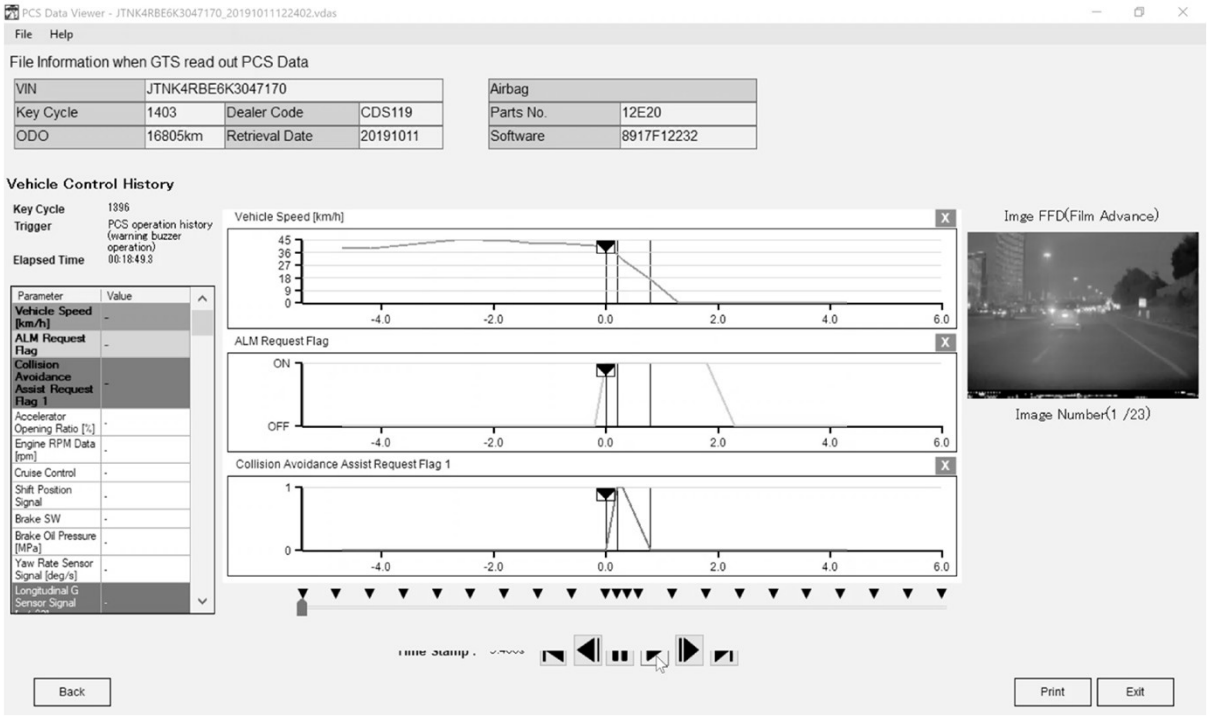
Toyota TechStream – PCS Data Viewer



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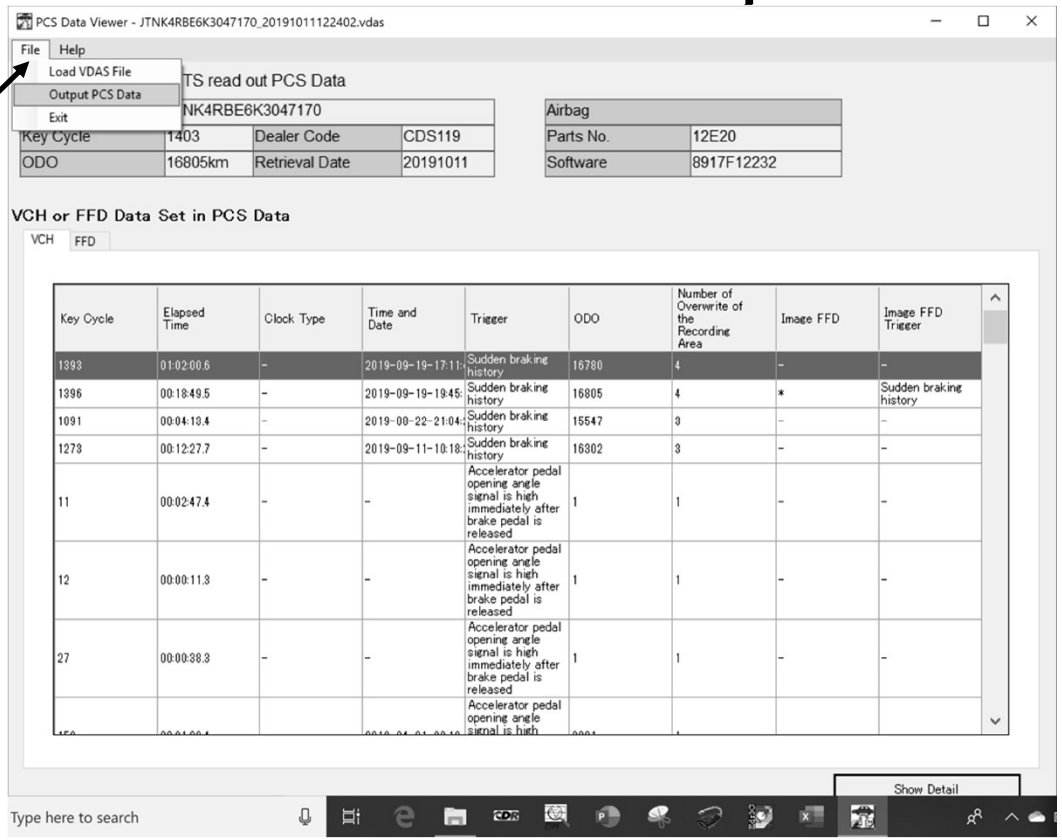
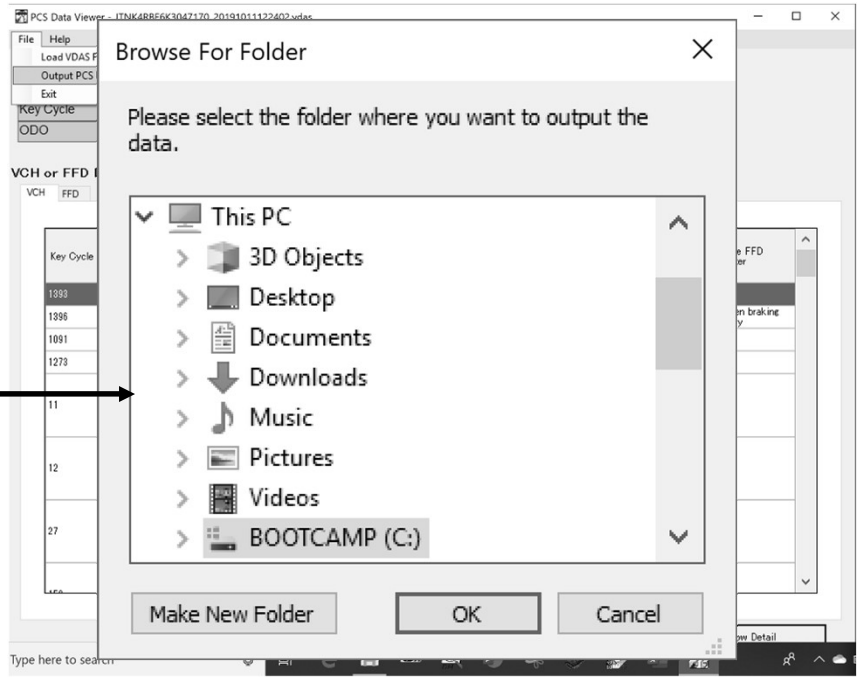
Toyota TechStream – PCS Data Output

- To output the VCH or FFD data to event specific .CSV files and Images to separate image files:
- Click File / Output



Toyota TechStream – PCS Data Output

- To output the VCH or FFD data to event specific .CSV files and Images to separate image files:
- Click File / Output
- This you will open a Dialogue for you to chose where you want to save the output files



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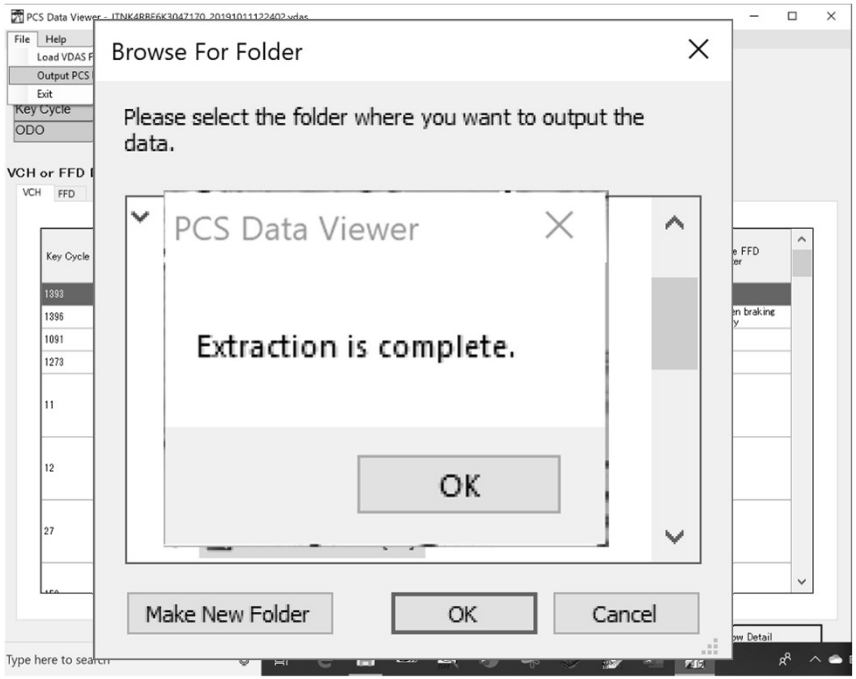
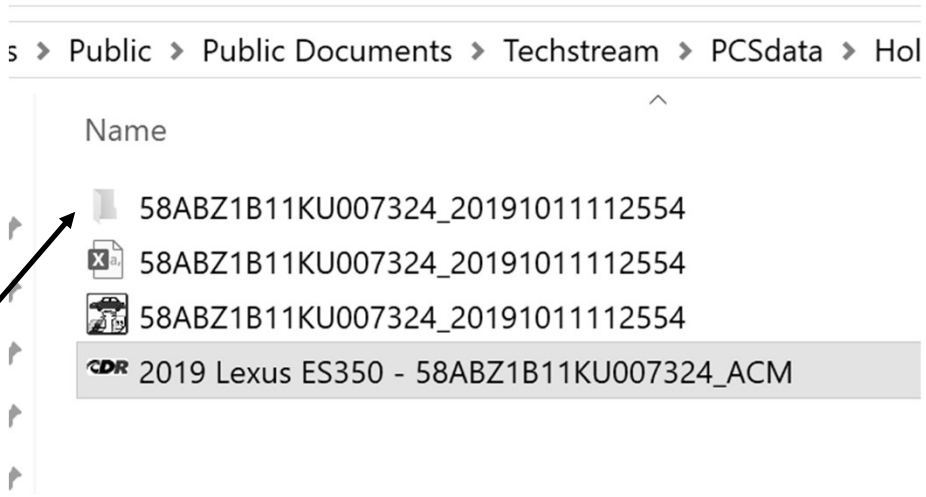
Toyota TechStream – PCS Data Output

- To output the VCH or FFD data to event specific .CSV files and Images to separate image files:
- Click File / Output
- This you will open a Dialogue for you to chose where you want to save the output files
- Once Extraction is complete you can view the output files



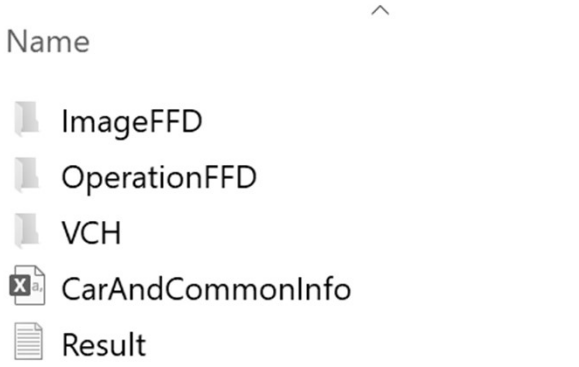
Toyota TechStream – PCS Data Output

- Navigate to where you saved the output in the previous step.
- Here, it was saved in the same location as the .VDAS file and the .CDRX was also copied to this location
- Double-Click the VDAS output directory



Toyota TechStream – PCS Data Output

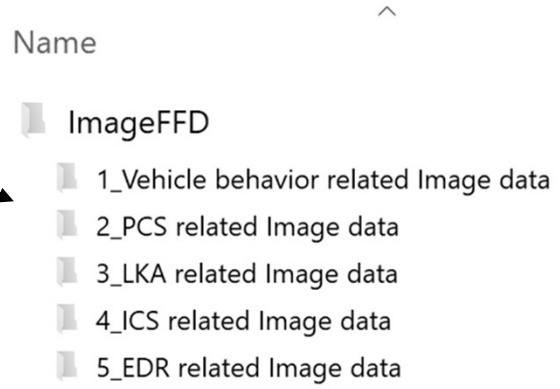
- Output will be contained in 3 Directories
  - PCS Images
  - FFD .CSV data files
  - VCH .CSV files
- Car and Common info contains some basic vehicle and tool related data
- Result.txt ~~displays~~ which VCH areas have data

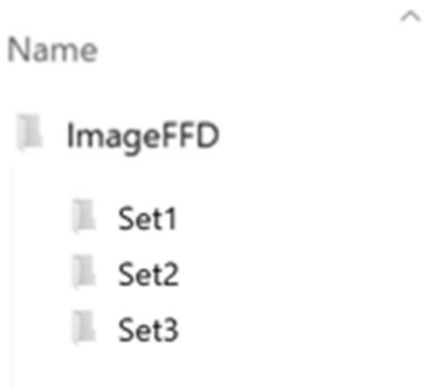


1	Car info	VIN	58ABZ1B11KU007324		
2		ECU info	Airbag	Parts No.	6880
3				software	8917F06451
4		Key Cycle	135		
5		Odo meter	163km		
6	Common inf	Dealer code	CDS LTD INC		
7		Retrieval Dat	20190405		

Toyota TechStream – PCS Data Output

- For TSS 2.0 supported vehicles, separate directories will contain images based on PCS trigger type
  - Images may be duplicated for concurrent triggers
- For TSS 1.0 there may be up to 3 sets if images only



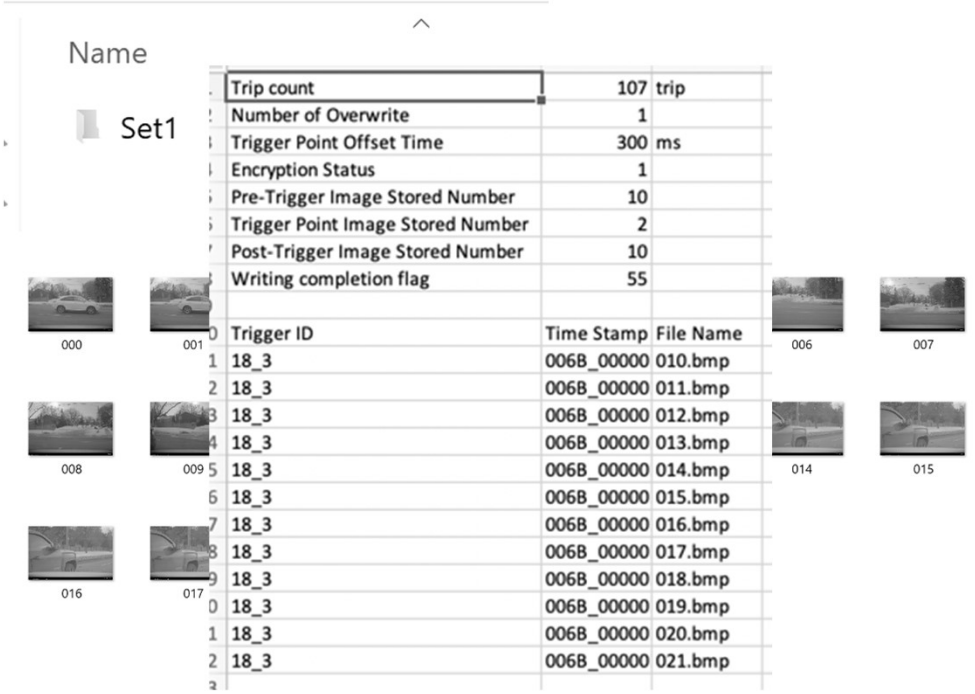


Toyota TechStream – PCS Data Output

- Images
  - Each Category may contain from 1 to 3 sets of images
- Number if images before / after trigger depend on the event trigger type

Toyota TechStream – PCS Data Output

- Images
  - Each Category may contain from 1 to 3 sets of images
- Number if images before / after trigger depend on the event trigger type
- The “Setx” .CSV file contains information to assist in linking to CDR data for TSS 2.0
- TSS 1.0 does not create this file



	Trip count	107	trip
	Number of Overwrite	1	
	Trigger Point Offset Time	300	ms
	Encryption Status	1	
	Pre-Trigger Image Stored Number	10	
	Trigger Point Image Stored Number	2	
	Post-Trigger Image Stored Number	10	
	Writing completion flag	55	
0	Trigger ID	Time Stamp	File Name
1	18_3	006B_00000	010.bmp
2	18_3	006B_00000	011.bmp
3	18_3	006B_00000	012.bmp
4	18_3	006B_00000	013.bmp
5	18_3	006B_00000	014.bmp
6	18_3	006B_00000	015.bmp
7	18_3	006B_00000	016.bmp
8	18_3	006B_00000	017.bmp
9	18_3	006B_00000	018.bmp
0	18_3	006B_00000	019.bmp
1	18_3	006B_00000	020.bmp
2	18_3	006B_00000	021.bmp

Toyota TechStream – PCS Data Output

- Vehicle Control History events are sorted and written in “Areas”
- Each event type has an Area classification
- Each Classification has overwriting rules and a number of events that can be stored of that type

Toyota TechStream – PCS Data Output

- Vehicle Control History events are sorted and written in “Areas”
- Each event type has an Area classification
- Each Classification has overwriting rules and a number of events that can be stored of that type
- Due to overwriting file No. may not be chronological
- .CSV Files written for each VCH Event Trigger

Toyota TechStream – PCS Data Output

- Vel
  are
- Ea
  cla
- Ea
  ove
  eve
  typ
- Du
  not

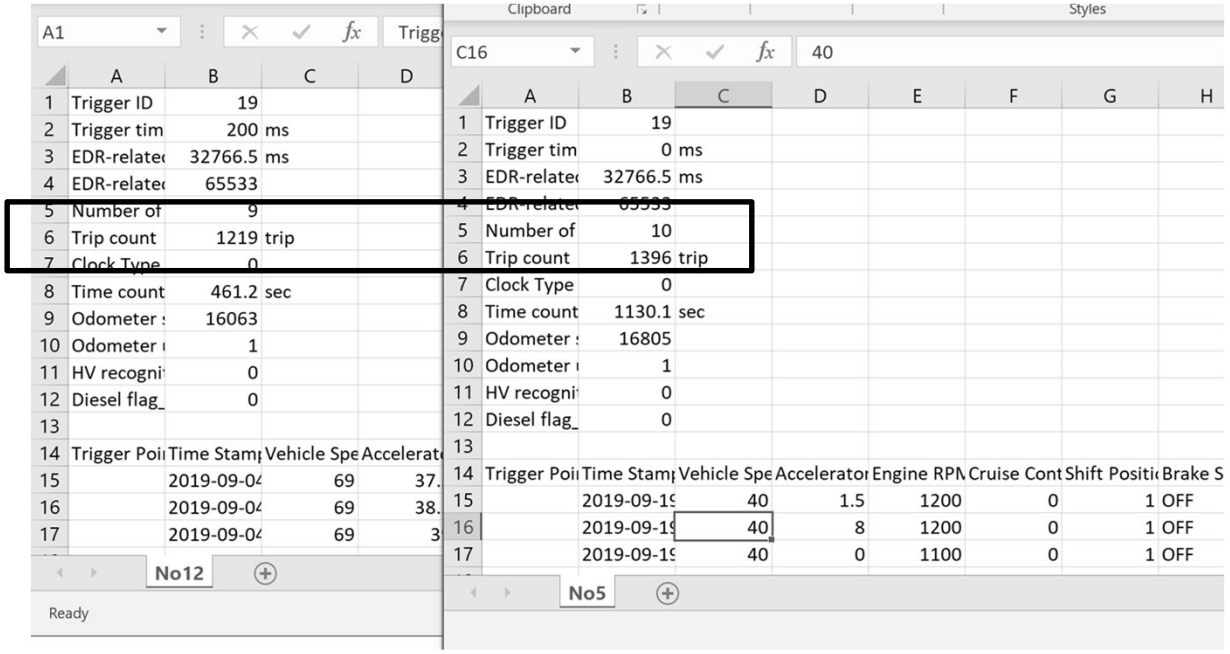
A1 | Trigger ID

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Trigger ID	15												
2	Trigger tim	50	ms											
3	EDR-relate	25	ms											
4	EDR-relate	2												
5	Number of	1												
6	Trip count	107	trip											
7	Clock Type	0												
8	Time count	604.3	sec											
9	Odometer	162												
10	Odometer	1												
11	HV recogni	0												
12	Diesel flag_	0												
13														
14	Trigger Poi	Time Stamp	Vehicle Spe	Accelerator	Engine RPM	Cruise Cont	Shift Positi	Brake SW	Brake Oil P	Yaw Rate S	Longitudina	Lateral G S	Throttle Op	Steering Sig
15		2019-02-08	0	16	800	0	1	OFF	0	-1.464	0.93314	0	2.5	-121.5
16		2019-02-08	0	11	900	0	1	OFF	0	-1.952	1.00492	-0.07178	0.5	-121.5
17		2019-02-08	0	4.5	900	0	1	OFF	0	-2.44	1.00492	-0.07178	1.5	-121.5
18		2019-02-08	1	0	800	0	1	OFF	0	-2.44	0.64602	0	0.5	-112.5
19		2019-02-08	2	0	700	0	1	OFF	0	-2.44	0.57424	0.14356	0	-108
20		2019-02-08	2	0	700	0	1	OFF	0	-2.928	0.50246	0	0	-103.5
21		2019-02-08	2	0	600	0	1	OFF	0	-2.928	0.21534	-0.07178	0	-99
22		2019-02-08	2	0	600	0	1	OFF	0	-2.928	0.21534	0.14356	0	-81

No17

Toyota TechStream – PCS Data Output

- Due to overwriting rules, the file No. may not be chronological!!
- Here, the Trip Count for Set No. 5? is HIGHER than the Trip Count for Set No. 12? indicating over-writes of this VCH data area



	A	B	C	D
1	Trigger ID	19		
2	Trigger tim	200	ms	
3	EDR-relate	32766.5	ms	
4	EDR-relate	65533		
5	Number of	9		
6	Trip count	1219	trip	
7	Clock Type	0		
8	Time count	461.2	sec	
9	Odometer	16063		
10	Odometer	1		
11	HV recogni	0		
12	Diesel flag_	0		
13				
14	Trigger Poi	Time Stam	Vehicle Spe	Accelerat
15		2019-09-0	69	37.
16		2019-09-0	69	38.
17		2019-09-0	69	3

No12

Ready

	A	B	C	D	E	F	G	H
1	Trigger ID	19						
2	Trigger tim	0	ms					
3	EDR-relate	32766.5	ms					
4	EDR-relate	65533						
5	Number of	10						
6	Trip count	1396	trip					
7	Clock Type	0						
8	Time count	1130.1	sec					
9	Odometer	16805						
10	Odometer	1						
11	HV recogni	0						
12	Diesel flag_	0						
13								
14	Trigger Poi	Time Stam	Vehicle Spe	Accelerator	Engine RPM	Cruise Cont	Shift Positi	Brake S
15		2019-09-19	40	1.5	1200	0	1	OFF
16		2019-09-19	40	8	1200	0	1	OFF
17		2019-09-19	40	0	1100	0	1	OFF

No5

Linking To a CDR Event

- Link “Time Count”
- Link ”Time Count”

System Status at Event (Most Recent Event, TRG 1)

TRG Count (times)	1
Event Type	Frontal/Rear/Side Crash
Previous Crash Type	No Event
Time from Previous TRG (msec)	32767 or greater
Time from Time Zero to TRG (msec)	84.0
Event Establishment Factor	Frontal Crash
TRG Establishment Factor	Frontal Crash
Freeze Signal	OFF
Freeze Signal Factor	None
Recording Status , Front/Rear and Side Crash Info.	Complete
Odometer signal (miles [km])	10,442 [16,805]
Trip count (times)	1396
Time count (msec)	1,130,100
Time count input system	Normal

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Trigger ID	19												
2	Trigger tim	0	ms											
3	EDR-relate	32766.5	ms											
4	EDR-relate	65533												
5	Number of	10												
6	Trip count	1396	trip											
7	Clock Type	0												
8	Time count	1130.1	sec											
9	Odometer	16805												
10	Odometer	1												
11	HV recogni	0												
12	Diesel flag_	0												
13														
14	Trigger Point	Time Stamp	Vehicle Speed [km/h]	Accelerator Opening Ratio [%]	Engine RPM Data [rpm]	Cruise Control	Shift Position Signal	Brake SW	Brake Oil Pressure [MPa]	Yaw Rate Sensor Signal [deg/s]	Longitudinal G Sensor Signal [m/s^2]	Lateral G Sensor Signal [m/s^2]	Throttle Opening Ratio [%]	Steering Signal [deg]	HV Re n I
15		2019-09-19	40	1.5	1200	0	1	OFF	0	-1.464	-0.28712	0.14356	0	0	
16		2019-09-19	40	8	1200	0	1	OFF	0	-1.464	0.21534	0.50246	2.5	0	

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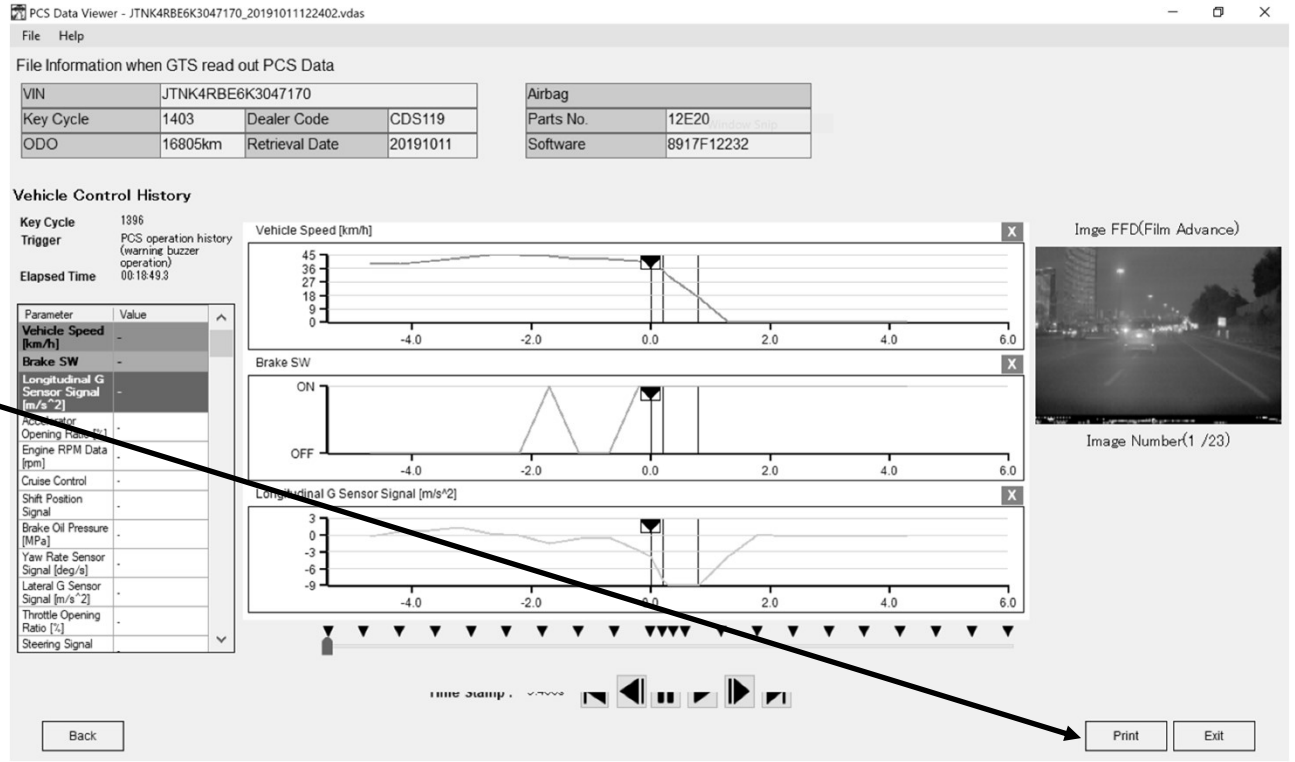
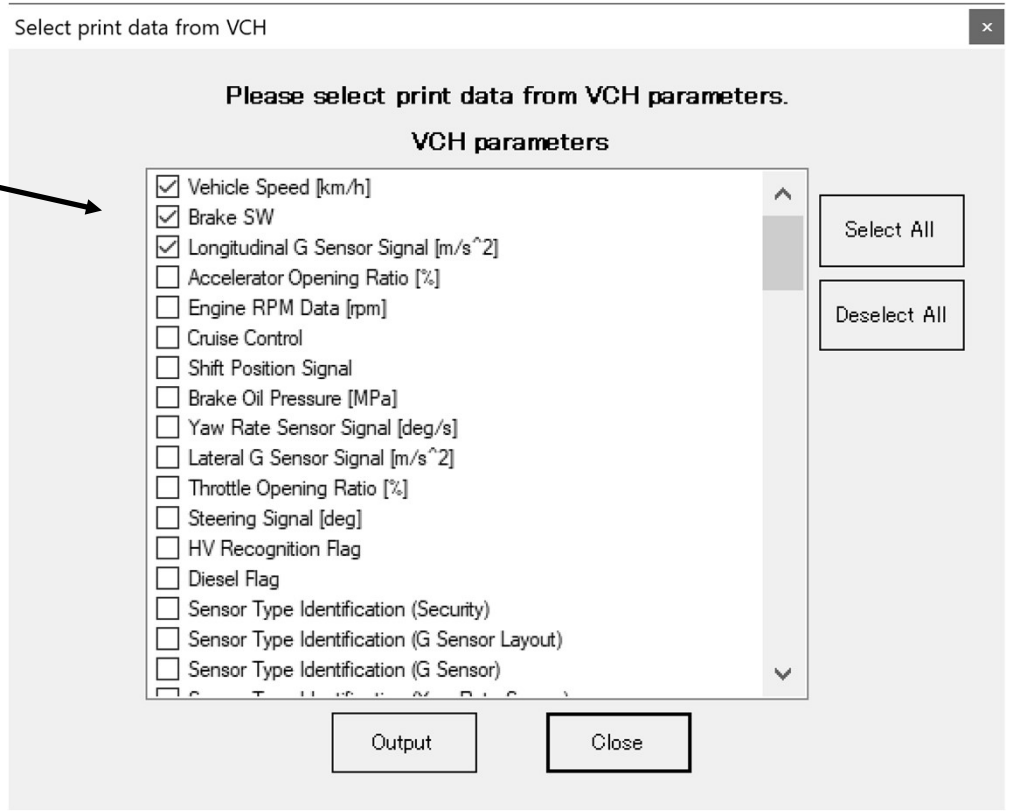
Toyota TechStream – PCS Data Printing

- Output can be printed to generate a full report
- Click “Print”
- You do NOT need to have items in the viewing area



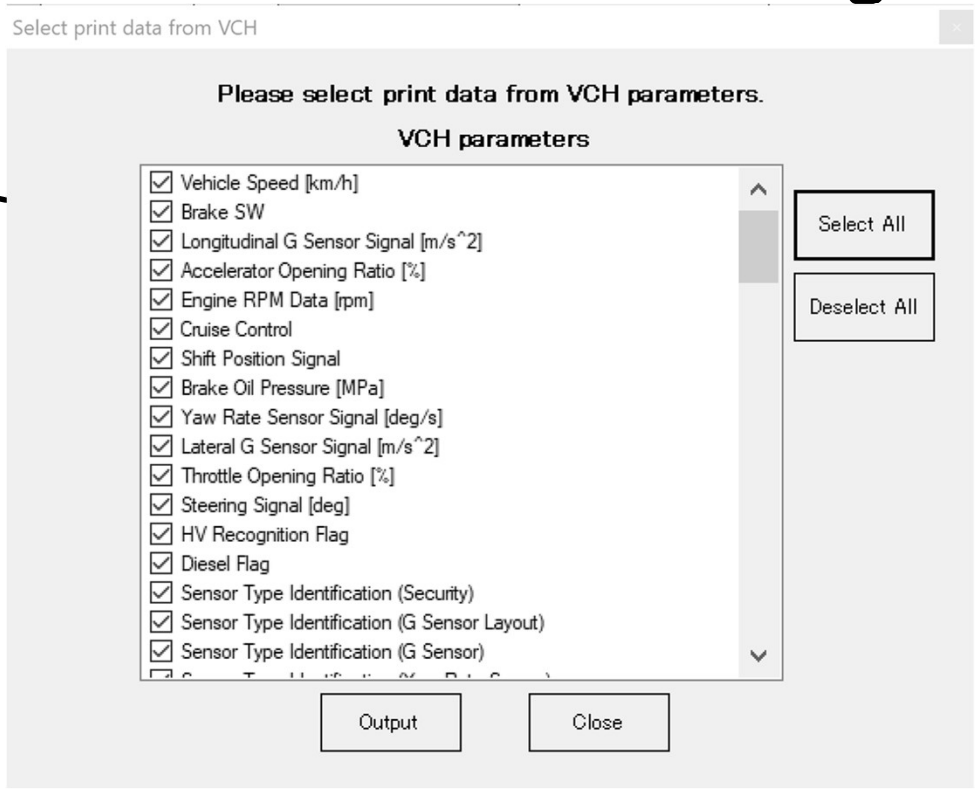
Toyota TechStream – PCS Data Printing

- You can select all, or some of the parameters to print
- Note: Images will be printed at bottom of report



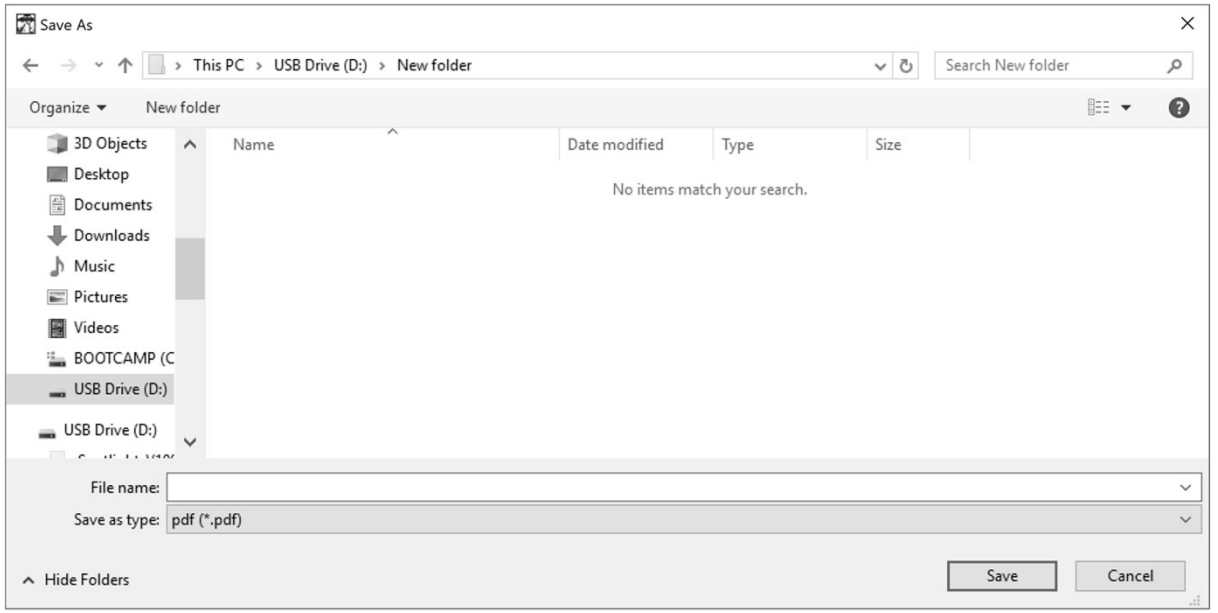
Toyota TechStream – PCS Data Printing

- You can select all, or some of the parameters to print
- Note: Images will be printed at bottom of report



Toyota TechStream – PCS Data Printing

- You can select all, or some of the parameters to print
- Note: Images will be printed at bottom of report
- File Explorer window opens, navigate to Save location of choice



Toyota TechStream – PCS Data Printing

PCS Data Report

This report created by PCS Data Viewer.
PCS Data Viewer data output

1. General information
2. File information
3. User selected VCH data and graph
4. Image FFD
5. Hexadecimal Data

1.General Information

1.1.Data Limitations

- Due to limitations of the data recorded, such as the resolution, data range, sampling interval, time period of the recording, and the items recorded, the information provided by this data may not be sufficient to capture the entire event.
- The recorded data is recorded in discrete intervals.

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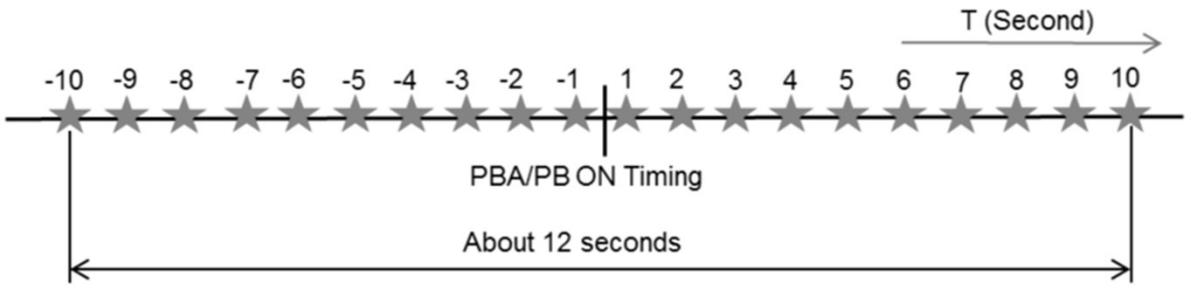
Toyota TechStream – PCS Data Printing

- The following PCS data (Operation FFD) is recorded upon the PCS operation.
  - Vehicle velocity
  - Accelerator pedal position
  - Brake pedal(ON/OFF)
  - The operation situation of each function of PCS
  - Information such as the distance with objects such as preceding vehicles and their relative velocity
- PCS data (Image FFD) is recorded upon the pre-collision brake operation (PBA, PB).
  - In addition to the above, system records image information (Image FFD) of the camera sensor.
- Image information, Image FFD, is recorded 20 shots×3 events in TSS-P/LSS+, 2 shots×3events in TSS-C and 20 shots×10 events in LSS+A, and also maximum 10 shots at every trigger including the first trigger each event if another trigger occurs after the first trigger for six seconds.

Image FFD
- TSS-P/LSS+ PCS : 20shots
- LSS+A PCS : 20 shots+ Maximum 10 shots at every trigger including the first trigger

a) Pattern A



Toyota TechStream – PCS Data Printing

1.3 Appendix : The meaning of Abbreviation

Abbreviation	Meaning
ABS	Antilock Brake System
ACC	Adaptive Cruise Control
AEB	Autonomous Emergency Braking
AHB	Automatic High-Beam
AIM	AIM (Camera axis)
ALM	ALarM
ARS	Adaptive Radar Sensor
ASL	Adaptive Speed Limiter
AVS	Adaptive Variable Suspension
CAA	Collision Avoidance Assist

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Toyota TechStream – PCS Data Printing

3.User Selected VCH or FFD Data

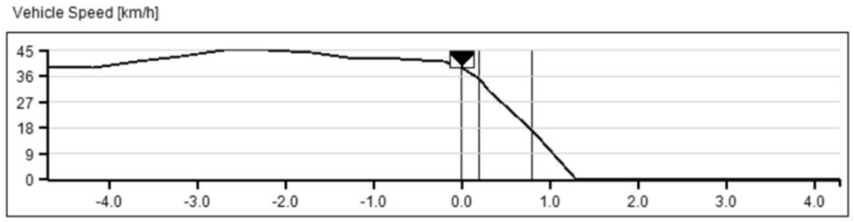
Trigger Name:PCS operation history (warning buzzer operation)

Elapsed Time: 00:18:49.3

3.1 Vehicle Speed [km/h]

Time Stamp	Parameter Value
-4.700	39
-4.200	39
-3.700	41
-3.200	43
-2.700	45
-2.200	45
-1.700	44
-1.200	42
-0.700	42
-0.200	41
0.000	39
0.000	39
0.200	35
0.200	35
0.300	31
0.800	17
0.800	17
1.300	0
1.800	0
2.300	0
2.800	0
3.300	0
3.800	0
4.300	0

Time Stamp	Parameter Value



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Toyota TechStream – PCS Data Printing

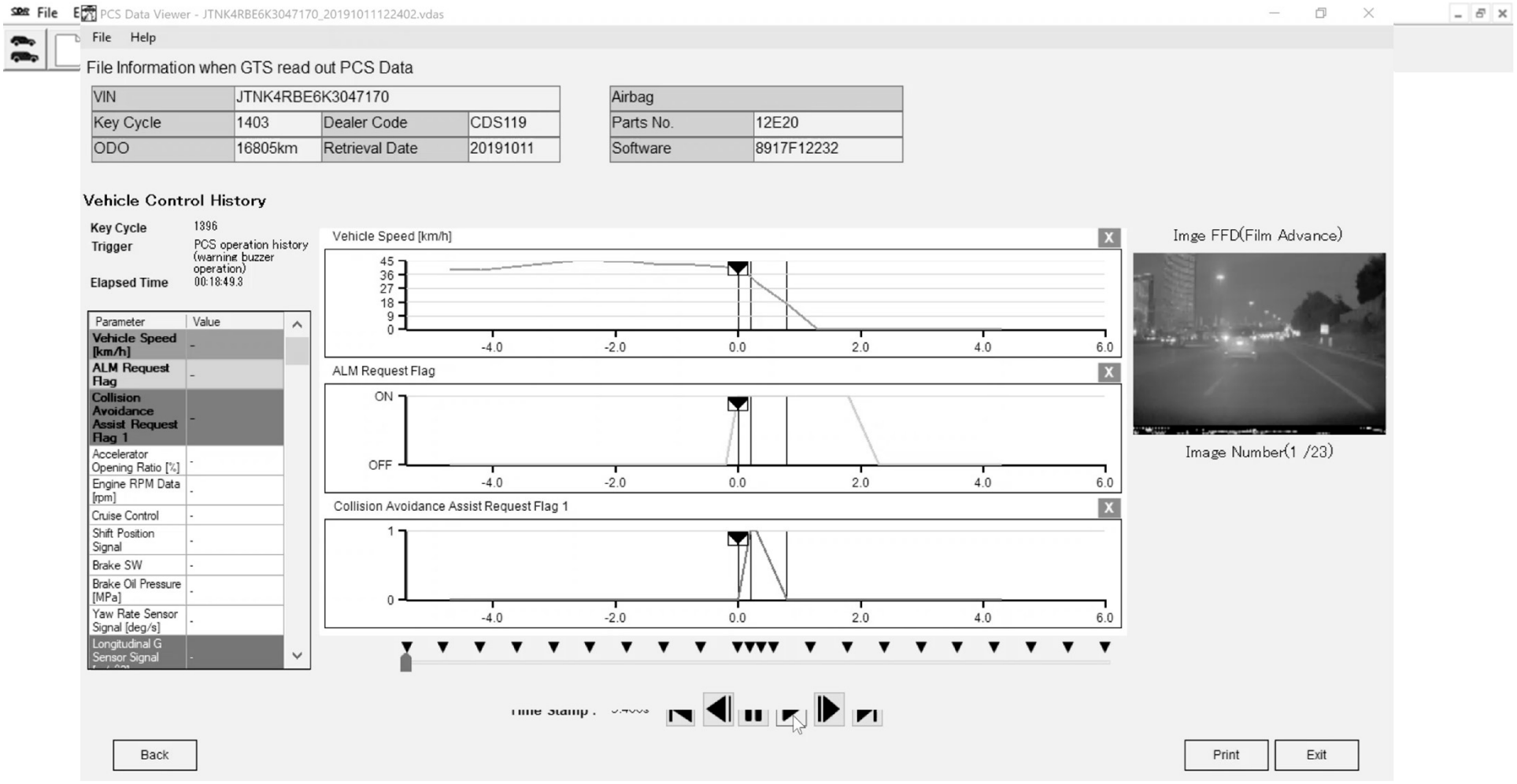
4.Image FFD

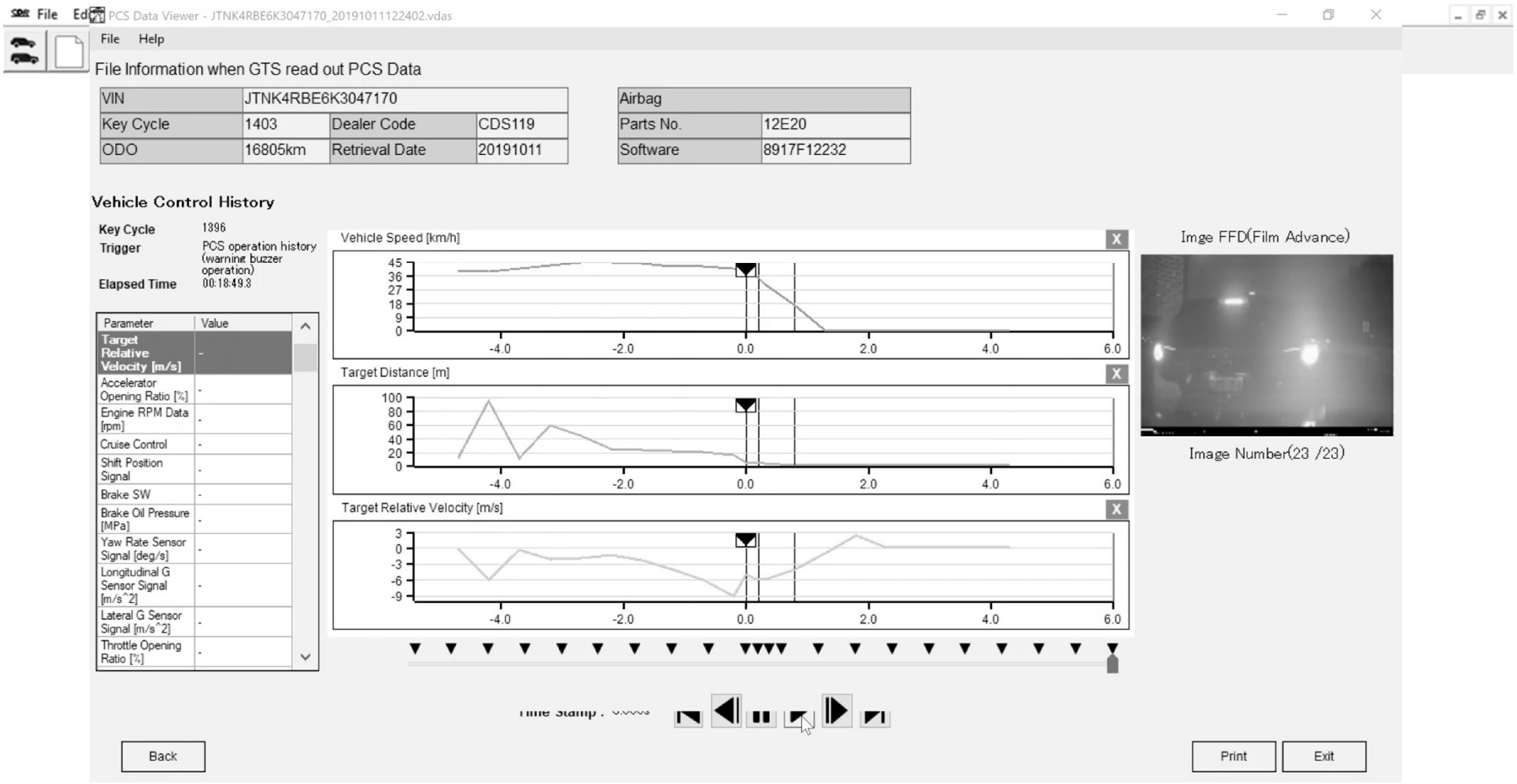
-5.400s

-3.000s

-4.800s

-2.400s


File E
PCS Data Viewer - JTNK4RBE6K3047170_20191011122402.vdas
File Help
File Information when GTS read out PCS Data
VIN JTNK4RBE6K3047170
Key Cycle 1403 Dealer Code CDS119
ODO 16805km Retrieval Date 20191011
Airbag
Parts No. 12E20
Software 8917F12232
Vehicle Control History
Key Cycle 1396
Trigger PCS operation history (warning buzzer operation)
Elapsed Time 00:18:49.3
Parameter Value
Vehicle Speed [km/h]
ALM Request Flag
Collision Avoidance Assist Request Flag 1
Accelerator Opening Ratio [%]
Engine RPM Data [rpm]
Cruise Control
Shift Position Signal
Brake SW
Brake Oil Pressure [MPa]
Yaw Rate Sensor Signal [deg/s]
Longitudinal G Sensor Signal
Vehicle Speed [km/h]
45 36 27 18 9 0
-4.0 -2.0 0.0 2.0 4.0 6.0
ALM Request Flag
ON
OFF
Collision Avoidance Assist Request Flag 1
1 0
Imge FFD(Film Advance)
Image Number(1 /23)
Back
Print
Exit


File Edit PCS Data Viewer - JTNK4RBE6K3047170_20191011122402.vdas
File Help
File Information when GTS read out PCS Data
VIN JTNK4RBE6K3047170
Key Cycle 1403 Dealer Code CDS119
ODO 16805km Retrieval Date 20191011
Airbag
Parts No. 12E20
Software 8917F12232
Vehicle Control History
Key Cycle 1396
Trigger PCS operation history (warning buzzer operation)
Elapsed Time 00:18:49.3
Parameter Value
Target Relative Velocity [m/s] -
Accelerator Opening Ratio [%] -
Engine RPM Data [rpm] -
Cruise Control -
Shift Position Signal -
Brake SW -
Brake Oil Pressure [MPa] -
Yaw Rate Sensor Signal [deg/s] -
Longitudinal G Sensor Signal [m/s^2] -
Lateral G Sensor Signal [m/s^2] -
Throttle Opening Ratio [%] -
Vehicle Speed [km/h]
45 36 27 18 9 0
-4.0 -2.0 0.0 2.0 4.0 6.0
Target Distance [m]
100 80 60 40 20 0
-4.0 -2.0 0.0 2.0 4.0 6.0
Target Relative Velocity [m/s]
3 0 -3 -6 -9
-4.0 -2.0 0.0 2.0 4.0 6.0
Imge FFD(Film Advance)
Image Number(23 /23)
Back
Print
Exit

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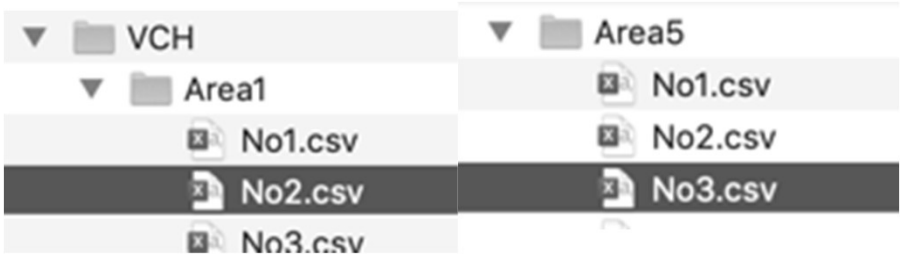
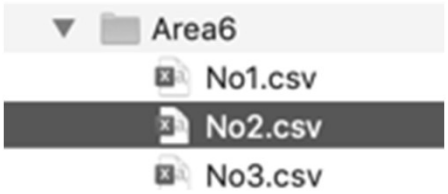
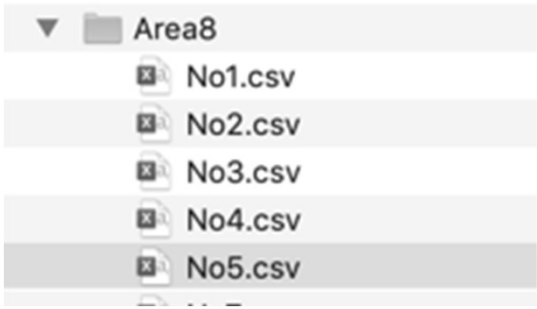
VCH Areas Related to EDR Record

- Checking though the available “Areas” under VCH in the extracted PCS data from the VDAS file finds 4 possible related .CSV files







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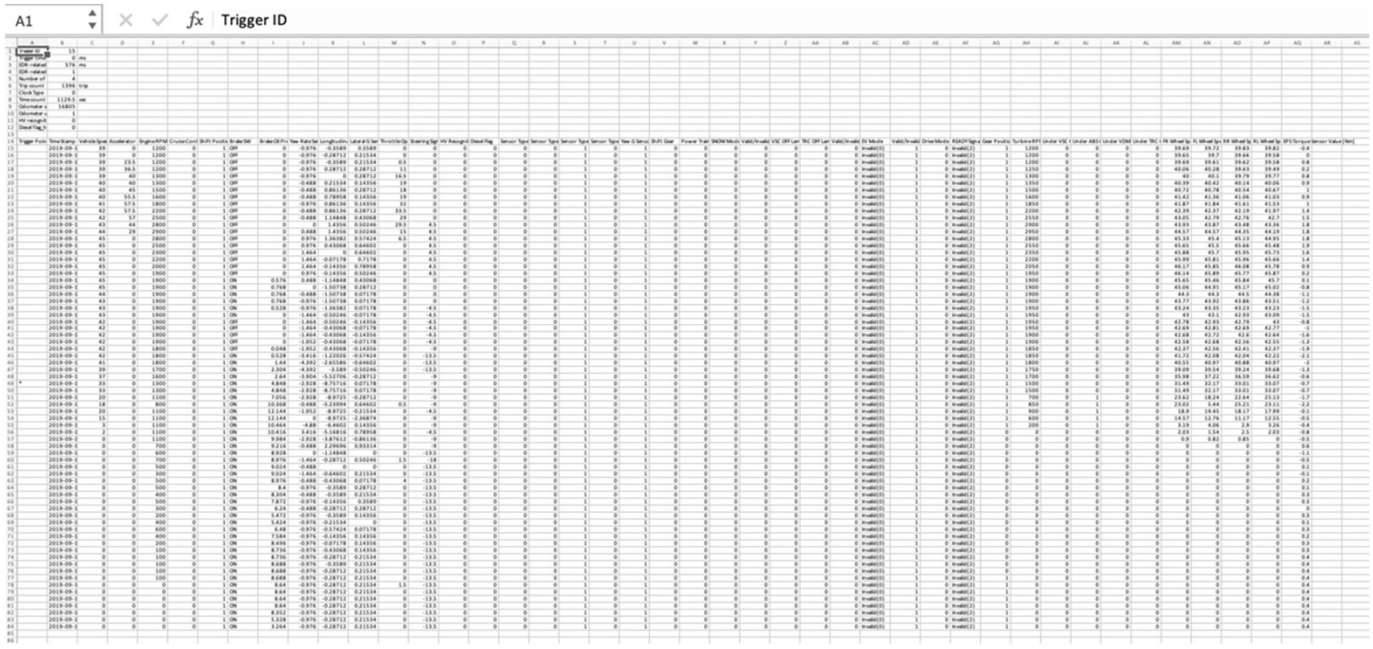
VCH .CSV Files

- .CSV – Comma Separated Value format
- Work in Excel or similar



VCH .CSV Files

- .CSV – Comma Separated Value format
- Work in Excel or similar
- Can be further analyzed or graphed

1	Trigger ID	15	
2	Trigger time	0	ms
3	EDR-related	576	ms
4	EDR-related	1	
5	Number of C	4	
6	Trip count	1396	trip
7	Clock Type	0	
8	Time count	1129.5	sec
9	Odometer si	16805	
10	Odometer ur	1	
11	HV recogniti	0	
12	Diesel flag_H	0	
13			

14 Trigger Point	Time Stamp	Vehicle Speed [km/h]	Accelerator Opening Ratio [%]	Engine RPM Data [rpm]	Cruise Control	Shift Position Signal	Brake SW	Brake Oil Pressure [MPa]	Yaw Rate Sensor Signal [deg/s]	Longitudinal G Sensor Signal [m/s^2]	Lateral G Sensor Signal [m/s^2]	Throttle Opening Ratio [%]
15	2019-09-19-19:45:14.700	39	0	1200	0	1	OFF	0	-0.976	-0.3589	0.3589	
16	2019-09-19-19:45:14.850	39	0	1200	0	1	OFF	0	-0.976	-0.28712	0.21534	
17	2019-09-19-19:45:15.000	39	23.5	1200	0	1	OFF	0	-0.976	-0.3589	0.21534	0
18	2019-09-19-19:45:15.150	39	36.5	1200	0	1	OFF	0	-0.976	0.28712	0.28712	1
19	2019-09-19-19:45:15.300	39	40	1300	0	1	OFF	0	-0.976	0	0.28712	16
20	2019-09-19-19:45:15.450	40	40	1300	0	1	OFF	0	-0.488	0.21534	0.14356	1
21	2019-09-19-19:45:15.600	40	45	1500	0	1	OFF	0	-0.488	0.86136	0.28712	1
22	2019-09-19-19:45:15.750	40	55.5	1600	0	1	OFF	0	-0.488	0.78958	0.14356	1
23	2019-09-19-19:45:15.900	41	57.5	1800	0	1	OFF	0	-0.976	0.86136	0.14356	3
24	2019-09-19-19:45:16.050	42	57.5	2200	0	1	OFF	0	-0.488	0.86136	0.28712	33

VCH Areas Related to EDR Record

- Areas 1 and 5 (and 11) relate to the Brake control system on the vehicle.
- These events may be sudden braking / tractions control / vehicle stability system events
- Different over-writing rules for these sections

Brake control	Area 5 (Approximately 5 times)	VSC operation history	When VSC operation starts.	-	Only for vehicles with VSC
		TRC operation history	When TRC operation starts.	-	Only for vehicles with TRC
		ABS operation history	When ABS operation starts.	-	-
	Area 11				Only for
	Area 1 (Approximately 4 times)	Sudden braking history	Detects forward and backward acceleration above a certain level.	○	Only for vehicles with VSC
		Sudden turning history	Detects lateral acceleration above a certain level.	○	Only for vehicles with VSC

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VCH Areas Related to EDR Record

- Area 6 Relates to the Pre-Collision Detection, Warning and Auto-Braking System
- Up to ten records may be stored before over-writing

		operation)			with PCS
Pre-collision	Area 6 (Approximately 10 times)	PCS operation history (pre-collision seat belt operation)	PCS (PCS seat belt) operates.	-	Only for vehicles with PCS
		PCS operation history (pre-pump operation)	PCS (PCS brake assist standby) operates.	-	Only for vehicles with PCS
		PCS operation history (PBR operation)	PCS (stop light illumination) operates.	-	Only for vehicles with PCS
		PCS operation history (PBH operation)	PCS (after operation brake hold) operates.	-	Only for vehicles with PCS
		PCS operation history (avoidance support)	PCS (evasion support) operates.	-	Only for vehicles with PCS
		PCS operation			

ıs Related to EDR Record

Pre-collision	Area 6 (Approximately 10 times)	PCS operation history (warning buzzer operation)	PCS (warning buzzer) operates.	-	Only for vehicles with PCS
		PCS operation history (warning brake operation)	PCS (brake alarm) operates.	-	Only for vehicles with PCS (w/ Driver Monitor Camera)
		PCS operation history (pre-collision brake assist operation)	PCS (PCS brake assist) operates.	○	Only for vehicles with PCS
		PCS operation history (prior brake operation)	PCS (preliminary braking) operates.	○	Only for vehicles with PCS
		PCS operation history (pre-collision brake operation)	PCS (PCS brake) operates.	○	Only for vehicles with PCS
		PCS operation history (pre-collision seat belt operation)	PCS (PCS seat belt) operates.	-	Only for vehicles with PCS
		PCS operation history (pre-pump operation)	PCS (PCS brake assist standby) operates.	-	Only for vehicles with PCS
		PCS operation history (PBR operation)	PCS (stop light illumination) operates.	-	Only for vehicles with PCS
		PCS operation history (PBH operation)	PCS (after operation brake hold) operates.	-	Only for vehicles with PCS
		PCS operation history (avoidance support)	PCS (evasion support) operates.	-	Only for vehicles with PCS
		PCS operation history (when acceleration above certain amount is detected)	Forward and backward acceleration above a certain amount is detected.	○	Only for vehicles with PCS
		PCS operation history (request combined steering)	Steering assist is requested.	-	Only for vehicles with PCS
		PCS operation history (caution operation)	PCS warning operates.	-	Only for vehicles with PCS
		Pedestrian protection system over constant pressure sensor signal input	Pressure sensor signal input above a certain amount is detected.	○	Only for vehicles with Pedestrian Protection System
		PCS operation history (side airbag operation request)	Side airbag prevention-linked function operates.	-	Only for vehicles with PCS

		operation)		with PCS
Pre-collision	Area 6 (Approximately 10 times)	PCS operation history (pre-collision seat belt operation)	PCS (PCS seat belt) operates.	-	Only for vehicles with PCS
		PCS operation history (pre-pump operation)	PCS (PCS brake assist standby) operates.	-	Only for vehicles with PCS
		PCS operation history (PBR operation)	PCS (stop light illumination) operates.	-	Only for vehicles with PCS
		PCS operation history (PBH operation)	PCS (after operation brake hold) operates.	-	Only for vehicles with PCS
		PCS operation history (avoidance support)	PCS (evasion support) operates.	-	Only for vehicles with PCS
		PCS operation			

VCH Areas Related to EDR Record

- Area 8 Relates to the Lane Departure Alert / Lane Keep Assist system
- This area may hold up to 12 records before over-writing

Lane departure alert/Lane-keeping assist	Area 8 (Approximately 12 times)	Torque sensor signal above certain amount is detected during LKA/LDA operation	When the LKA/LDA/LTA system is ON, a torque sensor signal above a certain amount is detected.	○	Only for vehicles with LKA/LDA (w/ Steering Control)/LTA
		Lane change assist operation history	Lane change assist operates.	-	Only for vehicles with LCA

Lexus Case – Lane Change > Crash

Data Element Sign Convention:
The following table provides an explanation of the sign notation for data elements that may be included in this CDR report.

Data Element Name	Positive Sign Notation Indicates
Maximum Delta-V, Longitudinal	Forward
Delta-V, Longitudinal	Forward
Delta-V, Lateral	Left to Right
Lateral Acceleration, Side Satellite Sensor 1	Left to Right
Lateral Acceleration, Side Satellite Sensor 2	Left to Right
Lateral Acceleration, Side Satellite Sensor 3	Left to Right
Lateral Acceleration, Side Satellite Sensor 4	Left to Right
Rate of Change of Pressure, Side Satellite Sensor 1	The pressure of a door interior is applied.
Rate of Change of Pressure, Side Satellite Sensor 2	The pressure of a door interior is applied.
Rate of Change of Pressure, Side Satellite Sensor 3	The pressure of a door interior is applied.
Rate of Change of Pressure, Side Satellite Sensor 4	The pressure of a door interior is applied.
Lateral Acceleration for Side Crash, Floor Sensor	Left to Right
Roll Angle Peak	Clockwise Rotation
Roll Angle at the Time of TRG	Clockwise Rotation
Roll Rate	Clockwise Rotation
Lateral Acceleration for Rollover, Floor Sensor	Left to Right
Longitudinal Acceleration , VSC Sensor	Forward
Yaw Rate	Left Turn
Steering Input	Left Turn

Toyota 17EDR – System Status at Retrieval

System Status at Retrieval

ECU Part Number	89170-12E20
EDR Generation	17EDR
Complete File Recorded	Yes
Ignition Cycle, Download (times)	1387
Multi-Event, Number of Events (times)	1
Time from Event 1 to 2 (sec)	N/A
Diagnostic Trouble Codes Exist	No
Location of Side Satellite Sensor 1, L	front door
Location of Side Satellite Sensor 2, L	B-Pillar
Location of Side Satellite Sensor 3, L	Not Equipped
Location of Side Satellite Sensor 4, L	C-Pillar
Location of Side Satellite Sensor 1, R	front door
Location of Side Satellite Sensor 2, R	B-Pillar
Location of Side Satellite Sensor 3, R	Not Equipped
Location of Side Satellite Sensor 4, R	C-Pillar
Location of Floor Sensor	Airbag ECU

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash Recording Status	Diagnostic Data Recording Status	Occupant Data Recording Status	Crash Info Recording Status	Time Series Recording Status
Most Recent Event	1	Frontal/Rear/Side Crash	0	Complete	Complete	Complete	Complete	Complete

Toyota 17EDR – System Status at Event

System Status at Event (Most Recent Event, TRG 1)

TRG Count (times)	1
Event Type	Frontal/Rear/Side Crash
Previous Crash Type	No Event
Time from Previous TRG (msec)	32767 or greater
Time from Time Zero to TRG (msec)	84.0
Event Establishment Factor	Frontal Crash
TRG Establishment Factor	Frontal Crash
Freeze Signal	OFF
Freeze Signal Factor	None
Recording Status , Front/Rear and Side Crash Info.	Complete
Odometer signal (miles [km])	10,442 [16,805]
Trip count (times)	1396
Time count (msec)	1,130,100
Time count input system	Normal

Linking VCH to EDR

System Status at Event (Most Recent Event, TRG 1)

TRG Count (times)				1
Event Type	Trigger ID	18_1		Frontal/Rear/Side Crash
Previous Crash Type	Trigger time information	300	ms	No Event
Time from Previous TRG (ms	EDR-related information	826	ms	32767 or greater
Time from Time Zero to TRG	EDR-related information	1		84.0
Event Establishment Factor	Number of Overwrite of the Recording Area	4		Frontal Crash
TRG Establishment Factor	Trip count	1396	trip	Frontal Crash
Freeze Signal	Clock Type	0		OFF
Freeze Signal Factor	Time count	1129.3	sec	None
Recording Status , Front/Rea	Odometer signal	16805		Complete
Odometer signal (miles [km])	Odometer unit signal	1		10,442 [16,805]
Trip count (times)	HV recognition flag_header	0		1396
Time count (msec)	Diesel flag_header	0		1,130,100
Time count input system				Normal

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Toyota 17EDR – Deployment Command Data

Deployment Command Data (Most Recent Event, TRG 1)

Active Head Restraint, Time to Deploy, Driver (msec)	SNA
Active Head Restraint, Time to Deploy, Front Passenger (msec)	SNA
Frontal Airbag Deployment, Time to 1st Stage Deployment, Driver (msec)	No
Frontal Airbag Deployment, Time to 1st Stage Deployment, Front Passenger (msec)	No
Frontal Airbag Deployment, Time to 2nd Stage, Driver (msec)	No
Frontal Airbag Deployment, Time to 2nd Stage, Front Passenger (msec)	No
Pretensioner Deployment, Time to Fire, 1st Seat, Driver (msec)	No
Pretensioner Deployment, Time to Fire, 1st Seat, Passenger (msec)	No
Pretensioner Deployment, Time to Fire, 2nd Seat, Driver (msec)	SNA
Pretensioner Deployment, Time to Fire, 2nd Seat, Passenger (msec)	SNA
Rear Window Airbag Deployment, Time to Deploy (msec)	SNA
Side Airbag Deployment, Time to Deploy, 1st Seat, Driver (msec)	No
Side Airbag Deployment, Time to Deploy, 1st Seat, Passenger (msec)	No
Side Airbag Deployment, Time to Deploy, 2nd Seat, Driver (msec)	SNA
Side Airbag Deployment, Time to Deploy, 2nd Seat, Passenger (msec)	SNA
Side Curtain Airbag Deployment, Time to Deploy, Driver (msec)	No
Side Curtain Airbag Deployment, Time to Deploy, Passenger (msec)	No

Toyota 17EDR – DTC and Pre-Crash -1 Second

DTCs Present at Time of Event (Most Recent Event, TRG 1)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (Most Recent Event, TRG 1)

Recording Status, Occupant	Complete
Recording Status, Pre-Crash	Complete
Time from Pre-Crash to TRG (msec)	300
Safety Belt Status, Driver	ON
Safety Belt Status, Front Passenger	ON
Occupant Size Classification, Front Passenger	AM50 (Not Child)
Frontal Airbag Suppression Switch Status, Front Passenger	SNA
RSCA Disable Switch	SNA
Seat Track Position Switch, Foremost, Status, Driver	No
Airbag Warning Lamp, On/Off	OFF
Ignition Cycle, Crash (times)	1382

Longitudinal/Lateral Crash Pulse (Most Recent Event, TRG 1)

Recording Status , Time Series Data	Complete
Power Supply Status at the time of Max. Delta-V	ON
Maximum Delta-V, Longitudinal (MPH [km/h])	-8.1 [-13.0]
Time, Maximum Delta-V, Longitudinal (msec)	166.0
Clipping Time, Longitudinal Delta-V (msec)	No
Clipping Time, Lateral Delta-V (msec)	No
Length of Delta-V (msec)	160

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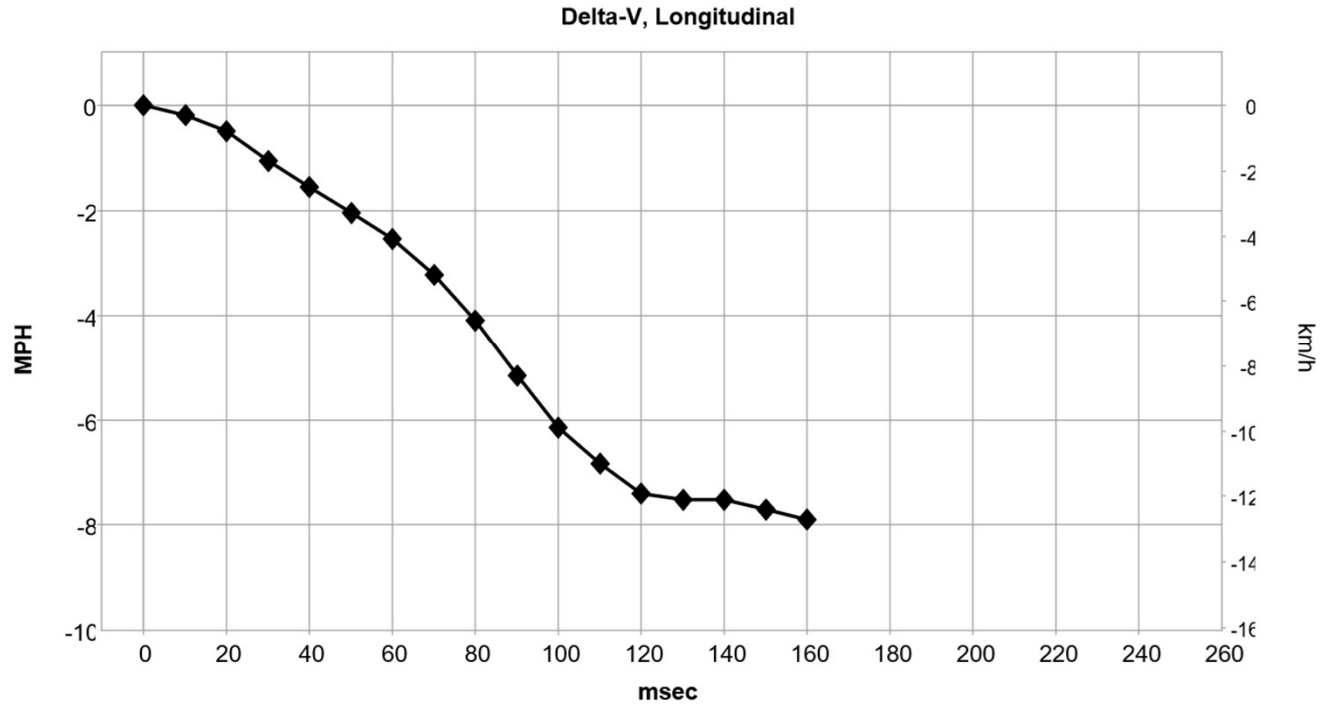
Toyota 17EDR – Longitudinal Crash Pulse



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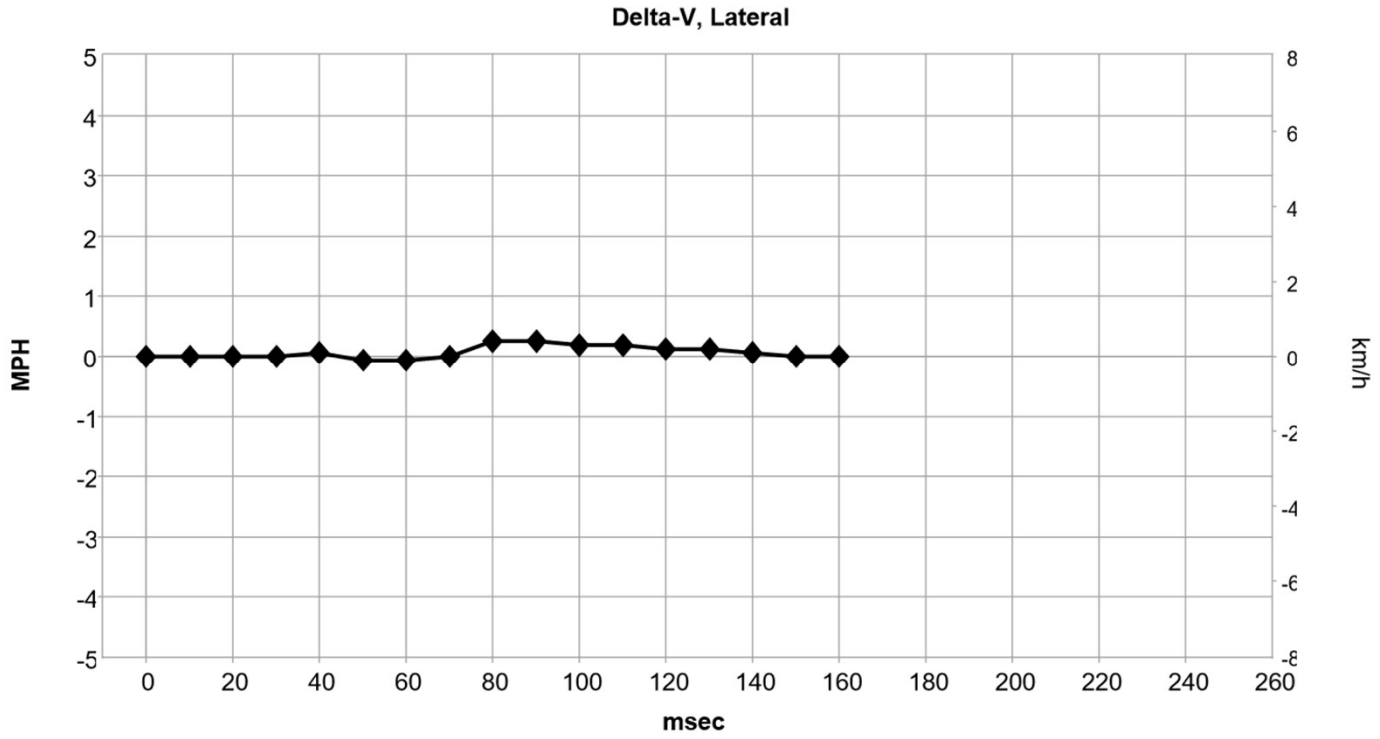
Toyota 17EDR – Lateral Crash Pulse



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Toyota 17EDR –Crash Pulse / Power Supply

Longitudinal/Lateral Crash Pulse (Most Recent Event, TRG 1)

Time (msec)	Delta-V, Longitudinal (MPH [km/h])	Delta-V, lateral (MPH [km/h])	Power Supply Status
0	0.0 [0.0]	0.0 [0.0]	ON
10	-0.2 [-0.3]	0.0 [0.0]	ON
20	-0.5 [-0.8]	0.0 [0.0]	ON
30	-1.1 [-1.7]	0.0 [0.0]	ON
40	-1.6 [-2.5]	0.1 [0.1]	ON
50	-2.1 [-3.3]	-0.1 [-0.1]	ON
60	-2.5 [-4.1]	-0.1 [-0.1]	ON
70	-3.2 [-5.2]	0.0 [0.0]	ON
80	-4.1 [-6.6]	0.2 [0.4]	ON
90	-5.2 [-8.3]	0.2 [0.4]	ON
100	-6.2 [-9.9]	0.2 [0.3]	ON
110	-6.8 [-11.0]	0.2 [0.3]	ON
120	-7.4 [-11.9]	0.1 [0.2]	ON
130	-7.5 [-12.1]	0.1 [0.2]	ON
140	-7.5 [-12.1]	0.1 [0.1]	ON
150	-7.7 [-12.4]	0.0 [0.0]	ON
160	-7.9 [-12.7]	0.0 [0.0]	ON
170	0.0 [0.0]	0.0 [0.0]	ON
180	0.0 [0.0]	0.0 [0.0]	ON
190	0.0 [0.0]	0.0 [0.0]	ON
200	0.0 [0.0]	0.0 [0.0]	ON
210	0.0 [0.0]	0.0 [0.0]	ON
220	0.0 [0.0]	0.0 [0.0]	ON
230	0.0 [0.0]	0.0 [0.0]	ON
240	0.0 [0.0]	0.0 [0.0]	ON
250	0.0 [0.0]	0.0 [0.0]	ON

Toyota 17EDR – Pre-Crash Table 1

Pre-Crash Data -5 to 0 Seconds (Most Recent Event, TRG 1) - Table 1 of 4

Time (sec)	Vehicle Speed (MPH [km/h])	Accelerator Pedal, % Full (%)	Percentage of Engine Throttle (%)	Fuel Injection Quantity (mm^3/st)	Engine RPM (RPM)	Motor RPM (RPM)	Service Brake, ON/OFF
-4.80	24.9 [40]	45.0	18.0	Invalid	1,500	Invalid	OFF
-4.30	26.1 [42]	57.5	32.5	Invalid	2,300	Invalid	OFF
-3.80	27.3 [44]	0.0	6.5	Invalid	2,900	Invalid	OFF
-3.30	28.0 [45]	0.0	0.0	Invalid	2,200	Invalid	OFF
-2.80	28.0 [45]	0.0	0.0	Invalid	1,900	Invalid	ON
-2.30	26.7 [43]	0.0	0.0	Invalid	1,900	Invalid	ON
-1.80	26.1 [42]	0.0	0.0	Invalid	1,900	Invalid	OFF
-1.30	26.1 [42]	0.0	0.0	Invalid	1,800	Invalid	OFF
-0.80	23.6 [38]	0.0	0.0	Invalid	1,700	Invalid	ON
-0.30	10.6 [17]	0.0	0.5	Invalid	800	Invalid	ON
TRG(0)	10.6 [17]	0.0	0.0	Invalid	1,100	Invalid	ON

Linking VCH to EDR

Pre-Crash Data -5 to 0 Seconds (Most Recent E ; 1) - Table 1

Time (sec)	Vehicle Speed (MPH [km/h])	Accelerator Pedal, % Full (%)	Percentage of Engine Throttle (%)	Engine RPM (RPM)
-4.80	24.9 [40]	45.0	18.0	1,500
-4.30	26.1 [42]	57.5	32.5	2,300
-3.80	27.3 [44]	0.0	6.5	2,900
-3.30	28.0 [45]	0.0	0.0	2,200
-2.80	28.0 [45]	0.0	0.0	1,900
-2.30	26.7 [43]	0.0	0.0	1,900
-1.80	26.1 [42]	0.0	0.0	1,900
-1.30	26.1 [42]	0.0	0.0	1,800
-0.80	23.6 [38]	0.0	0.0	1,700
-0.30	10.6 [17]	0.0	0.5	800
TRG(0)	10.6 [17]	0.0	0.0	1,100

Trigger Point	Time Stamp	Vehicle Speed [km/h]	Accelerator Opening Ratio [%]	Engine RPM Data [rpm]
	2019-09-19-19:45:14.900	39	0	1200
	2019-09-19-19:45:15.400	39	40	1300
	2019-09-19-19:45:15.900	41	57.5	1700
	2019-09-19-19:45:16.400	43	44	2800
	2019-09-19-19:45:16.900	45	0	2400
	2019-09-19-19:45:17.400	45	0	1900
	2019-09-19-19:45:17.900	44	0	1900
	2019-09-19-19:45:18.400	42	0	1900
	2019-09-19-19:45:18.900	42	0	1900
	2019-09-19-19:45:19.400	41	0	1800
*	2019-09-19-19:45:19.600	39	0	1700
multi-trigger	2019-09-19-19:45:19.600	39	0	1700
multi-trigger	2019-09-19-19:45:19.800	35	0	1400
multi-trigger	2019-09-19-19:45:19.800	35	0	1400
	2019-09-19-19:45:19.900	31	0	1200
multi-trigger	2019-09-19-19:45:20.400	17	0	1100
	2019-09-19-19:45:20.400	17	0	1100
	2019-09-19-19:45:20.900	0	0	1100
	2019-09-19-19:45:21.400	0	0	600
	2019-09-19-19:45:21.900	0	0	400
	2019-09-19-19:45:22.400	0	0	300
	2019-09-19-19:45:22.900	0	0	500
	2019-09-19-19:45:23.400	0	0	100
	2019-09-19-19:45:23.900	0	0	100

Toyota 17EDR – Pre-Crash Table 2

Pre-Crash Data -5 to 0 Seconds (Most Recent Event, TRG 1) - Table 2 of 4

Time (sec)	ABS Control Status	BOS Control Status	Brake Oil Pressure (Mpa)	Longitudinal Acceleration , VSC Sensor (m/s^2)	Yaw Rate (deg/s)	Steering Input (degrees)	Shift Position
-4.80	OFF	OFF	0.00	0.790	-0.49	0.0	D
-4.30	OFF	OFF	0.00	0.933	-0.49	1.5	D
-3.80	OFF	OFF	0.00	1.220	0.49	6.0	D
-3.30	OFF	OFF	0.00	-0.072	1.46	7.5	D
-2.80	OFF	OFF	0.62	-1.220	0.49	3.0	D
-2.30	OFF	OFF	0.72	-1.436	-0.98	-3.0	D
-1.80	OFF	OFF	0.00	-0.431	-1.46	-4.5	D
-1.30	OFF	OFF	0.10	-0.502	-1.95	-12.0	D
-0.80	OFF	OFF	2.40	-4.307	-4.39	-15.0	D
-0.30	ON	OFF	10.94	-7.106	-0.49	-10.5	D
TRG(0)	ON	OFF	12.14	-8.973	2.44	-10.5	D

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Linking VCH t[illegible] EDR

Resolution 4.5 deg

Resolution 1.5 deg

Pre-Crash D[illegible] **[illegible]vent, TRG 1) - Table 2 of 4**

Time (sec)	Yaw Rate Sensor Signal [deg/s]	Longitudinal G Sensor Signal [m/s^2]	Lateral G Sensor Signal [m/s^2]	Throttle Opening Ratio [%]	Steering Signal [deg]	Longitudinal Acceleration , VSC Sensor (m/s^2)	Yaw Rate (deg/s)	Steering Input (degrees)	Sh[illegible]ion
-4.80	-0.976	-0.28712	0.21534	0	0	0.790	-0.49	0.0	D
-4.30	-0.976	0.64602	0.14356	14.5	0	0.933	-0.49	1.5	D
-3.80	-0.976	0.86136	0.14356	31	0	1.220	0.49	6.0	D
-3.30	0	1.4356	0.50246	29.5	4.5	-0.072	1.46	7.5	D
-2.80	0.976	0.3589	0.7178	0	4.5	-1.220	0.49	3.0	D
-2.30	1.464	-0.07178	0.64602	0	4.5	-1.436	-0.98	-3.0	D
-1.80	-0.488	-1.50738	0.07178	0	0	-0.431	-1.46	-4.5	D
-1.30	-1.464	-0.50246	0	0	-4.5	-0.502	-1.95	-12.0	D
-0.80	-1.464	-0.43068	-0.14356	0	-4.5	-4.307	-4.39	-15.0	D
-0.30	-4.392	-2.65586	-0.64602	0	-13.5	-7.106	-0.49	-10.5	D
TRG(0)	-4.392	-3.73256	-0.43068	0	-13.5	-8.973	2.44	-10.5	D
	-4.392	-3.73256	-0.43068	0	-13.5				
	-3.416	-7.8958	0.07178	0	-9				
	-3.416	-7.8958	0.07178	0	-9				
	-2.928	-8.9725	0.21534	0	-9				
	0	-8.9725	0.78958	0	-9				
	0	-8.9725	0.78958	0	-9				
	-2.928	-3.87612	-0.86136	0	-9				
	-0.488	0.14356	0.50246	0	-13.5				
	-0.976	-0.28712	0	0	-13.5				
	-0.488	-0.28712	0.28712	0	-13.5				
	-0.976	-0.14356	0.28712	0	-13.5				
	-0.976	-0.28712	0.21534	0	-13.5				
	-0.976	-0.28712	0.21534	0	-13.5				

Toyota 17EDR – Pre-Crash Table 3

Pre-Crash Data -5 to 0 Seconds (Most Recent Event, TRG 1) - Table 3 of 4

Time (sec)	Sequential Shift Range	Cruise Control Status	VSC Control Status	READY Signal	Drive Mode, Power Train	Drive Mode, Snow	Drive Mode, EV
-4.80	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
-4.30	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
-3.80	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
-3.30	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
-2.80	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
-2.30	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
-1.80	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
-1.30	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
-0.80	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
-0.30	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
TRG(0)	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid

Toyota 17EDR – Pre-Crash Table 4

Pre-Crash Data -5 to 0 Se

Time (sec)	Drive mode select signal
-4.80	Normal
-4.30	Normal
-3.80	Normal
-3.30	Normal
-2.80	Normal
-2.30	Normal
-1.80	Normal
-1.30	Normal
-0.80	Normal
-0.30	Normal
TRG(0)	Normal

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Vehicle Control History

- Available on:
- 2013+ Rav 4
- 2014 IS250 / 350
- Requires Toyota TechStream Diagnostic Tool
  - Available with the “Lite” Version

Vehicle Control History

- Specific Triggers result in CASA (ACM) Storing Pre / Post Trigger Data
- 26 Different “Behaviours” will result in Trigger
- 3 Categories:
  - Vehicle Input - Accelerator Pedal Input
  - Control System Activation (VSC / ABS etc..)
  - Vehicle Motion (Long. / Lat. g load)

Vehicle Control History

- Up to 140 VCH Events can be stored
  - Stored in “FIFO” method – first in, first out
  - Recording memory is “partitioned”
    - Prevents rare triggers from being overwritten by more frequent ones
- 5-8 seconds of Pre & Post trigger data
  - Trigger Type determines time frame
- Up to 40 parameters including Odometer
- May include Date & Time

General Parameter List Overview

General data* *(Recorded for all Triggers)		Powertrain system	Brake control system	Driver Assistance system
•Vehicle speed •Num. of ignition cycles •Time from IG-ON •Odometer •Accelerator pedal % •Shift position •Brake SW •Brake fluid pressure •Steering angle	•Fuel injection command •Longitudinal G sensor •Lateral G sensor •Yaw rate	•Engine RPM •Gear position •Turbo RPM •Ready-ON (HV) •% of Engine throttle •Engine load factor	•VSC/ABS/TRC activation •VSC/TRC switch •Wheel speed sensors	•Cruise control SW •Cruise control operation •Pre Collision System activation •Lane Keep Assist •Target distance •Target relative velocity •Drive mode

Vehicle Control History

- From: techinfo.toyota.com
- Search on subject vehicle to see if VCH is supported
- Full listing of Trigger ID's / Memory Areas (Set Areas) / Image Triggers
- May be vehicle specific

SUPPLEMENTAL RESTRAINT SYSTEMS: AIRBAG SYSTEM (for TMC Made ...2019 - 2020 MY Corolla Corolla Hatchback [06/2018 -] 2019-04-08, 8:09 AM

Last Modified: 02-22-2019	6.8:8.0.48	**Doc ID:** RM100000001CZRE
Model Year Start: 2019	**Model:** Corolla Hatchback	**Prod Date Range:** [06/2018 -]
Title: SUPPLEMENTAL RESTRAINT SYSTEMS: AIRBAG SYSTEM (for TMC Made Gasoline Model): VEHICLE CONTROL HISTORY; 2019 - 2020 MY Corolla Corolla Hatchback [06/2018 -]		

VEHICLE CONTROL HISTORY

Function Overview

(a) The vehicle control history is a function that records control data (record data) when triggered by specific vehicle behavior. When DTCs are not detected according to information provided by customers, by checking the vehicle control history, it is possible to gain a quantitative grasp on diagnostic information and the history can be used as a reference to proceed with malfunction diagnosis.

(b) The vehicle control history also stores image data when a specific trigger is set.

(c) The vehicle control history is recorded in different storage areas designated by trigger group and it is possible to save up to 140 items in total. If storage space runs out, the data is cleared from the oldest data first for the respective storage area and new data is recorded.

(d) Each trigger receives time information from the main body ECU (time elapsed since engine switch turned on (IG)) or the navigation system* (absolute time).

*: Only for factory-installed navigation systems (factory option)

Trigger ID's

- Extensive List of ID numbers and recording time, intervals and numbe of samples surrounding trigger point

TRIGGER ITEM		RECORD DATA				
		BEFORE DETECTION		AFTER DETECTION		SAMPLING CYCLE (SEC.)
ID NO.	ITEM	RECORDING TIME (SEC.)	NUMBER OF POINTS	RECORDING TIME (SEC.)	NUMBER OF POINTS	
2	Accelerator pedal opening angle signal is high during low speed	7.5	15	7.5	16	0.5
3	Accelerator high position in mid to high speed	7.5	15	7.5	16	0.5
4	Accelerator pedal opening angle signal is high immediately after brake pedal is released	7.5	15	7.5	16	0.5
5-1	Accelerator pedal opening angle is medium or higher immediately after shifting to R	5	10	5	11	0.5
5-2	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position					
5-3	Accelerator pedal opening angle is medium or higher immediately after shifting to driving position					

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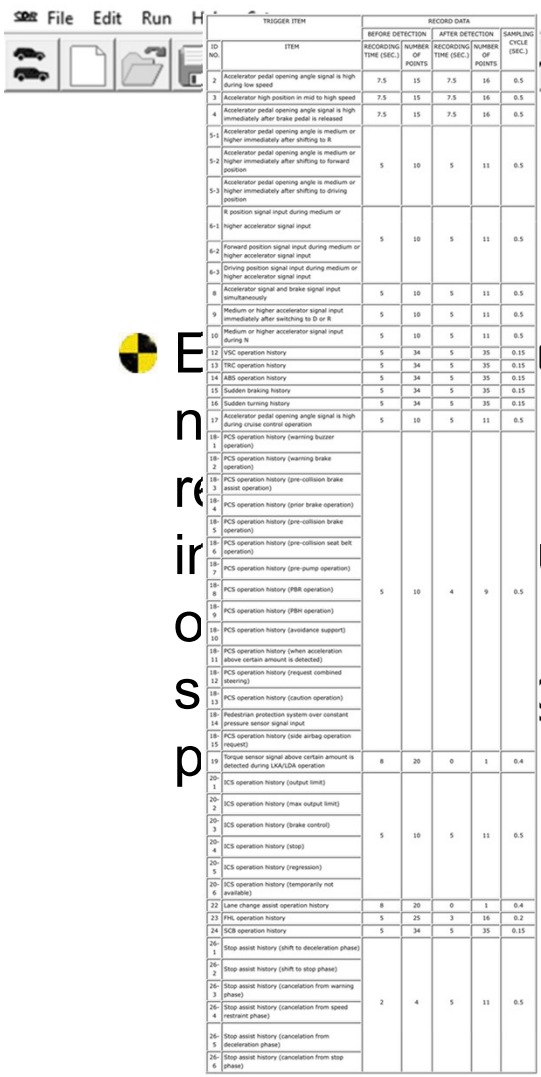

TRIGGER ITEM		RECORD DATA				
		BEFORE DETECTION		AFTER DETECTION		SAMPLING CYCLE (SEC.)
ID NO.	ITEM	RECORDING TIME (SEC.)	NUMBER OF POINTS	RECORDING TIME (SEC.)	NUMBER OF POINTS	
2	Accelerator pedal opening angle signal is high during low speed	7.5	15	7.5	16	0.5
3	Accelerator high position in mid to high speed	7.5	15	7.5	16	0.5
4	Accelerator pedal opening angle signal is high immediately after brake pedal is released	7.5	15	7.5	16	0.5
5-1	Accelerator pedal opening angle is medium or higher immediately after shifting to R	5	10	5	11	0.5
5-2	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position					
5-3	Accelerator pedal opening angle is medium or higher immediately after shifting to driving position					
6-1	R position signal input during medium or higher accelerator signal input	5	10	5	11	0.5
6-2	Forward position signal input during medium or higher accelerator signal input					
6-3	Driving position signal input during medium or higher accelerator signal input					
8	Accelerator signal and brake signal input simultaneously	5	10	5	11	0.5
9	Medium or higher accelerator signal input immediately after switching to D or R	5	10	5	11	0.5
10	Medium or higher accelerator signal input during N	5	10	5	11	0.5
12	VSC operation history	5	34	5	35	0.15
13	TRC operation history	5	34	5	35	0.15
14	ABS operation history	5	34	5	35	0.15
15	Sudden braking history	5	34	5	35	0.15
16	Sudden turning history	5	34	5	35	0.15
17	Accelerator pedal opening angle signal is high during cruise control operation	5	10	5	11	0.5
18-1	PCS operation history (warning buzzer operation)	5	10	4	9	0.5
18-2	PCS operation history (warning brake operation)					
18-3	PCS operation history (pre-collision brake assist operation)					
18-4	PCS operation history (prior brake operation)					
18-5	PCS operation history (pre-collision brake operation)					
18-6	PCS operation history (pre-collision seat belt operation)					
18-7	PCS operation history (pre-pump operation)					
18-8	PCS operation history (PBR operation)					
18-9	PCS operation history (PBH operation)					
18-10	PCS operation history (avoidance support)					
18-11	PCS operation history (when acceleration above certain amount is detected)					
18-12	PCS operation history (request combined steering)					
18-13	PCS operation history (caution operation)					
18-14	Pedestrian protection system over constant pressure sensor signal input					
18-15	PCS operation history (side airbag operation request)					
19	Torque sensor signal above certain amount is detected during LKA/LDA operation	8	20	0	1	0.4
20-1	ICS operation history (output limit)	5	10	5	11	0.5
20-2	ICS operation history (max output limit)					
20-3	ICS operation history (brake control)					
20-4	ICS operation history (stop)					
20-5	ICS operation history (regression)					
20-6	ICS operation history (temporarily not available)					
22	Lane change assist operation history	8	20	0	1	0.4
23	FHL operation history	5	25	3	16	0.2
24	SCB operation history	5	34	5	35	0.15
26-1	Stop assist history (shift to deceleration phase)	2	4	5	11	0.5
26-2	Stop assist history (shift to stop phase)					
26-3	Stop assist history (cancelation from warning phase)					
26-4	Stop assist history (cancelation from speed restraint phase)					
26-5	Stop assist history (cancelation from deceleration phase)					
26-6	Stop assist history (cancelation from stop phase)					

Trigger ID's

- E… of ID n… re… ir… umbe… o… s… gger p…

Image Recording

- Listing of which Trigger ID's may have images associated to them
- # of images before & after trigger
- Recording time
- Sampling interval

Image Data Specifications

TRIGGER ITEM		RECORD DATA				
		BEFORE DETECTION		AFTER DETECTION		SAMPLING CYCLE (SEC.)
ID NO.	ITEM	RECORDING TIME (SEC.)	NUMBER OF POINTS	RECORDING TIME (SEC.)	NUMBER OF POINTS	
15	Sudden braking history	6	10	6	10	0.6
16	Sudden turning history	6	10	6	10	0.6
18-3	PCS operation history (pre-collision brake assist operation)	6	10	6	10	0.6
18-4	PCS operation history (prior brake operation)					
18-5	PCS operation history (pre-collision brake operation)					
18-11	PCS operation history (when acceleration above certain amount is detected)					
18-14	Pedestrian protection system over constant pressure sensor signal input					
19	Torque sensor signal above certain amount is detected during LKA/LDA operation	6	10	6	10	0.6
20-2	ICS operation history (max output limit)	6	10	6	10	0.6
20-3	ICS operation history (brake control)					

Parameter vs. Trigger ID listing

- Full listing of which Trigger ID's a given parameter will be recorded under
- Very long list
- This is a sample of what's available
- See: techinfo.toyota.com for a full listing for subject vehicle

Record Data Chart

ITEM NAME	TRIGGER ID NO.
Vehicle Speed	2, 3, 4, 5-1, 5-2, 5-3, 6-1, 6-2, 6-3, 8, 9, 10, 12, 13, 14, 15, 16, 17, 18-1, 18-2, 18-3, 18-4, 18-5, 18-6, 18-7, 18-8, 18-9, 18-10, 18-11, 18-12, 18-13, 18-14, 18-15, 19, 20-1, 20-2, 20-3, 20-4, 20-5, 20-6, 22, 23, 24, 26-1, 26-2, 26-3, 26-4, 26-5, 26-6
Accelerator Opening Ratio	2, 3, 4, 5-1, 5-2, 5-3, 6-1, 6-2, 6-3, 8, 9, 10, 12, 13, 14, 15, 16, 17, 18-1, 18-2, 18-3, 18-4, 18-5, 18-6, 18-7, 18-8, 18-9, 18-10, 18-11, 18-12, 18-13, 18-14, 18-15, 19, 20-1, 20-2, 20-3, 20-4, 20-5, 20-6, 22, 23, 24, 26-1, 26-2, 26-3, 26-4, 26-5, 26-6
Engine RPM Data	2, 3, 4, 5-1, 5-2, 5-3, 6-1, 6-2, 6-3, 8, 9, 10, 12, 13, 14, 15, 16, 17, 18-1, 18-2, 18-3, 18-4, 18-5, 18-6, 18-7, 18-8, 18-9, 18-10, 18-11, 18-12, 18-13, 18-14, 18-15, 19, 20-1, 20-2, 20-3, 20-4, 20-5, 20-6, 22, 23, 24, 26-1, 26-2, 26-3, 26-4, 26-5, 26-6

VCH – Parameter Definition / Range / Units

- Not all parameters are shown
- Refer to Vehicle Control History section of on-line Service Manual
- Techinfo.Toyota.com*
- *Subscription Required

ITEM NAME	ITEM EXPLANATION	DISPLAYED VALUE	REMARKS
Vehicle Speed	Displays the vehicle speed.	0 to 200 km/h	-
Accelerator Opening Ratio	Displays the accelerator position sensor opening angle.	0 to 100%	-
Engine RPM Data	Displays the engine speed.	0 to 12800 rpm	-
Cruise Control	Displays whether the cruise control is operating.	ON: Operating OFF: Not Operating / Unequipped Vehicles	Unequipped vehicles display OFF.
Shift Position Signal	Displays the shift lever position.	N / D / R / P / 5 / 4 / 3 / 2 / LO / B / SD / Sequential Mode / Not Specified	-
Brake SW	Displays the stop light switch status.	ON: Stop light illuminated OFF: Not Illuminated	-
Brake Oil Pressure	Displays the fluid pressure inside the master cylinder.	0 to 12.144 MPa	-
Yaw Rate Sensor Signal	Displays the angular velocity when turning.	-61 to 61 deg/s	Becomes a positive value when turning left.
Longitudinal G Sensor Signal	Displays the front and rear acceleration detected by the sensor.	-8.9725 to 8.9725 m/s^2	• Acceleration becomes a positive value • Displays "INVALID" if below the smallest value
Lateral G Sensor Signal	Displays the right and left acceleration detected by the sensor.	-8.9725 to 8.9725 m/s^2	Becomes a positive value when turning left.
Throttle Opening Ratio	Displays the engine throttle opening amount.	0 to 100%	Only displayed for diesel vehicles ("INVALID" is displayed for other vehicles)
Command Value of Fuel Injection Amount	Displays the fuel injection volume (ECU command value).	0 to 120 mm^3/st	Only displayed for diesel vehicles.
Steering Signal	Displays the steering angle.	-567 to 567 deg	Becomes a positive value when turning left.
HV Recognition Flag		HV: Hybrid vehicle	-

VCH – Parameter Definition / Range / Units

ITEM NAME	ITEM EXPLANATION	DISPLAYED VALUE	REMARKS
	Displays whether the vehicle is a hybrid vehicle.	Other: Gasoline or diesel vehicle	
Diesel Flag	Displays whether the vehicle is a diesel vehicle.	Diesel: Diesel vehicle Other: Gasoline or hybrid vehicle	-
Sensor Type Identification (Security)	Displays whether a clinometer sensor is installed.	INSTALLED / NOT INSTALLED	Displays a fixed value for each vehicle.
Sensor Type Identification (G Sensor Layout)	Displays the acceleration sensor position angle for the vehicle forward direction.	45° / 90°	Displays a fixed value for each vehicle.
Sensor Type Identification (G Sensor)	Displays the number of acceleration sensors.	1 / 2	Displays a fixed value for each vehicle.
Sensor Type Identification (Yaw Rate Sensor)	Displays the number of yaw rate sensors.	1 / 2	Displays a fixed value for each vehicle.
Yaw G Sensor Installation Information	Displays normal direction or reverse direction (180°opposite) for the sensor position angle (acceleration sensor position angle).	NORMAL DIRECTION / REVERSE DIRECTION	Displays a fixed value for each vehicle.
Shift Gear	Displays the shift step (meter display value) when in sequential mode.	OFF: Sequential mode off GEARS 1 to 8: Shift step on meter display	-

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VCH – Parameter Definition / Range / Units

Engine Load Factor	Displays the engine load.	0 to 100%	Hybrid and diesel vehicles display "INVALID"
READY Signal	Displays whether driving is possible.	ON / OFF	Conventional and diesel vehicles display "INVALID".
Gear Position Signal	Displays the current gear step.	1 / 2 / 3 / 4 / 5 / 6 / 7 / 8	• Hybrid vehicles display "INVALID" • CVT normally displays "1"
Turbine RPM/Input Shaft RPM	Displays the torque converter output axis speed (input shaft speed).	0 to 12650 rpm	-
FR Wheel Speed	Displays the front right wheel speed.	-67.67 to 260km/h	-
FL Wheel Speed	Displays the front left wheel speed.	-67.67 to 260km/h	-
RR Wheel Speed	Displays the rear right wheel speed.	-67.67 to 260km/h	-
RL Wheel Speed	Displays the rear left wheel speed.	-67.67 to 260km/h	-
ACC Target Distance	Displays the vehicle-to-vehicle distance to the preceding vehicle targeted by ACC (vehicle-to-vehicle distance mode).	0 to 251 m	-

VCH – Parameter Definition / Range / Units

ITEM NAME	ITEM EXPLANATION	DISPLAYED VALUE	REMARKS
ACC Target Condition		PRECEDING VEHICLE EXISTS / NO PRECEDING VEHICLE / FIXED SPEED MODE / RADAR MALFUNCTION	
ACC Relative Speed	Displays the relative vehicle speed of the preceding vehicle targeted by ACC (vehicle-to-vehicle distance mode).	-128 to 127km/h	-
PCS Reception Data Invalid Flag	Displays the VALID/INVALID status of PCS related data.	VALID / INVALID	-
ALM Request Flag	Displays whether an operation request was output to the warning function from PCS.	ON: Operation request OFF: No operation request	-
ABK Request Flag	Displays whether an operation request was output to the warning brake function from PCS.	ON: Operation request OFF: No operation request	Only displayed for vehicles with a driver monitor camera
PBA Request Flag	Displays whether an operation request was output according to the pre-collision brake assist function from PCS.	ON: Operation request OFF: No operation request	-

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VCH – Parameter Definition / Range / Units

Target Distance	Displays the PCS control target vehicle-to-vehicle distance.	0 to 163.83 m	If a target is recognized, but PCS control is determined unnecessary, the largest value is displayed.
Target Relative Velocity	Displays the PCS control target vehicle relative speed.	-51.175 to 0 m/s	If a target is recognized, but PCS control is determined unnecessary, the largest value is displayed.
Target Lateral Position	Displays the PCS control target side position.	-5.12 to 5.08 m	• If a target is recognized, but PCS control is determined unnecessary, the largest value is displayed. • If the target is on the right side of the vehicle a positive value is displayed, and if the target is on the left side a negative value is displayed.
Target Number	Displays the target number determined by PCS control for targets recognized by the sensors (maximum 8 targets).	1 to 8	When the target number is 1 and the related record data (control target vehicle-to-vehicle distance, relative speed and side position) all display the largest values, PCS control has been determined unnecessary.

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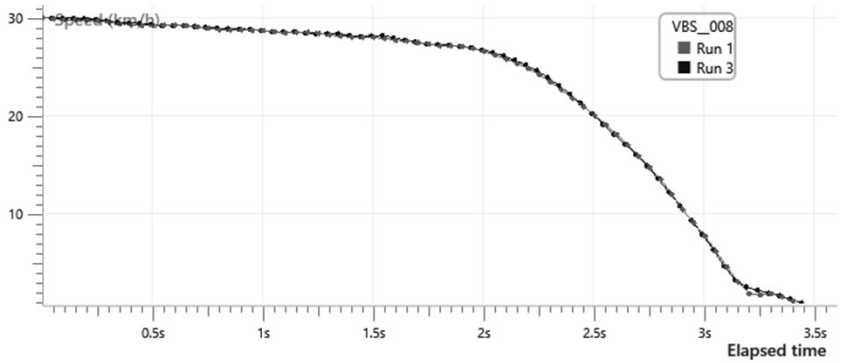
Braking Values

Toyota Factory Dash Cam



VBox Sport Data in VBox Test Suite

VBS_008

Run	Time (s)	Distance (m)	Avg Decel (g)	Centre Line Deviation (m)	MFDD (g)	AMS Distance (m)
1	3.45	21.57	0.16	-0.02	0.61	5.83
2	20.57	79.56	0.04	6.81	0.03	110.47
3	3.46	21.62	0.16	-0.05	0.63	5.60
Avg	3.45	21.60	0.16	-0.03	0.62	5.72
Max	3.46	21.62	0.16	-0.02	0.63	5.83
Min	3.45	21.57	0.16	-0.05	0.61	5.60

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Braking Values – Work In Process!

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1		Misc Information					Samsung cell phone	Measurements	Vericom data								
2	Test #	Vehicle	Sun position	Type of Test	Verbal speed relayed to Steve	Starting Position	Time	distance to target	VC Run #	Vericom Time (orig)	Vericom Adjust Time	OBDII speed	GPS Speed	Time	Speed	Distance	Average gx
3	0	Toyota															
4	1	Toyota	Facing Target	Avoidance	30.00 km/h	65.00 m	10:25	46.00 inch	1	10:18	10:25	26.00 km/h	26.68 km/h	1.12 s	25.45 km/h	4.66 m	-0.644
5	2	Toyota	Facing Target	Avoidance	30.00 km/h	82.00 m	11:07	28.00 inch	2	10:34	10:41	0.00 km/h	25.17 km/h	1.10 s	24.26 km/h	4.28 m	-0.625
6	3	Toyota	Facing Target	Avoidance	40.00 km/h	82.00 m	12:07	0.00 inch	3	11:41	11:48	33.00 km/h	34.25 km/h	1.20 s	34.05 km/h	5.93 m	-0.804
7	4	Toyota	Facing Target	Avoidance	30.00 km/h	82.00 m	12:11	57.00 inch	4	12:05	12:12	26.00 km/h	25.90 km/h	2.16 s	27.33 km/h	6.05 m	-0.358
8	5	Toyota	Facing Target	Avoidance	30.00 km/h	82.00 m	12:17	0.00 inch	5			0.00 km/h	51.20 km/h	1.79 s	49.98 km/h	12.71 m	-0.791
9	6	Toyota	Behind Target	Avoidance	30.00 km/h	n/a	12:25	0.00 inch	6	12:18	12:25	23.00 km/h	23.69 km/h	0.91 s	24.36 km/h	3.19 m	-0.758
10	7	Toyota	Behind Target	Avoidance	30.00 km/h	n/a	1:20	n/a	7	1:13	1:20	22.00 km/h	22.84 km/h	1.57 s	21.45 km/h	4.15 m	-0.387
11	8	Toyota	Behind Target	Avoidance	30.00 km/h	n/a	1:21	n/a	8	1:16	1:23	29.00 km/h	29.87 km/h	1.21 s	28.14 km/h	5.54 m	-0.659
12	9	Toyota	Behind Target	Avoidance	30.00 km/h	n/a		45.00 inch	9	1:23	1:30	25.00 km/h	24.28 km/h	1.10 s	24.10 km/h	4.14 m	-0.621
13	10	Toyota	Behind Target	Avoidance	30.00 km/h	n/a	1:34	40.00 inch	10	1:27	1:34	23.00 km/h	22.28 km/h	1.08 s	21.95 km/h	3.85 m	-0.576
14	11	Toyota	Behind Target	Avoidance	30.00 km/h	n/a	2:17	n/a									
15	12	Toyota	Behind Target	Avoidance	30.00 km/h	n/a	2:24	40.00 inch	11	2:09	2:16	0.00 km/h	21.51 km/h	1.00 s	20.35 km/h	3.38 m	-0.576
16	13	Chevy	Behind Target	Avoidance	32.00 km/h	n/a	2:34	27.00 inch									
17	14	Chevy	Behind Target	Avoidance	33.00 km/h	n/a	2:39	36.00 inch	12	2:32	2:39	0.00 km/h	30.26 km/h	1.31 s	31.10 km/h	6.65 m	-0.672
18	15	Chevy	Behind Target	Avoidance	40.00 km/h	n/a	2:41	0.00 inch	13	2:34	2:41	0.00 km/h	32.97 km/h	1.44 s	33.86 km/h	7.86 m	-0.666
19	16	Chevy	Behind Target	Avoidance	32.00 km/h	n/a	2:43	42.00 inch	14	2:37	2:44	0.00 km/h	29.21 km/h	1.25 s	30.43 km/h	6.17 m	-0.689
20	17	Chevy	Behind Target	Avoidance	33.00 km/h	n/a	2:45	15.00 inch	15	2:38	2:45	0.00 km/h	29.72 km/h	1.29 s	30.36 km/h	6.37 m	-0.667
21	18	Chevy		Brake Run	63.90 km/h	n/a	2:47	n/a	16	2:41	2:48	0.00 km/h	63.15 km/h	1.94 s	63.94 km/h	17.40 m	-0.934
22	19	Chevy		Brake Run	66.70 km/h	n/a	2:49	n/a	17	2:42	2:49	0.00 km/h	63.33 km/h	2.05 s	66.74 km/h	19.07 m	-0.922
23	20	Toyota	Behind Target	Avoidance	30.00 km/h	n/a	3:03	36.00 inch	18	2:56	3:03	24.00 km/h	24.80 km/h	1.20 s	24.97 km/h	4.65 m	-0.589
24	21	Toyota	Behind Target	Avoidance	30.00 km/h	n/a	3:05	0.00 inch	19	2:58	3:05	23.00 km/h	23.64 km/h	0.84 s	22.49 km/h	2.60 m	-0.758
25	22	Toyota		Brake Run	n/a	n/a	3:12	n/a	20	3:05	3:12	69.00 km/h	73.62 km/h	2.46 s	72.07 km/h	25.16 m	-0.83
26	23	Toyota		Brake Run	n/a	n/a	3:13	n/a	21	3:06	3:13	67.00 km/h	69.56 km/h	2.33 s	69.34 km/h	22.84 m	-0.843
27	24	Toyota		Lane Keeping	n/a	n/a	3:15	n/a	24	3:32	3:29	57.00 km/h	58.40 km/h	1.94 s	57.73 km/h	15.91 m	-0.843
28	25	Ford		Brake Run	n/a	n/a	n/a	n/a	25	3:34	3:34	65.00 km/h	66.73 km/h	2.41 s	66.86 km/h	22.79 m	-0.786
29	26	Ford		Brake Run	n/a	n/a	n/a	n/a	26	3:35	3:35	67.00 km/h	69.07 km/h	2.52 s	68.69 km/h	24.52 m	-0.772
30	27	Ford		Brake Run	n/a	n/a	n/a	n/a	27	3:39	3:39	78.00 km/h	80.42 km/h	2.54 s	78.87 km/h	28.61 m	-0.88
31	28	Ford		Brake Run	n/a	n/a	n/a	n/a	28	3:41	3:41	70.00 km/h	72.24 km/h	2.38 s	72.64 km/h	24.39 m	-0.865

Does it Always Work? NO!

Toyota Factory Dash Cam

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Does it Always Work? NO!

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PCS Activation – Trigger 18-3

PCS Activation – VCH Data

Time Stamp	Vehicle Speed [km/h]	Accelerator Opening Ratio [%]	Engine RPM Data [rpm]	Cruise Control	Shift Position Signal	Brake SW	Brake Oil Pressure [MPa]	Yaw Rate Sensor Signal [deg/s]	Longitudinal G Sensor Signal [m/s^2]	Lateral G Sensor Signal [m/s^2]	Throttle Opening Ratio [%]
2019-02-08-15:59:02.000	87	0	1900	0	1	ON	1.536	-0.976	-2.08162	0.21534	0
2019-02-08-15:59:02.500	84	0	1600	0	1	ON	1.056	-0.976	-1.72272	0.21534	0
2019-02-08-15:59:03.000	81	0	1700	0	1	ON	1.056	-1.464	-1.72272	0.14356	0
2019-02-08-15:59:03.500	78	0	1600	0	1	ON	1.2	-1.464	-1.86628	0	0
2019-02-08-15:59:04.000	75	0	1500	0	1	ON	1.296	-0.976	-1.86628	0.21534	0
2019-02-08-15:59:04.500	72	0	1500	0	1	ON	1.344	-0.488	-1.93806	0.3589	0
2019-02-08-15:59:05.000	68	0	1400	0	1	ON	2.16	-0.488	-2.72764	0.43068	0
2019-02-08-15:59:05.500	62	0	1200	0	1	ON	4.464	-0.488	-4.6657	0.57424	0
2019-02-08-15:59:06.000	54	0	1100	0	1	ON	4.368	0	-5.09638	0.3589	0
2019-02-08-15:59:06.500	46	0	900	0	1	ON	4.368	-1.464	-4.88104	0.07178	0
2019-02-08-15:59:06.900	38	0	800	0	1	ON	5.328	-1.464	-5.88596	0.14356	0
2019-02-08-15:59:06.900	38	0	800	0	1	ON	5.328	-1.464	-5.88596	0.14356	0
2019-02-08-15:59:07.000	36	0	800	0	1	ON	5.472	-1.464	-6.02952	0.14356	0
2019-02-08-15:59:07.500	26	0	700	0	1	ON	5.376	-0.976	-5.3835	0.14356	0
2019-02-08-15:59:08.000	16	0	800	0	1	ON	5.328	-0.976	-5.81418	0.28712	0
2019-02-08-15:59:08.500	7	0	600	0	1	ON	4.848	-0.976	-5.59884	0.21534	0
2019-02-08-15:59:09.000	0	0	500	0	1	ON	2.448	-0.976	-0.07178	0.21534	0
2019-02-08-15:59:09.500	0	0	500	0	1	ON	1.488	-0.976	0.28712	0.21534	0
2019-02-08-15:59:10.000	0	0	500	0	1	ON	0.912	-0.976	0	0.14356	0
2019-02-08-15:59:10.500	0	0	600	0	1	ON	0.96	-0.976	0	0.21534	0
2019-02-08-15:59:11.000	0	0	600	0	1	ON	0.864	-0.976	0	0.21534	0

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LKA / ICS / EDR Related

EDR (CDR) vs. VCH Data (#19 – Set 10)

System Status at Event (Most Recent Event, TRG 3)

TRG Count (times)	3
Event Type	Rollover
Previous Crash Type	Frontal/Rear/Side Crash
Time from Previous TRG (msec)	37.5
Freeze Signal	OFF
Freeze Signal Factor	None
Recording Status, Rollover Crash Info.	Complete
Odometer signal (miles [km])	101 [162]
Trip count (times)	107
Time count	604,400
Time count input system	Normal

Trigger ID	19	
Trigger time information	200	ms
EDR-related information	12.5	ms
EDR-related information	3	
Number of Overwrite of the Recording Area	1	
Trip count	107	trip
Clock Type	0	
Time count	604.4	sec
Odometer signal	162	
Odometer unit signal	1	
HV recognition flag_header	0	
Diesel flag_header	0	

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CDR Data

Pre-Crash Data -5 to 0 Seconds (Most Recent Event, TRG 3) - Table 1 of 4

Time (sec)	Vehicle Speed (MPH [km/h])	Accelerator Pedal, % Full (%)	Percentage of Engine Throttle (%)	Fuel Injection Quantity (mm^3/st)	Engine RPM (RPM)	Motor RPM (RPM)	Service Brake, ON/OFF
-4.80	0.0 [0]	0.0	0.5	Invalid	900	Invalid	OFF
-4.30	1.2 [2]	0.0	0.0	Invalid	600	Invalid	OFF
-3.80	1.9 [3]	1.5	0.0	Invalid	600	Invalid	OFF
-3.30	1.9 [3]	19.5	5.0	Invalid	700	Invalid	OFF
-2.80	3.7 [6]	40.0	9.0	Invalid	1,300	Invalid	OFF
-2.30	5.6 [9]	40.0	9.0	Invalid	1,600	Invalid	OFF
-1.80	7.5 [12]	41.0	11.5	Invalid	1,900	Invalid	OFF
-1.30	9.3 [15]	39.5	12.0	Invalid	2,000	Invalid	OFF
-0.80	11.2 [18]	44.0	20.5	Invalid	2,500	Invalid	OFF
-0.30	10.6 [17]	0.0	0.0	Invalid	1,200	Invalid	ON
TRG(0)	5.6 [9]	0.0	0.0	Invalid	900	Invalid	ON

Pre-Crash Data -5 to 0 Seconds (Most Recent Event, TRG 3) - Table 2 of 4

Time (sec)	ABS Control Status	BOS Control Status	Brake Oil Pressure (Mpa)	Longitudinal Acceleration , VSC Sensor (m/s^2)	Yaw Rate (deg/s)	Steering Input (degrees)	Shift Position
-4.80	OFF	OFF	0.00	0.861	-2.44	-120.0	D
-4.30	OFF	OFF	0.00	0.215	-2.44	-102.0	D
-3.80	OFF	OFF	0.00	0.359	-2.44	-67.5	D
-3.30	OFF	OFF	0.00	0.359	-2.44	-52.5	D
-2.80	OFF	OFF	0.00	1.866	-1.46	-31.5	D
-2.30	OFF	OFF	0.00	-0.431	-1.95	-15.0	D
-1.80	OFF	OFF	0.00	1.507	0.00	18.0	D
-1.30	OFF	OFF	0.00	2.010	12.20	142.5	D
-0.80	OFF	OFF	0.00	1.651	30.74	265.5	D
-0.30	OFF	OFF	4.94	-4.737	-0.98	-33.0	D
TRG(0)	ON	OFF	10.56	-8.973	-61.00	-166.5	D

Corresponding VCH Data

Time Stamp	Vehicle Speed [km/h]	Accelerator Opening Ratio [%]	Engine RPM Data [rpm]	Brake SW	Brake Oil Pressure [MPa]	Yaw Rate Sensor Signal [deg/s]	Longitudinal G Sensor Signal [m/s^2]	Lateral G Sensor Signal [m/s^2]	Throttle Opening Ratio [%]	Steering Signal [deg]
2019-02-08-16:02:22.400	0	0	600	ON	2.592	-0.976	-0.07178	0.07178	0	-103.5
2019-02-08-16:02:22.800	0	0	500	ON	2.544	-0.976	-0.07178	0.07178	0	-103.5
2019-02-08-16:02:23.200	0	0	600	ON	1.008	-0.976	-0.07178	0.07178	0	-103.5
2019-02-08-16:02:23.600	0	0	600	ON	0	-0.976	-0.07178	0.07178	0	-112.5
2019-02-08-16:02:24.000	0	0	600	OFF	0	-0.976	-0.07178	0.07178	0	-112.5
2019-02-08-16:02:24.400	0	0	600	OFF	0	-0.976	-0.07178	0.07178	0	-121.5
2019-02-08-16:02:24.800	0	14	700	OFF	0	-0.976	0	0.07178	3	-121.5
2019-02-08-16:02:25.200	0	15	900	OFF	0	-1.464	0.93314	-0.07178	2.5	-121.5
2019-02-08-16:02:25.600	1	0	800	OFF	0	-2.44	0.64602	0	0.5	-112.5
2019-02-08-16:02:26.000	2	0	600	OFF	0	-2.928	0.21534	-0.07178	0	-99
2019-02-08-16:02:26.400	3	0	600	OFF	0	-2.44	0.28712	0	0	-67.5
2019-02-08-16:02:26.800	3	11	600	OFF	0	-2.44	0.14356	0.21534	3.5	-49.5
2019-02-08-16:02:27.200	4	38.5	1100	OFF	0	-2.44	1.36382	0.07178	6	-40.5
2019-02-08-16:02:27.600	7	40	1400	OFF	0	-3.416	1.86628	0.14356	8	-31.5
2019-02-08-16:02:28.000	10	40	1600	OFF	0	-1.464	1.22026	0.86136	9	-13.5
2019-02-08-16:02:28.400	12	41	1800	OFF	0	0	1.50738	0.14356	11	13.5
2019-02-08-16:02:28.800	15	39.5	2100	OFF	0	3.904	1.14848	2.65586	12.5	81
2019-02-08-16:02:29.200	16	58.5	2200	OFF	0	21.96	0.78958	4.01968	16	229.5
2019-02-08-16:02:29.600	19	0	2000	OFF	0	28.304	1.86628	0.93314	0.5	193.5
2019-02-08-16:02:30.000	16	0	1100	ON	6	-4.88	-5.52706	-3.01476	0	-58.5
2019-02-08-16:02:30.200	11	0	900	ON	10.56	-61	-8.9725	-8.9725	0	-153
2019-02-08-16:02:30.400	0	0	600	ON	7.728	-61	-1.65094	5.7424	0	-229.5

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A Different Scenario….

System Status at Retreival

System Status at Retrieval

ECU Part Number	89170-12E20
EDR Generation	17EDR
Complete File Recorded	Yes
Ignition Cycle, Download (times)	405
Multi-Event, Number of Events (times)	N/A - 2 dependent events follow 1 independent
Time from Event 1 to 2 (sec)	N/A
Diagnostic Trouble Codes Exist	No
Location of Side Satellite Sensor 1, L	front door
Location of Side Satellite Sensor 2, L	B-Pillar
Location of Side Satellite Sensor 3, L	Not Equipped
Location of Side Satellite Sensor 4, L	C-Pillar
Location of Side Satellite Sensor 1, R	front door
Location of Side Satellite Sensor 2, R	B-Pillar
Location of Side Satellite Sensor 3, R	Not Equipped
Location of Side Satellite Sensor 4, R	C-Pillar
Location of Floor Sensor	Airbag ECU

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash Recording Status	Diagnostic Data Recording Status	Occupant Data Recording Status	Crash Info Recording Status	Time Series Recording Status
Most Recent Event	6	Frontal/Rear/Side Crash	0	Complete	Complete	Complete	Complete	Complete
1st Prior Event	5	Side Crash	29	N/A	Complete	N/A	N/A	Complete
2nd Prior Event	4	Side Crash	-284.5	N/A	Complete	N/A	N/A	Complete
3rd Prior Event	3	Side Crash	-528.5	N/A	Complete	N/A	N/A	Complete
4th Prior Event	2	Rollover	-536	Complete	Complete	Complete	Complete	Complete
5th Prior Event	1	Frontal/Rear/Side Crash	-609.5	Complete	Complete	Complete	Complete	Complete

VCH / FFD Data

From VCH Area Files

Trigger ID	18_11	
Trigger time information	350	ms
EDR-related information	16	ms
EDR-related information	3	
Number of Overwrite of the Recording Area	1	
Trip count	413	trip
Clock Type	0	
Time count	881.6	sec
Odometer signal	4853	
Odometer unit signal	1	
HV recognition flag_header	0	
Diesel flag_header	0	

From FFD Images Set Files Area

Trip count	413	trip
Number of Overwrite	2	
Trigger Point Offset Time	400	ms
Encryption Status	1	
Pre-Trigger Image Stored Number	10	
Trigger Point Image Stored Number	1	
Post-Trigger Image Stored Number	10	
Writing completion flag	55	

From VCH Trigger Definitions

18-11	PCS operation history (when acceleration above certain amount is detected)
18-	PCS operation history (request

System Status at Event (Most Recent Event, TRG 6)

TRG Count (times)	6
Event Type	Frontal/Rear/Side Crash
Previous Crash Type	Side Crash
Time from Previous TRG (msec)	39.0
Time from Time Zero to TRG (msec)	68.0
Event Establishment Factor	Frontal Crash
TRG Establishment Factor	Frontal Crash
Freeze Signal	OFF
Freeze Signal Factor	None
Recording Status , Front/Rear and Side Crash Info.	Complete
Odometer signal (miles [km])	3,016 [4,853]
Trip count (times)	413
Time count (msec)	882,100
Time count input system	Normal

Trigger ID	18_11	
Trigger time information	350	ms
EDR-related information	16	ms
EDR-related information	3	
Number of Overwrite of the Recording Area	1	
Trip count	413	trip
Clock Type	0	
Time count	881.6	sec
Odometer signal	4853	
Odometer unit signal	1	
HV recognition flag_header	0	
Diesel flag_header	0	

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Pre-Crash – 5 To 0

Pre-Crash Data -5 to 0 Seconds (Most Recent Event, TRG 6) - Table 1 of 4

Time (sec)	Vehicle Speed (MPH [km/h])	Accelerator Pedal, % Full (%)	Percentage of Engine Throttle (%)	Fuel Injection Quantity (mm^3/st)	Engine RPM (RPM)	Motor RPM (RPM)	Service Brake, ON/OFF
-5.00	50.3 [81]	28.5	6.5	Invalid	1,300	Invalid	OFF
-4.50	50.3 [81]	28.5	7.0	Invalid	1,300	Invalid	OFF
-4.00	50.3 [81]	28.5	7.0	Invalid	1,300	Invalid	OFF
-3.50	50.3 [81]	28.5	7.0	Invalid	1,300	Invalid	OFF
-3.00	50.3 [81]	28.5	7.0	Invalid	1,300	Invalid	OFF
-2.50	50.3 [81]	28.5	7.0	Invalid	1,300	Invalid	OFF
-2.00	49.7 [80]	28.5	7.0	Invalid	1,300	Invalid	OFF
-1.50	45.4 [73]	57.5	7.5	Invalid	1,100	Invalid	OFF
-1.00	42.9 [69]	100.0	21.5	Invalid	2,000	Invalid	OFF
-0.50	29.2 [47]	0.0	2.5	Invalid	2,000	Invalid	ON
TRG(0)	13.7 [22]	0.0	0.0	Invalid	900	Invalid	ON

Pre-Crash Data -5 to 0 Seconds (Most Recent Event, TRG 6) - Table 2 of 4

Time (sec)	ABS Control Status	BOS Control Status	Brake Oil Pressure (Mpa)	Longitudinal Acceleration , VSC Sensor (m/s^2)	Yaw Rate (deg/s)	Steering Input (degrees)	Shift Position
-5.00	OFF	OFF	0.00	-0.144	-0.98	0.0	D
-4.50	OFF	OFF	0.00	-0.359	-0.98	0.0	D
-4.00	OFF	OFF	0.00	-0.431	-0.98	0.0	D
-3.50	OFF	OFF	0.00	-0.359	-0.98	0.0	D
-3.00	OFF	OFF	0.00	-0.072	-0.98	0.0	D
-2.50	OFF	OFF	0.00	0.000	-0.98	0.0	D
-2.00	OFF	OFF	0.00	-0.287	-1.46	0.0	D
-1.50	ON	OFF	0.00	-8.973	7.32	16.5	D
-1.00	ON	OFF	0.00	-0.072	-5.86	24.0	D
-0.50	ON	OFF	6.67	-8.973	30.26	33.0	N
TRG(0)	ON	OFF	4.70	-8.973	-13.18	49.5	N

VCH Data

Vehicle Speed [km/h]	Accelerator Opening Ratio [%]	Engine RPM Data [rpm]	Cruise Control	Shift Position Signal	Brake SW	Brake Oil Pressure [MPa]	Yaw Rate Sensor Signal [deg/s]	Longitudinal G Sensor Signal [m/s^2]	Lateral G Sensor Signal [m/s^2]	Throttle Opening Ratio [%]	Steering Signal [deg]
81	28.5	1300	0	1	OFF	0	-0.976	-0.14356	0.28712	7	0
81	28.5	1300	0	1	OFF	0	-0.976	0	0.28712	6.5	0
81	28.5	1300	0	1	OFF	0	-0.976	-0.07178	0.43068	7	0
81	28.5	1300	0	1	OFF	0	-0.488	-0.21534	0.43068	7	0
81	28.5	1300	0	1	OFF	0	-0.976	-0.3589	0.50246	7	0
80	28.5	1300	0	1	OFF	0	-0.976	-0.3589	0.43068	7	0
80	28.5	1300	0	1	OFF	0	-0.488	-0.14356	0.3589	7	0
80	28.5	1300	0	1	OFF	0	-0.976	-0.07178	0.3589	7	0
60	100	1100	0	1	OFF	0	8.784	-5.0246	-1.22026	99	22.5
75	0	3100	0	1	OFF	0	-3.904	-0.57424	0	0.5	54
71	0	2900	0	1	ON	7.008	2.928	-8.9725	-1.0767	0	54
27	0	1400	0	0	ON	7.584	-17.08	-7.68046	-1.36382	0	40.5
9	0	500	0	0	ON	5.28	-16.592	-7.24978	1.86628	1	63
1	0	100	0	0	ON	11.952	-1.952	-1.4356	0.50246	0	40.5
0	0	0	0	0	ON	8.64	-0.976	-0.3589	0.21534	0	-4.5
0	0	0	0	0	ON	9.072	-0.976	-0.3589	0.21534	0	-4.5
0	0	0	0	0	ON	9.264	-0.976	-0.3589	0.21534	0	0
0	0	0	0	0	ON	9.264	-0.976	-0.3589	0.21534	0	0
0	0	0	0	0	ON	9.216	-0.976	-0.3589	0.21534	0	0
0	0	0	0	0	ON	9.264	-0.976	-0.3589	0.21534	0	0

Texas Case

- 2015 Toyota Camry in one direction
- Tractor / Float Trailer with Temporary building in opposite direction
- Wide load escort – not be escorted properly
- Construction area
- Impact between Camry and Building being transported
- Witness: “I saw the car turn into the trailer”
  - Did he?? Let’s see!

File Edit Run Help

File Edit Run Help Setup

System Status at Time of Retrieval

ECU Part Number	89170-06730
EDR Generation	13EDR
Complete File Recorded	Yes
Freeze Signal	ON
Freeze Signal Factor	Front Airbag Deployment
Diagnostic Trouble Codes Exist	No
Ignition Cycle ,Download (times)	947
Multi-event, number of events (times)	2 or greater
Time from event 1 to 2 (s)	0.26
Time from Previous Pre Crash TRG (msec)	16381 or greater
Latest Pre-Crash Page	0
Contains Unlinked Pre-Crash Data	No

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash & DTC Data Recording Status	Event & Crash Pulse Data Recording Status
Most Recent Event	4	Side Crash	0	Complete (Page 0)	Complete (Side Page 1)
1st Prior Event	3	Rollover	-260	Complete (Page 0)	Complete (Rollover Page 0)
2nd Prior Event	2	Side Crash	-266	Complete (Page 0)	Complete (Side Page 0)
3rd Prior Event	1	Front/Rear Crash	-272	Complete (Page 0)	Complete (Front/Rear Page 0)

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Data Element Sign Convention

Data Element Sign Convention:
The following table provides an explanation of the sign notation for data elements that may be included in this CDR report.

Data Element Name	Positive Sign Notation Indicates
Maximum Delta-V, Longitudinal	Forward
Delta-V, Longitudinal	Forward
Lateral Acceleration for Frontal/Rear Crash, Floor Sensor	Left to Right
Lateral Acceleration, Side Satellite Sensor 1	Left to Right
Lateral Acceleration, Side Satellite Sensor 2	Left to Right
Lateral Acceleration, Side Satellite Sensor 3	Left to Right
Lateral Acceleration, Side Satellite Sensor 4	Left to Right
Lateral Acceleration for Side Crash, Floor Sensor	Left to Right
Roll Angle Peak	Clockwise Rotation
Roll Angle at the Time of TRG	Clockwise Rotation
Roll Rate	Clockwise Rotation
Lateral Acceleration for Rollover, Floor Sensor	Left to Right
Longitudinal Acceleration , VSC Sensor	Forward
Yaw Rate	Left Turn
Steering Input	Left Turn

System Status at Event (3rd Prior Event, TRG 1)

Recording Status, Front/Rear Crash Info.	Complete
Crash Type	Front/Rear Crash
TRG Count (times)	1
Previous Crash Type	No Event
Time from Pre-Crash TRG (msec)	0
Linked Pre-Crash Page	0
Frontal Airbag Deployment, Time to 1st Stage Deployment, Driver (msec)	4
Frontal Airbag Deployment, Time to 1st Stage Deployment, Front Passenger (msec)	No
Pretensioner Deployment, Time to Fire, Driver (msec)	3
Pretensioner Deployment, Time to Fire, Front Passenger (msec)	3
Frontal Airbag Deployment, Time to 2nd Stage, Driver (msec)	34
Frontal Airbag Deployment, Time to 2nd Stage, Front Passenger (msec)	N/A
Active Head Restraint, Time to Deploy, Driver (msec)	SNA
Active Head Restraint, Time to Deploy, Front Passenger (msec)	SNA
Side Curtain Airbag Deployment, Time to Deploy, Driver (msec)	4
Side Curtain Airbag Deployment, Time to Deploy, Passenger (msec)	4
Side Airbag Deployment, Time to Deploy, Driver (msec)	4
Side Airbag Deployment, Time to Deploy, Passenger (msec)	4
Rear Window Airbag Deployment, Time to Deploy (msec)	SNA

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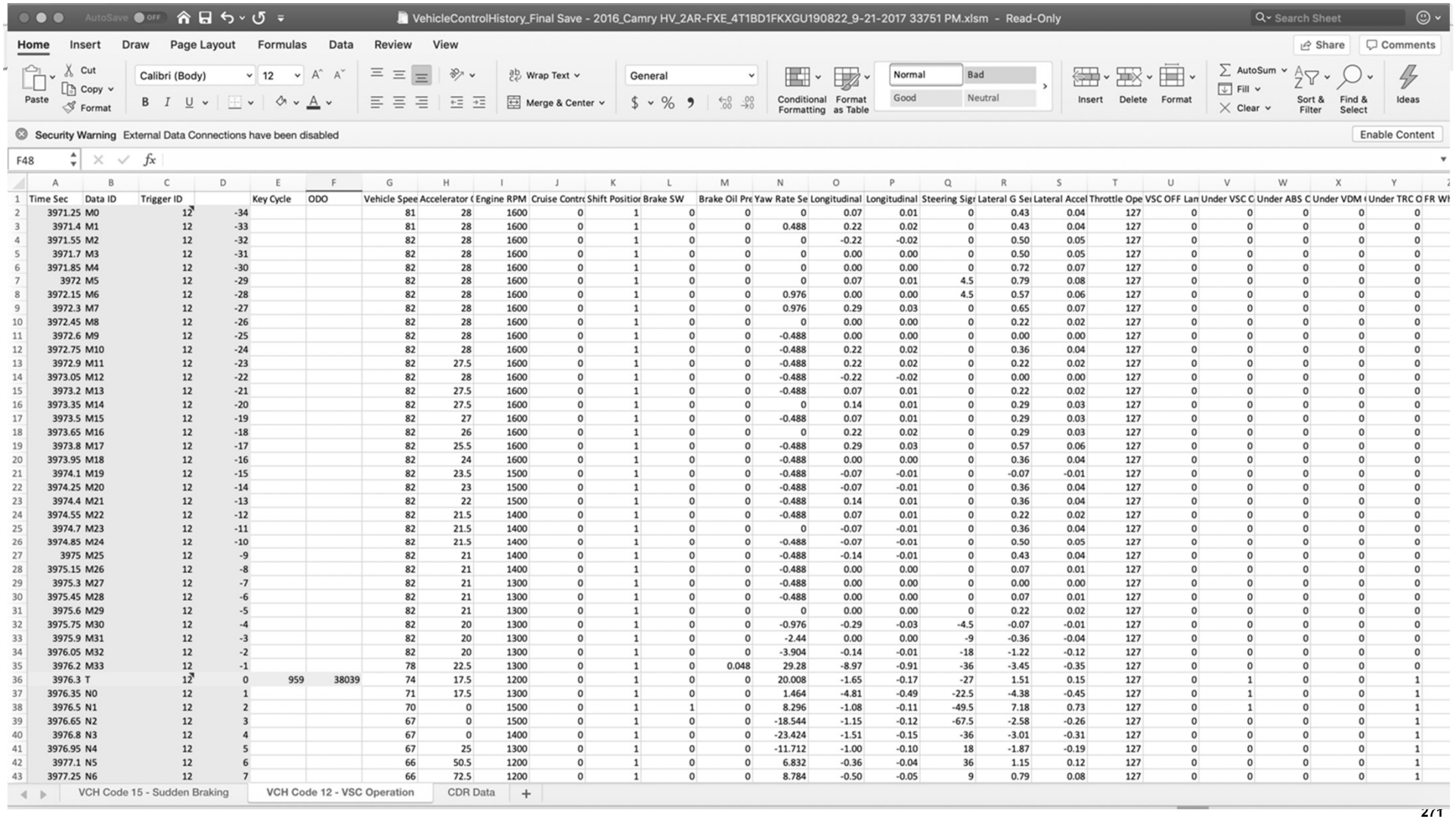
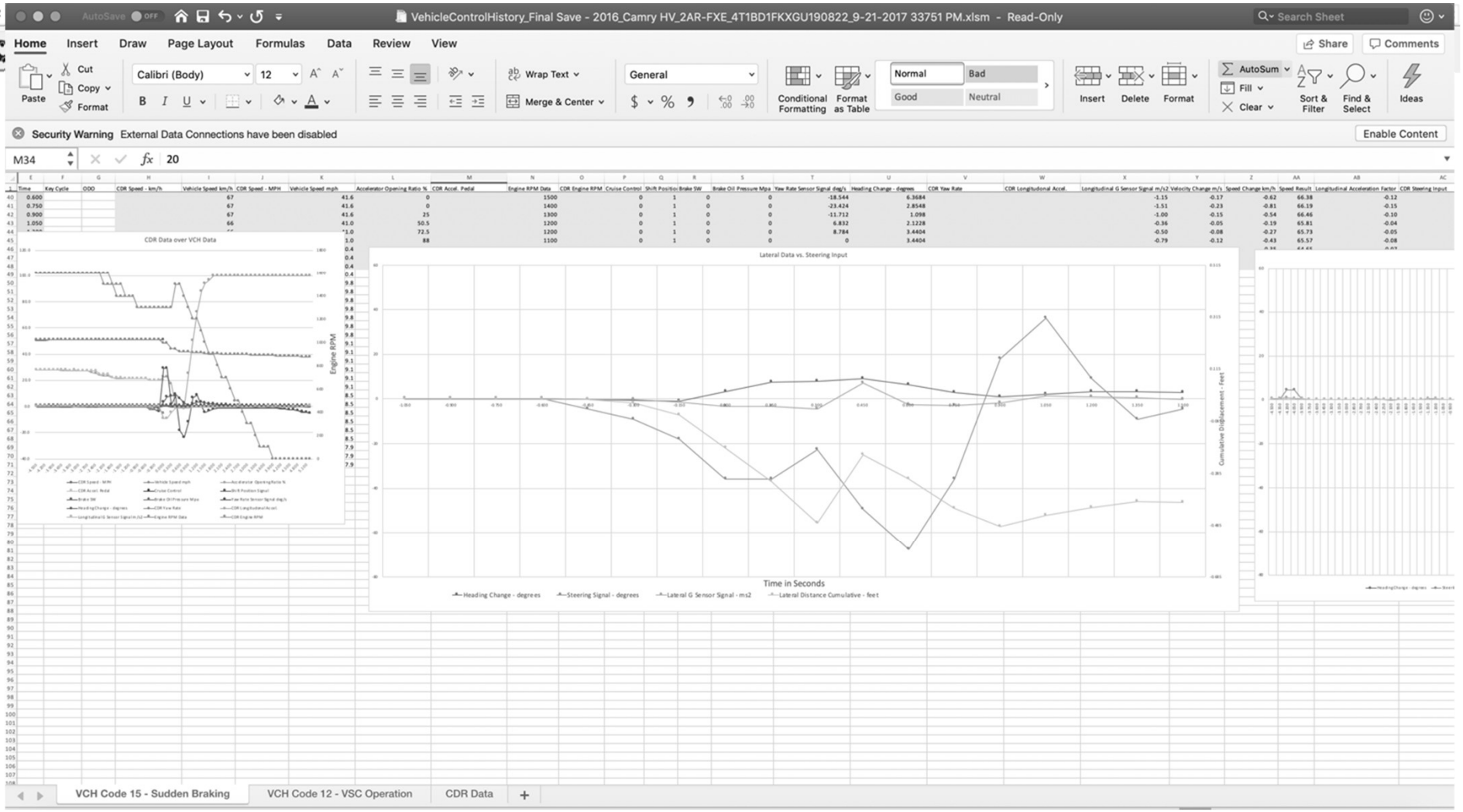
Pre-Crash Data, -5 to 0 seconds (Most Recent Event, TRG 4)

Time (sec)	-4.75	-4.25	-3.75	-3.25	-2.75	-2.25	-1.75	-1.25	-0.75	-0.25	0 (TRG)
Vehicle Speed (MPH [km/h])	50.3 [81]	51 [82]	51 [82]	51 [82]	51 [82]	51 [82]	51 [82]	51 [82]	51 [82]	51 [82]	51 [82]
Accelerator Pedal, % Full (%)	28.0	28.0	27.5	27.5	27.0	24.5	22.0	21.0	21.0	20.0	20.0
Percentage of Engine Throttle (%)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Engine RPM (RPM)	1,600	1,600	1,600	1,600	1,600	1,600	1,500	1,400	1,300	1,300	1,300
Motor RPM (RPM)	5,900	5,900	5,900	6,000	6,000	5,900	6,000	6,000	6,000	6,000	5,900
Service Brake, ON/OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Brake Oil Pressure (Mpa)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Longitudinal Acceleration , VSC Sensor (m/sec^2)	0.215	0.072	0.215	0.215	0.000	-0.072	0.144	-0.144	-0.144	-0.072	-8.973
Yaw Rate (deg/sec)	0.49	0.00	0.49	-0.49	0.00	-0.49	-0.49	-0.49	-0.49	-2.44	-4.88
Steering Input (degrees)	1.5	3.0	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-10.5	-30.0
Shift Position	D	D	D	D	D	D	D	D	D	D	D
Sequential Shift Range	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined
Cruise Control Status	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
1	Time Sec	Data ID	Trigger ID		Key Cycle	ODO	Vehicle Spee	Accelerator (	Engine RPM	Cruise Contro	Shift Positior	Brake SW	Brake Oil Pre	Yaw Rate Se	Longitudinal	Longitudinal	Steering Sigr	Lateral G Sei	Lateral Accel	Throttle Ope	VSC OFF Lan	Under VSC C	Under ABS C	Under VDM (	Under TRC O	FR WI
2	3971.25	M0	12	-34			81	28	1600	0	1	0	0	0	0.07	0.01	0	0.43	0.04	127	0	0	0	0	0	
3	3971.4	M1	12	-33			81	28	1600	0	1	0	0	0.488	0.22	0.02	0	0.43	0.04	127	0	0	0	0	0	
4	3971.55	M2	12	-32			82	28	1600	0	1	0	0	0	-0.22	-0.02	0	0.50	0.05	127	0	0	0	0	0	
5	3971.7	M3	12	-31			82	28	1600	0	1	0	0	0	0.00	0.00	0	0.50	0.05	127	0	0	0	0	0	
6	3971.85	M4	12	-30			82	28	1600	0	1	0	0	0	0.00	0.00	0	0.72	0.07	127	0	0	0	0	0	
7	3972	M5	12	-29			82	28	1600	0	1	0	0	0	0.07	0.01	4.5	0.79	0.08	127	0	0	0	0	0	
8	3972.15	M6	12	-28			82	28	1600	0	1	0	0	0.976	0.00	0.00	4.5	0.57	0.06	127	0	0	0	0	0	
9	3972.3	M7	12	-27			82	28	1600	0	1	0	0	0.976	0.29	0.03	0	0.65	0.07	127	0	0	0	0	0	
10	3972.45	M8	12	-26			82	28	1600	0	1	0	0	0	0.00	0.00	0	0.22	0.02	127	0	0	0	0	0	
11	3972.6	M9	12	-25			82	28	1600	0	1	0	0	-0.488	0.00	0.00	0	0.00	0.00	127	0	0	0	0	0	
12	3972.75	M10	12	-24			82	28	1600	0	1	0	0	-0.488	0.22	0.02	0	0.36	0.04	127	0	0	0	0	0	
13	3972.9	M11	12	-23			82	27.5	1600	0	1	0	0	-0.488	0.22	0.02	0	0.22	0.02	127	0	0	0	0	0	
14	3973.05	M12	12	-22			82	28	1600	0	1	0	0	-0.488	-0.22	-0.02	0	0.00	0.00	127	0	0	0	0	0	
15	3973.2	M13	12	-21			82	27.5	1600	0	1	0	0	-0.488	0.07	0.01	0	0.22	0.02	127	0	0	0	0	0	
16	3973.35	M14	12	-20			82	27.5	1600	0	1	0	0	0	0.14	0.01	0	0.29	0.03	127	0	0	0	0	0	
17	3973.5	M15	12	-19			82	27	1600	0	1	0	0	-0.488	0.07	0.01	0	0.29	0.03	127	0	0	0	0	0	
18	3973.65	M16	12	-18			82	26	1600	0	1	0	0	0	0.22	0.02	0	0.29	0.03	127	0	0	0	0	0	
19	3973.8	M17	12	-17			82	25.5	1600	0	1	0	0	-0.488	0.29	0.03	0	0.57	0.06	127	0	0	0	0	0	
20	3973.95	M18	12	-16			82	24	1600	0	1	0	0	-0.488	0.00	0.00	0	0.36	0.04	127	0	0	0	0	0	
21	3974.1	M19	12	-15			82	23.5	1500	0	1	0	0	-0.488	-0.07	-0.01	0	-0.07	-0.01	127	0	0	0	0	0	
22	3974.25	M20	12	-14			82	23	1500	0	1	0	0	-0.488	-0.07	-0.01	0	0.36	0.04	127	0	0	0	0	0	
23	3974.4	M21	12	-13			82	22	1500	0	1	0	0	-0.488	0.14	0.01	0	0.36	0.04	127	0	0	0	0	0	
24	3974.55	M22	12	-12			82	21.5	1400	0	1	0	0	-0.488	0.07	0.01	0	0.22	0.02	127	0	0	0	0	0	
25	3974.7	M23	12	-11			82	21.5	1400	0	1	0	0	0	-0.07	-0.01	0	0.36	0.04	127	0	0	0	0	0	
26	3974.85	M24	12	-10			82	21.5	1400	0	1	0	0	-0.488	-0.07	-0.01	0	0.50	0.05	127	0	0	0	0	0	
27	3975	M25	12	-9			82	21	1400	0	1	0	0	-0.488	-0.14	-0.01	0	0.43	0.04	127	0	0	0	0	0	
28	3975.15	M26	12	-8			82	21	1400	0	1	0	0	-0.488	0.00	0.00	0	0.07	0.01	127	0	0	0	0	0	
29	3975.3	M27	12	-7			82	21	1300	0	1	0	0	-0.488	0.00	0.00	0	0.00	0.00	127	0	0	0	0	0	
30	3975.45	M28	12	-6			82	21	1300	0	1	0	0	-0.488	0.00	0.00	0	0.07	0.01	127	0	0	0	0	0	
31	3975.6	M29	12	-5			82	21	1300	0	1	0	0	0	0.00	0.00	0	0.22	0.02	127	0	0	0	0	0	
32	3975.75	M30	12	-4			82	20	1300	0	1	0	0	-0.976	-0.29	-0.03	-4.5	-0.07	-0.01	127	0	0	0	0	0	
33	3975.9	M31	12	-3			82	20	1300	0	1	0	0	-2.44	0.00	0.00	-9	-0.36	-0.04	127	0	0	0	0	0	
34	3976.05	M32	12	-2			82	20	1300	0	1	0	0	-3.904	-0.14	-0.01	-18	-1.22	-0.12	127	0	0	0	0	0	
35	3976.2	M33	12	-1			78	22.5	1300	0	1	0	0.048	29.28	-8.97	-0.91	-36	-3.45	-0.35	127	0	0	0	0	0	
36	3976.3	T	12	0	959	38039	74	17.5	1200	0	1	0	0	20.008	-1.65	-0.17	-27	1.51	0.15	127	0	1	0	0	1	
37	3976.35	N0	12	1			71	17.5	1300	0	1	0	0	1.464	-4.81	-0.49	-22.5	-4.38	-0.45	127	0	1	0	0	1	
38	3976.5	N1	12	2			70	0	1500	0	1	1	0	8.296	-1.08	-0.11	-49.5	7.18	0.73	127	0	1	0	0	1	
39	3976.65	N2	12	3			67	0	1500	0	1	0	0	-18.544	-1.15	-0.12	-67.5	-2.58	-0.26	127	0	0	0	0	1	
40	3976.8	N3	12	4			67	0	1400	0	1	0	0	-23.424	-1.51	-0.15	-36	-3.01	-0.31	127	0	0	0	0	1	
41	3976.95	N4	12	5			67	25	1300	0	1	0	0	-11.712	-1.00	-0.10	18	-1.87	-0.19	127	0	0	0	0	1	
42	3977.1	N5	12	6			66	50.5	1200	0	1	0	0	6.832	-0.36	-0.04	36	1.15	0.12	127	0	0	0	0	1	
43	3977.25	N6	12	7			66	72.5	1200	0	1	0	0	8.784	-0.50	-0.05	9	0.79	0.08	127	0	0	0	0	1	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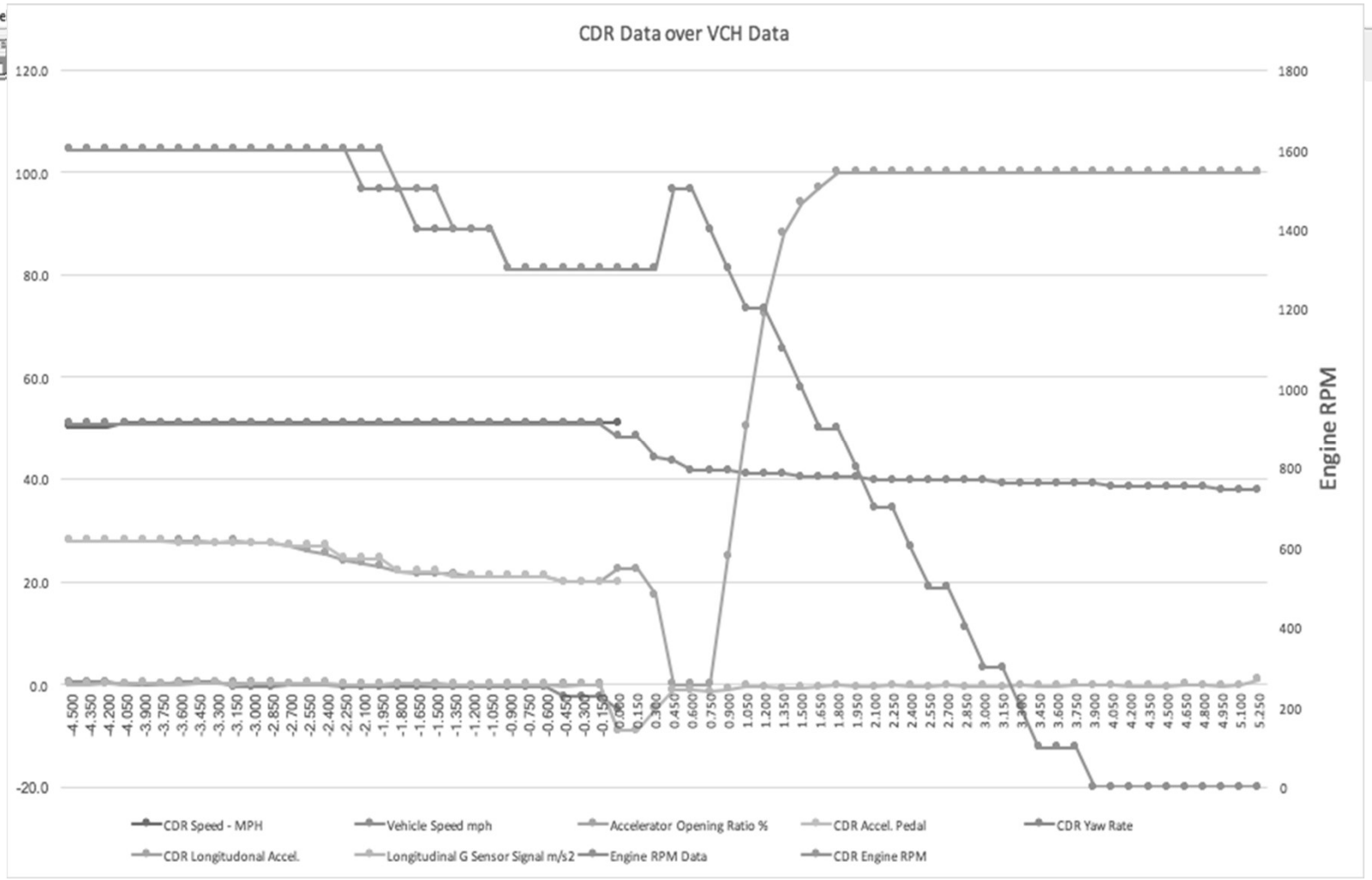
Security Warning External Data Connections have been disabled
Enable Content
CDR Data over VCH Data
Engine RPM
Lateral Data vs. Steering Input
Time in Seconds
Heading Change - degrees
Steering Signal - degrees
Lateral G Sensor Signal - ms2
Lateral Distance Cumulative - feet
VCH Code 15 - Sudden Braking
VCH Code 12 - VSC Operation
CDR Data

File Edit Run He



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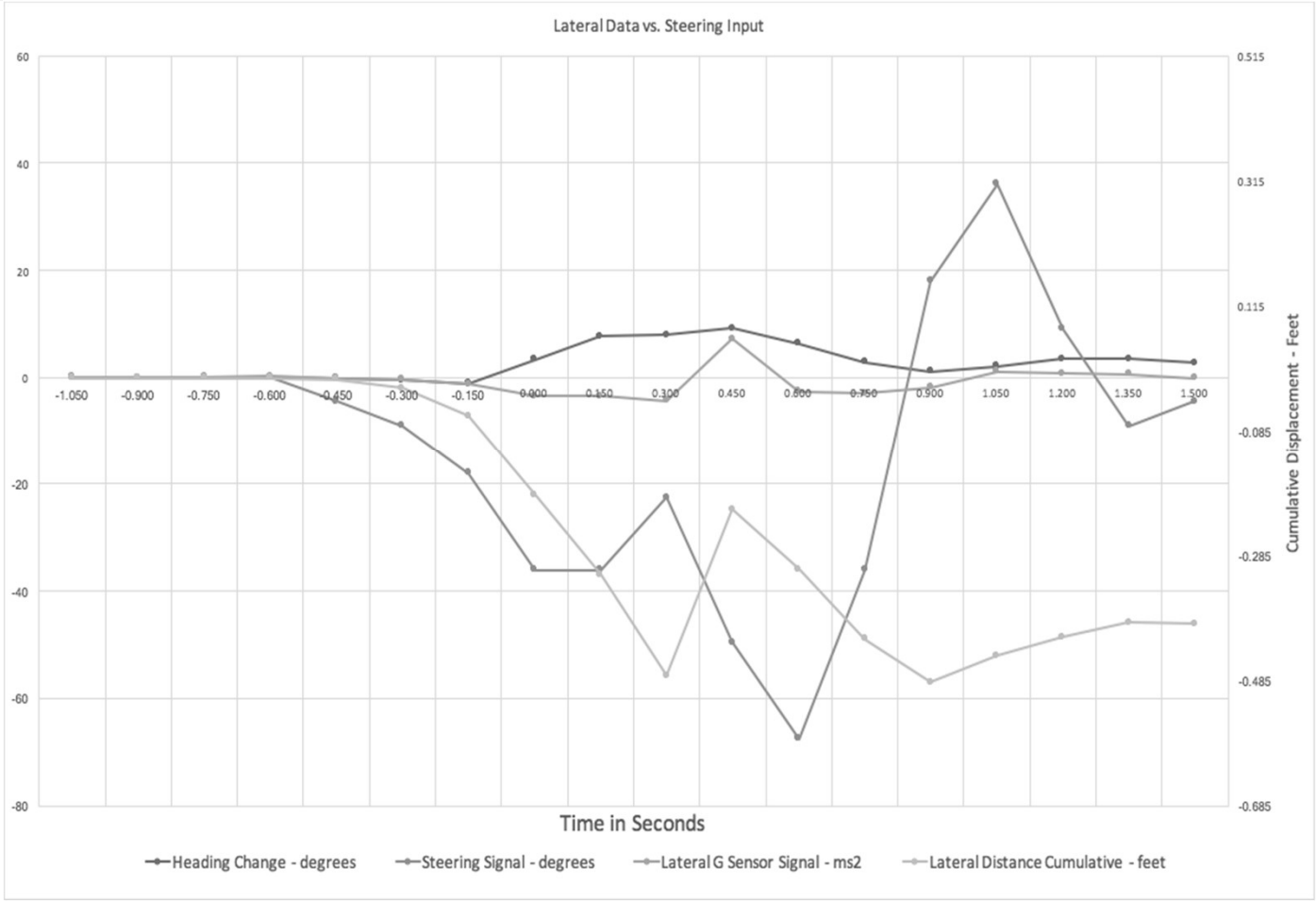

Airbag Timing – Very Important!